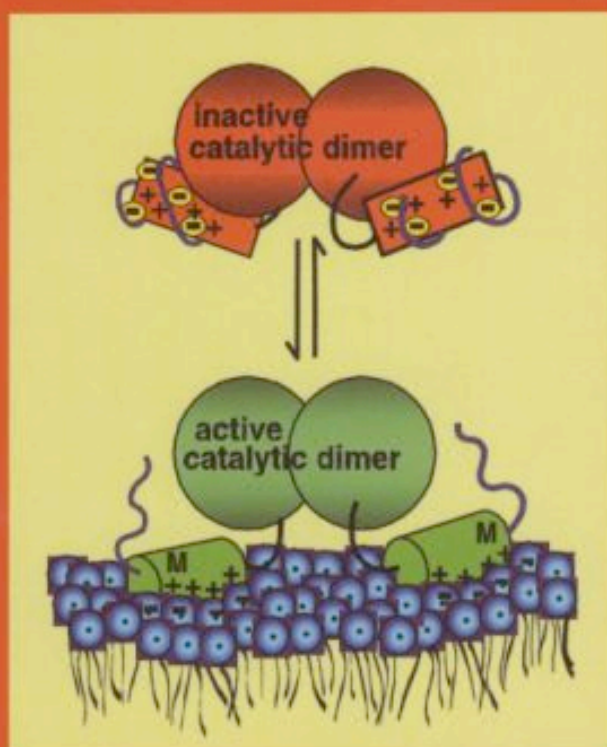


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Biochemistry of Lipids, Lipoproteins and Membranes, 4th edition

edited by
D.E. Vance and J.E. Vance



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Preface

The first edition of this textbook was published in 1985. However, research in the biochemistry and molecular biology of lipids and lipoproteins has experienced a remarkable rebirth within the past few years with the realization that lipids play important roles not only in membrane structure and the functioning of membrane proteins, but also in diseases such as heart disease, diabetes, obesity, stroke, cancer and neurological diseases. In addition, lipids are known to participate widely in signaling pathways which impact on all basic biological processes. We have therefore assembled the fourth edition of this textbook by taking account of these major advances in these fields.

The 4th edition has been written with two major objectives in mind. The first is to provide students and teachers with an advanced and up-to-date textbook covering the major areas in the fields of lipid, lipoprotein and membrane biochemistry and molecular biology. The chapters are written for students who have already taken an introductory course in biochemistry, who are familiar with the basic concepts and principles of biochemistry, and who have a general background in the area of lipid metabolism. This book should, therefore, provide the basis for an advanced course for students and teachers in the biochemistry of lipids, lipoproteins and membranes. The second objective of this book is to satisfy the need for a general reference and review book for scientists studying lipids, lipoproteins and membranes. Our goal was to provide a clear summary of these research areas for scientists presently working in, or about to enter, these and related fields. This book remains unique in that it is not a collection of exhaustive reviews on the various topics, but rather is a current, readable and critical summary of these areas of research. This book should allow scientists to become familiar with recent discoveries related to their own research interests, and should also help clinical researchers and medical students keep abreast of developments in basic science that are important for clinical advances in the future.

All of the chapters have been extensively revised since the third edition appeared in 1996. New chapters have been added on lipid modifications of proteins, bile acids, lipoprotein structure, and the relation between lipids and atherosclerosis. We have not attempted to describe in detail the structure and function of biological membranes or the mechanism of protein assembly into membranes since these topics are covered already in a number of excellent books. The first chapter, however, contains a summary of the principles of membrane structure as a basis for the subsequent chapters.

Excellent up-to-date reviews are available on all the topics included in this book and many of these reviews are cited in the relevant chapters. We have limited the number of references cited at the end of each chapter and have emphasized review articles. In addition, the primary literature is cited in the body of the text by providing the name of

one author and the year in which the work was published. Using this system, readers will readily be able to find the original citation via computer searching.

The editors and contributors assume full responsibility of the content of the various chapters and we would be pleased to receive comments and suggestions for future editions of this book.

Dennis and Jean Vance
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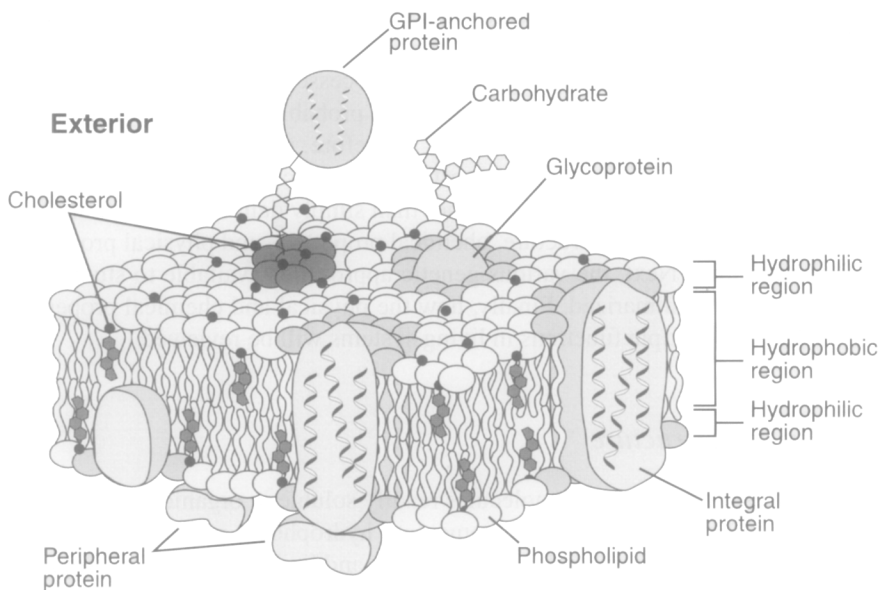
Functional roles of lipids in membranes

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1. Introduction and overview

Lipids as a class of molecules display a wide diversity in both structure and biological function. A primary role of lipids in cellular function is in the formation of the permeability barrier of cells and subcellular organelles in the form of a lipid bilayer (Fig. 1). Although the major lipid type defining this bilayer in almost all membranes is glycerol-based phospholipid, other lipids are important components and vary in their



Interior

Fig. 1. Model for membrane structure. This model of the plasma membrane of a eukaryotic cell is an adaptation of the origin model proposed by Singer and Nicholson [1]. The phospholipid bilayer is shown with integral membrane proteins largely containing α -helical transmembrane domains. Peripheral membrane proteins associate either with the lipid surface or with other membrane proteins. Lipid rafts (dark gray headgroups) are enriched in cholesterol and contain a glycosylphosphatidylinositol-linked (GPI) protein. The light gray headgroups depict lipids in close association with protein. The irregular surface and wavy acyl chains denote the fluid nature of the bilayer.

presence and amounts across the spectrum of organisms. Sterols are present in all eukaryotic cytoplasmic membranes and in a few bacterial membranes. The ceramide-based sphingolipids are also present in the membranes of all eukaryotes. Neutral glycerol-based glycolipids are major membrane-forming components in many Gram-positive bacteria and in the membranes of plants while Gram-negative bacteria utilize a glucosamine-based phospholipid (Lipid A) as a major structural component of the outer membrane. Additional diversity results in the variety of the hydrophobic domains of lipids. In eukaryotes and eubacteria these domains are usually long chain fatty acids or alkyl alcohols with varying numbers and positions of double bonds. In the case of archaeobacteria, the phospholipids have long chain reduced polyisoprene moieties, rather than fatty acids, in ether linkage to glycerol. If one considers a simple organism such as *Escherichia coli* with three major phospholipids and several different fatty acids along with many minor precursors and modified products, the number of individual phospholipid species ranges in the hundreds. In more complex eukaryotic organisms with greater diversity in both the phospholipids and fatty acids, the number of individual species is in the thousands.

If one or two phospholipids are sufficient to form a stable bilayer structure, why is the above diversity in lipid structures present in biological membranes [2]? The adaptability and flexibility in membrane structure necessitated by environment is possible only with a broad spectrum of lipid mixtures. The membrane is also the supporting matrix for a wide spectrum of proteins involved in many cellular processes. Approximately 20–35% of all proteins are integral membrane proteins, and probably half of the remaining proteins function at or near a membrane surface. Therefore, the physical and chemical properties of the membrane directly affect most cellular processes making the role of lipids dynamic with respect to cell function rather than simply defining a static barrier.

In this chapter, the diversity in structure, chemical properties, and physical properties of lipids will be outlined. Next, the various genetic approaches available to study lipid function in vivo will be summarized. Finally, how the physical and chemical properties of lipids relate to their multiple functions in living systems will be reviewed.

2. Diversity in lipid structure

Lipids are defined as those biological molecules readily soluble in organic solvents such as chloroform, ether, or toluene. However, some very hydrophobic proteins such as the F_0 subunits of ATP synthase are soluble in chloroform, and lipids with large hydrophilic domains such as lipopolysaccharide are not soluble in these solvents. Here we will consider only those lipids that contribute significantly to membrane structure or have a role in determining protein structure or function. The broad area of lipids as second messengers is covered in Chapters 12–14.

2.1. Glycerol-based lipids

The primary building blocks of most membranes are glycerol phosphate-containing lipids generally referred to as phospholipids (Fig. 2). The diacylglycerol backbone in

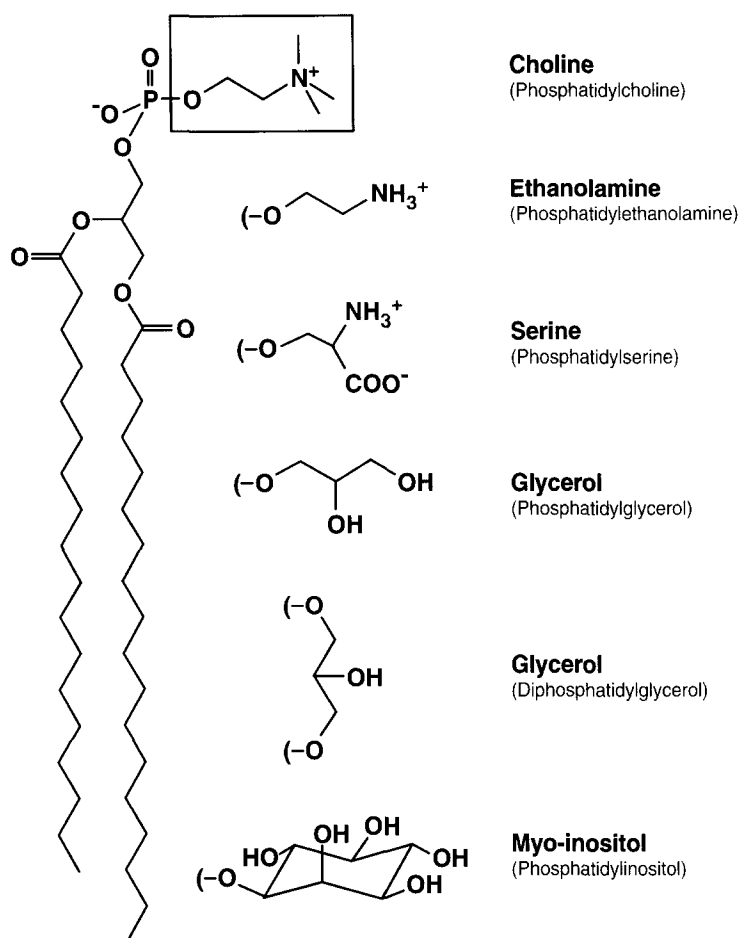


Fig. 2. Structure of glycerol phosphate-based lipids. The complete lipid structure shown is 1,2-distearoyl-*sn*-glycerol-3-phosphocholine or phosphatidylcholine (PC). Substitution of choline in the box with the headgroups listed below results in the other phospholipid structures. CDP-diacylglycerol has a CMP and phosphatidic acid has a hydroxyl group in place of choline (not shown). Diphosphatidylglycerol, which contains two phosphatidic acids joined by glycerol, is commonly referred to as cardiolipin (CL).

eubacteria and eukaryotes is *sn*-3-glycerol esterified at the 1- and 2-position with long chain fatty acids. In archaeobacteria (Fig. 3), *sn*-1-glycerol forms the lipid backbone and the hydrophobic domain is composed of phytanyl (a saturated isoprenyl) groups in ether linkage at the 2- and 3-position (an archaeol). In addition two *sn*-1-glycerol groups are found connected in ether linkage by two biphytanyl groups (dibiphytanyldiglycerophosphatetetraether) [3] to form a covalently linked bilayer. Some eubacteria (mainly hyperthermophiles) have dialkyl (long chain alcohols in ether linkage) glycerophosphate lipids and similar ether linkages are found in the plasmalogens of eukaryotes. The headgroups of the phospholipids (boxed area of Fig. 2) extend the diversity of lipids defining phosphatidic acid (PA, with OH), phosphatidylcholine (PC), phosphatidylserine

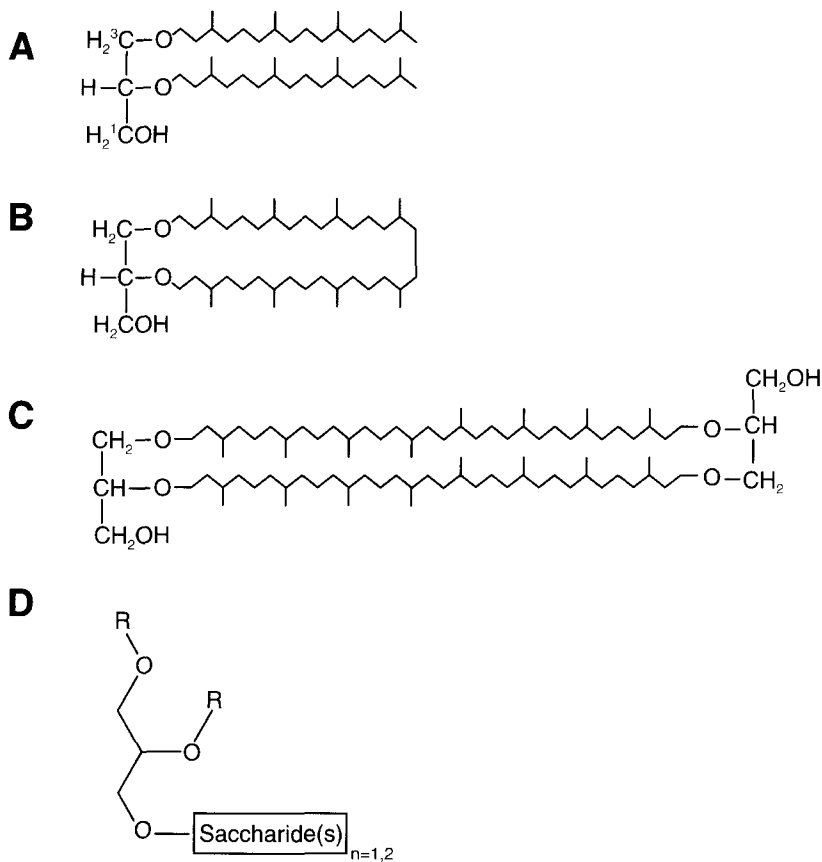


Fig. 3. Structure of dialkylglycerols in archaeobacteria. Archaeobacteria have phytanyl chains in ether linkage to the 2- and 3-positions of *sn*-1-glycerol (archaeol). The 1-position can be derivatized with phosphodiester. (A) Diphytanylglycerol (C20–C20 diether) with the stereochemistry of glycerol indicated. (B) Cyclic biphytanyl (C40) diether. (C) Biphytanyl diglycerol diether. (D) A glycolipid with either a mono- or disaccharide (glucose or galactose) at the 1-position of *sn*-1-glycerol. The R groups are ether-linked phytanyl chains. Similar glycolipids are found in eubacteria and plants with a *sn*-3-glycerol backbone and ester-linked fatty acid chains at the 1- and 2-positions.

(PS), phosphatidylglycerol (PG), phosphatidylinositol (PI), and cardiolipin (CL). Archaeobacteria analogues exist with headgroups of glycerol and glycerolmethylphosphate as well as all of the above except PC and CL (Chapter 3). Archaeobacteria also have neutral glycolipid derivatives in which mono- and disaccharides (glucose or galactose) are directly linked to *sn*-1-archaeol (Fig. 3). Plants (mainly in the thylakoid membrane) and many Gram-positive bacteria also have high levels of neutral glycolipids with mono- or disaccharides linked to the 3-carbon of *sn*-3-diacylglycerol (Chapter 4). Therefore, the diversity of glycerol-based lipids in a single organism is significant, but the diversity throughout nature is enormous. The lipid composition of various biological membranes is shown in Table 1.

Table 1
Lipid composition of various biological membranes

Lipid	Erythrocyte ^b	Myelin ^b	Mitochondria ^c	Endoplasmic reticulum ^c	<i>E. coli</i> ^d
Cholesterol	23	22	3	6	–
PE	18	15	35	17	70
PC	17	10	39	40	–
Sphingomyelin	18	8	–	5	–
PS	7	9	2	5	–
PG	–	–	–	–	20
CL	–	–	21	–	10
Glycolipid ^a	3	28	–	–	–
Others	13	8	–	27	–

The data are expressed as weight % of total lipid.

^a Ceramide based.

^b Human sources.

^c Rat liver. Inner and outer mitochondrial membrane.

^d Inner and outer membrane excluding Lipid A.

The majority of information on the chemical and physical properties of lipids comes from studies on the major phospholipid classes of eubacteria and eukaryotes with only limited information on the lipids from archaeobacteria. The biosynthetic pathways and the genetics of lipid metabolism have also been extensively studied in eubacteria (Chapter 3) and eukaryotes (Chapter 8). Clearly the archaeol lipids confer some advantage with respect to the environment of archaeobacteria. Many of these organisms exist in harsh environments that call for more chemically stable lipid bilayers which is afforded by the above lipids. How the physical properties of the more commonly studied lipids change with environment will be discussed later.

2.2. Diglucoseamine phosphate-based lipids

The outer membrane of Gram-negative bacteria (Fig. 4) contains a lipid made up of a headgroup derived from glucosamine phosphate (Chapter 3). The core lipid (Lipid A, see Fig. 5 and Chapter 3) in *E. coli* is a phospholipid containing two glucoseamine groups in $\beta(1-6)$ linkage that are decorated at positions 2, 3, 2' and 3' with *R*-3-hydroxymyristic acid (C14) and at positions 1 and 4' with phosphates. Further modification at position 6' with a KDO disaccharide (two 3-deoxy-D-manno-octulosonic acids in $\alpha(1-3)$ linkage) results in KDO₂-Lipid A that is further modified by an inner core, an outer core, and the O-antigen. Laboratory strains of *Salmonella typhimurium* and *E. coli* such as K-12 lack the O-antigen found in the wild-type and clinically important strains.

The complete structure either with or without O-antigen is referred to as lipopolysaccharide or LPS. The core Lipid A forms the outer monolayer of the outer membrane bilayer of Gram-negative bacteria; the inner monolayer of the outer membrane (Fig. 4) is made up of glycerophosphate-based lipids. The whole lipopolysaccharide structure defines the outer surface of Gram-negative bacteria, but only the KDO₂-Lipid A struc-

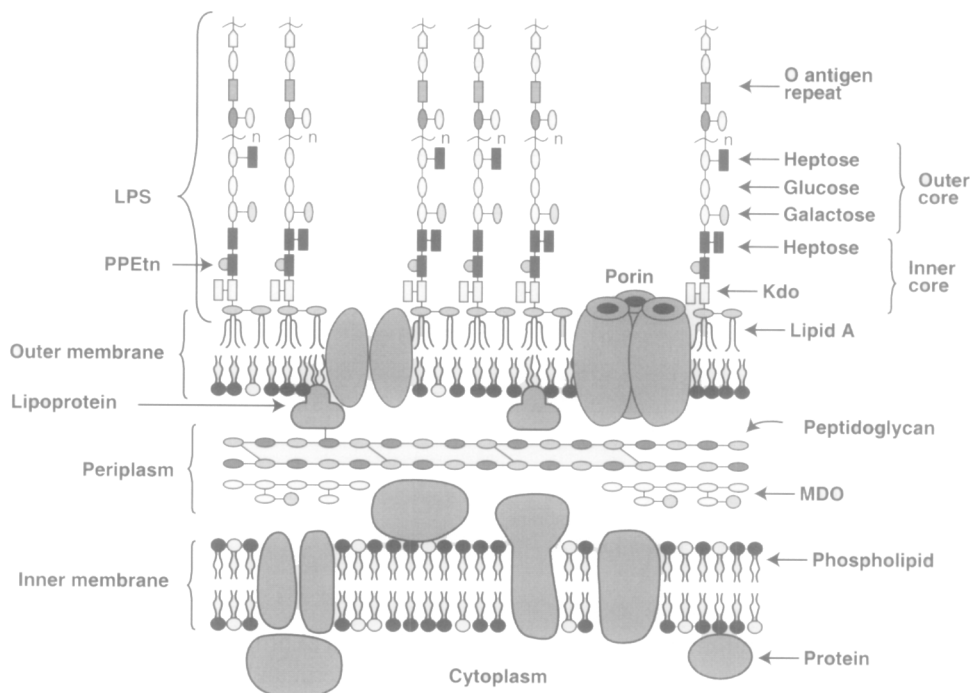


Fig. 4. *E. coli* cell envelope. The complete cell envelope of Gram-negative bacteria contains an inner membrane that is a typical phospholipid bilayer and is the permeability barrier of the cell. The outer membrane is composed of an inner monolayer of phospholipid and an outer monolayer of the Lipid A portion of lipopolysaccharide (LPS). The structure of KDO₂-Lipid A is shown in Fig. 5 and is connected to a polysaccharide to build up the inner core, outer core and the O antigen repeat. PPEtn is ethanolamine pyrophosphate. The outer membrane is a permeability barrier for molecules larger than 750–1000 Da that pass through various pores in the outer membrane. The periplasmic space contains many proteins and the membrane-derived oligosaccharide (MDO) that is one component of the osmolarity regulatory system. MDO is decorated with *sn*-glycerol-1-phosphate and ethanolamine phosphate derived from PG and PE, respectively. The amino acid–sugar crosslinked peptidoglycan gives structural rigidity to the cell envelope. One-third of the lipoproteins (*lpp* gene product) is covalently linked via its carboxyl terminus to the peptidoglycan and in complex with the remaining lipoproteins as trimers that associate with the outer membrane via covalently linked fatty acids at the amino terminus. The amino terminal cysteine is blocked with a fatty acid, derived from membrane phospholipids, in amide linkage and is derivatized with diacylglycerol, derived from PG, in thioether linkage. Figure is courtesy of C.R.H. Raetz.

ture is essential for viability of laboratory strains. However, the remainder of the lipopolysaccharide structure is important to survival of Gram-negative bacteria in their natural environment. This structure is modified post-assembly in response to environment including host fluids, temperature, ionic properties, and antimicrobial agents [4]. In addition, both enteric and non-enteric Gram-negative bacteria show a great diversity in all component parts of the LPS structure. Studies of Lipid A biosynthesis is of clinical importance because it is the primary antigen responsible for toxic shock syndrome caused by Gram-negative bacterial infection.

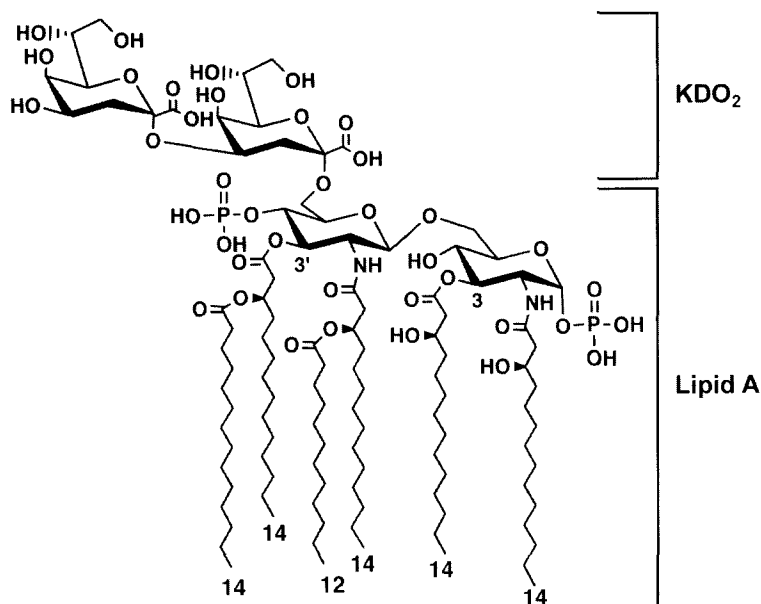


Fig. 5. Structure of KDO₂-Lipid A. Lipid A is a disaccharide of glucosamine phosphate that is multiply acylated in both amide and ester linkages with fatty acids of the chain lengths indicated (12 and 14). As illustrated in Fig. 4, Lipid A is attached to KDO₂ that is then elongated with the remainder of lipopolysaccharide structure. Figure is courtesy of C.R.H. Raetz.

3. Properties of lipids in solution

The matrix that defines a biological membrane is a lipid bilayer composed of a hydrophobic core excluded from water and an ionic surface that interacts with water and defines the hydrophobic–hydrophilic interface (Fig. 1). Much of our understanding of the physical properties of lipids in solution and the driving force for the formation of lipid bilayers comes from the concept of the ‘hydrophobic effect’ as developed by Charles Tanford [5]. The ‘fluid mosaic’ model for membrane structure further popularized these concepts [1]. This model, since extensively refined, envisioned membrane proteins as undefined globular structures freely moving in a homogeneous sea of lipids.

3.1. Why do polar lipids self-associate?

Polar lipids are amphipathic in nature containing both hydrophobic domains, which do not interact with water, and hydrophilic domains that readily interact with water. The basic premise of the hydrophobic effect is that the hydrocarbon domains of polar lipids disrupt the stable hydrogen bonded structure of water and therefore are at an energy minimum when such domains self associate to minimize the total surface area in contact with water. The polar domains of lipids interact either through hydrogen bonding or

ionic interaction with water and therefore are energetically stable in an aqueous environment. The structural organization that a polar lipid assumes in water is determined by its concentration and the law of opposing forces, i.e. hydrophobic forces driving self-association versus steric and ionic repulsive forces of the polar domains in opposing self-association. At low concentrations, amphipathic molecules exist as monomers in solution. As the concentration of the molecule increases, its stability in solution as a monomer decreases until the favorable interaction of the polar domain with water is outweighed by the unfavorable interaction of the hydrophobic domain with water. At this point, a further increase in concentration results in the formation of increasing amounts of self-associated monomers in equilibrium with a constant amount of free monomer. This point of self-association and the remaining constant free monomer concentration is the critical micelle concentration [6]. The larger the hydrophobic domain, the lower the critical micelle concentration due to the increased hydrophobic effect. However, the larger the polar domain, either because of the size of neutral domains or charge repulsion for ionic domains, the higher the critical micelle concentration due to the unfavorable steric hindrance in bringing these domains into close proximity. The critical micelle concentration of amphipathic molecules with a net charge is influenced by ionic strength of the medium due to dampening of the charge repulsion effect. Therefore, the critical micelle concentration of the detergent sodium dodecyl sulfate is reduced ten-fold when the NaCl concentration is raised from 0 to 0.5 M.

These physical properties and the shape of amphipathic molecules define three supramolecular structural organizations of polar lipids and detergents in solution (Fig. 6). Detergents, lysophospholipids (containing only one alkyl chain), and phospholipids with short alkyl chains (eight or fewer carbons) have an inverted cone-shape (large head group relative to a small hydrophobic domain) and self associate above the critical micelle concentration with a small radius of curvature to form micellar structures with a hydrophobic core excluding water. The micelle surface, rather than being a smooth spherical or elliptical structure with the hydrophobic domains completely sequestered inside a shell of polar residues that interact with water, is a very rough surface with many of the hydrophobic domains exposed to water. The overall structure reflects the packing of amphipathic molecules at an energy minimum by balancing the attractive force of the hydrophobic effect and the repulsive force of close headgroup association. The critical micelle concentration for most detergents ranges from micromolar to millimolar. Lysophospholipids also form micelles with critical micelle concentrations in the micromolar range. However, phospholipids with chain lengths of 14 and above self associate at a concentration around 10^{-10} M due to the hydrophobic driving force contributed by two alkyl chains. Phospholipids with long alkyl chains do not form micelles but organize into bilayer structures, which allow tight packing of adjacent side chains with the maximum exclusion of water from the hydrophobic domain. In living cells, phospholipids are not found free as monomers in solution, but are organized into either membrane bilayers or protein complexes. When long chain phospholipids are first dried to a solid from organic solvent and then hydrated, they spontaneously form large multilamellar bilayer sheets separated by water. Sonication disperses these sheets into smaller unilamellar bilayer structures that satisfy the hydrophobic nature of the ends of the bilayer by closing into sealed vesicles (also termed liposomes) defined by

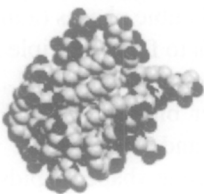
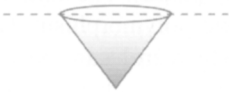
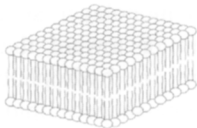
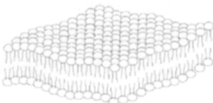
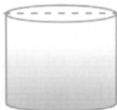
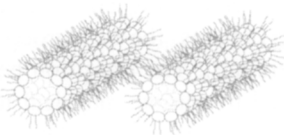
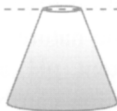
LIPIDS	PHASE	MOLECULAR SHAPE
Lysophospholipids Detergents	 Micellar	 Inverted Cone
Phosphatidylcholine Sphingomyelin Phosphatidylserine Phosphatidylinositol Phosphatidylglycerol Phosphatidic Acid Cardiolipin Digalactosyldiglyceride	 L_{β}  L_{α} Bilayer	 Cylindrical
Phosphatidylethanolamine Cardiolipin - Ca^{2+} Phosphatidic Acid - Ca^{2+} Phosphatidic Acid ($\text{pH} < 3.0$) Phosphatidylserine ($\text{pH} < 4.0$) Monogalactosyldiglyceride	 Hexagonal (H_{II})	 Cone

Fig. 6. Polymorphic phases and molecular shapes exhibited by lipids. The space filling model for the micellar phase is of β -D-octyl glucoside micelle (50 monomers). The polar portions of the detergent molecules (oxygen atoms are black) do not cover completely the micelle surface (hydrocarbons in gray) leaving substantial portions of the core exposed to bulk solvent. Inverted cone-shaped molecules form micelles. Model adapted with permission from Garavito and Ferguson-Miller [6]. Copyright 2001 The American Society for Biochemistry and Molecular Biology. Polar lipids with two long alkyl chains adopt a bilayer or non-bilayer (H_{II}) structure depending on the geometry of molecule (cylindrical or cone-shaped, respectively) and environmental conditions. The L_{β} (order gel) and L_{α} (liquid crystalline) bilayer phases differ in the order within the hydrophobic domain and in mobility of the individual molecules.

a continuous single bilayer and an aqueous core much like the membrane surrounding cells. Liposomes can also be made by physical extrusion of lamellar structures through a small orifice or by dilution of a detergent–lipid mixture below the critical micelle concentration of the detergent.

Cylindrical shaped lipids (head group and hydrophobic domains of similar diameter) such as PC form lipid bilayers. Cone-shaped lipids (small head groups relative to a large hydrophobic domain) such as PE (unsaturated fatty acids) favor an inverted micellar

structure where the headgroups sequester an internal aqueous core and the hydrophobic domains are oriented outward and self-associate in non-bilayer structures. These are denoted as the hexagonal II (H_{II}) and cubic phases (a more complex organization similar to the H_{II} phase). The ability of lipids to form multiple structural associations is referred to as lipid polymorphism. Lipids such as PE, PA, CL, and monosaccharide derivatives of diacylglycerol can exist in either bilayer or the H_{II} phase, depending on solvent conditions, alkyl chain composition, and temperature.

Both cone-shaped and inverted cone-shaped lipids are considered as non-bilayer-forming lipids and when mixed with the bilayer-forming lipids change the physical properties of the bilayer and introduce stress or strain in the bilayer structure. When bilayer-forming lipids are spread as a monolayer at an aqueous-air interface, they have no tendency to bend away from or toward the aqueous phase due to their cylindrical symmetry. In such a system, the hydrophobic domain orients toward the air. Monolayers of the asymmetric cone-shaped lipids (H_{II} -forming) tend to bend toward from the aqueous interface (negative radius of curvature) while monolayers of asymmetric inverted cone shaped lipids (micelle-forming) tend to bend away from the aqueous phase (positive radius of curvature). The significance of shape mis-match in lipid mixtures will be covered later.

3.2. *Physical properties of membrane bilayers*

The organization of diacylglycerol-containing polar lipids in solution (Fig. 6) is dependent on the nature of the alkyl chains, the headgroups, and the solvent conditions (i.e., ion content, pH, and temperature). The transition between these phases for pure lipids in solution can be measured by various physical techniques such as ^{31}P -NMR and microcalorimetry. The difference between the ordered gel (L_{β}) and liquid crystalline (L_{α}) phases is the viscosity or fluidity of the hydrophobic domains of the lipids which is a function of temperature and the alkyl chain structure. At any given temperature the 'fluidity' (the inverse of the viscosity) of the hydrocarbon core of the bilayer increases with increasing content of unsaturated or branched alkyl chain or with decreasing alkyl chain length. Due to the increased mobility of the fatty acid chains with increasing temperature, the fluidity and also space occupied by the hydrophobic domain of lipids also increases. A bilayer-forming lipid such as PC assumes a cylindrical shape over a broad temperature range and with different alkyl chain compositions. When analyzed in pure form, PC exists in either the L_{β} or L_{α} phase mainly dependent on the alkyl chain composition and the temperature. Non-bilayer-forming lipids such as PE exist at low temperatures in the L_{β} phase, at intermediate temperatures in the L_{α} phase, and at elevated temperatures in the H_{II} or cubic phase (Fig. 7). The last transition is temperature dependent but also depends on the shape of the lipid. The shape of lipids with relatively small head groups can change from cylindrical to conical (H_{II} phase) with increasing unsaturation or length of the alkyl chains or with increasing temperature. As can be seen from Fig. 7, the midpoint temperature (T_m) of the transition from the L_{β} to L_{α} phase increases with an increase in the length of the fatty acids, but the midpoint of the transition temperature (T_{LH}) between the L_{α} and H_{II} phases decreases with increasing chain length (or increasing unsaturation, not shown).

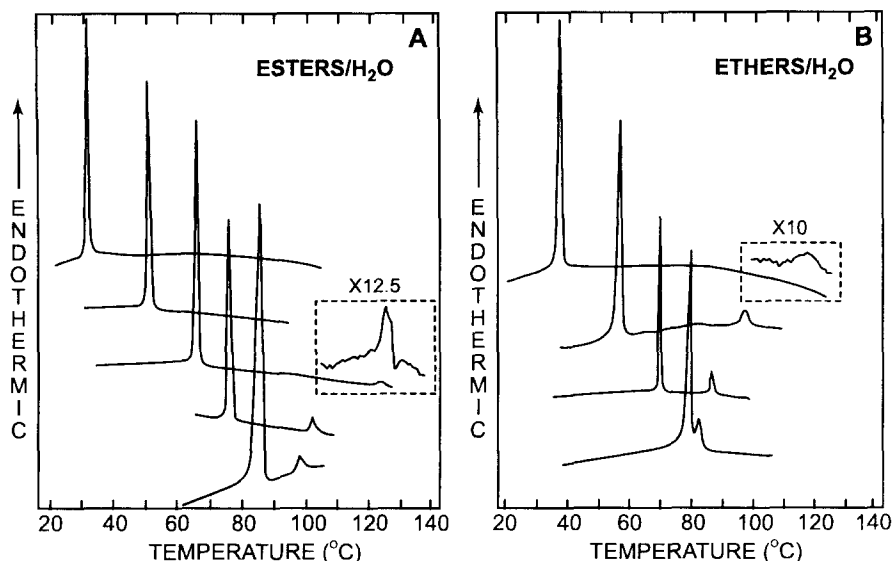


Fig. 7. Phase behavior of PE as a function of temperature and chain length. As hydrated lipids pass through a phase transition heat is absorbed as indicated by the peaks. The large peaks at the lower temperatures are due to the L_{β} to L_{α} transition and the smaller peaks at higher temperatures are due to the L_{α} to H_{II} transition. (A) Even numbered diacyl-PEs ranging from C12 to C20 top to bottom. (B) Even numbered dialkyl-PEs in ether linkage ranging from C12 to C18 top to bottom. The inserts indicate an expanded scale for the transition to H_{II} . Figure adapted with permission from Seddon et al. [7]. Copyright 1983 American Chemical Society.

Similar transition plots as well as complex phase diagrams have been generated with mixtures of lipids. The physical property of a lipid mixture is a collective property determined by each of the component lipids. A large number of studies indicate that the L_{α} state of the membrane bilayer is required for cell viability and cells adjust their lipid composition in response to many environmental factors so that the collective property of the membrane exhibits the L_{α} state. Addition of non-bilayer-forming lipids to bilayer-forming lipids can result in non-bilayer formation, but at a higher temperature than for the pure non-bilayer-forming lipid. Addition of non-bilayer-forming lipids also adds another parameter of tension between the two monolayers. These lipids in each half of the bilayer tend to reduce the radius of curvature of each monolayer that results in a tendency to pull the bilayer apart by curving the monolayers away from each other (see the end of Section 3.1). This process results in potential energy residing in the bilayer that is a function of the presence of non-bilayer lipids. Forcing non-bilayer-forming lipids into a bilayer structure also exposes the hydrophobic core to the aqueous phase. Mixtures of lipids with dissimilar phase properties can also generate phase separations with local domain formation. Such discontinuities in the bilayer structure may be required for many structural organizations and cellular processes such as accommodation of proteins into the bilayer, movement of macromolecules across the bilayer, cell division, and membrane fusion and fission events. The need for bilayer discontinuity may be the reason that all natural membranes contain a significant

proportion of non-bilayer-forming lipids even though the membrane under physiological conditions is in the L_{α} phase.

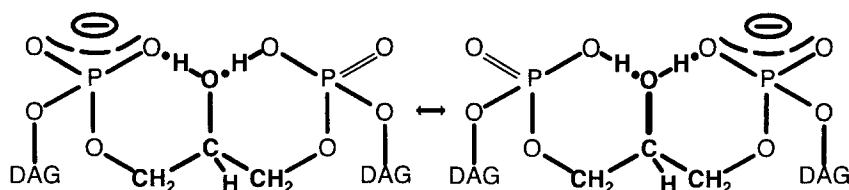
Addition of cholesterol to lipid mixtures has a profound effect on the physical properties of a bilayer. Increasing amounts of cholesterol inhibit the organization of lipids into the L_{β} phase and favor a less fluid but more ordered structure than the L_{α} phase resulting in the lack of a phase transition normally observed in the absence of cholesterol. The solvent surrounding the lipid bilayer also influences these transitions primarily by affecting the size of the headgroup relative to the hydrophobic domain. Ca^{2+} and other divalent cations (Mg^{2+} , Sr^{2+} , but not Ba^{2+}) reduce the effective size of the negatively charged headgroups of CL and PA allowing organization into the H_{II} phase. Low pH has a similar effect on the headgroup of PS. Since Ca^{2+} is an important signaling molecule that elicits many cellular responses, it is possible that part of its effects may be transmitted through changes in the physical properties of membranes. In eukaryotes, CL is found almost exclusively in the inner membrane of the mitochondria where Ca^{2+} fluxes play important regulatory roles.

3.3. Special properties of cardiolipin

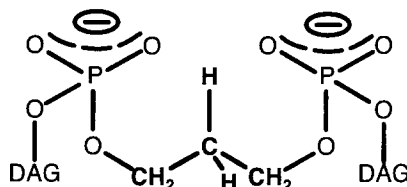
CL has the unique property of being both a bilayer and non-bilayer lipid depending on the absence or presence of divalent cations. CL is found almost exclusively in eukaryotic mitochondria and in bacteria that utilize oxidative phosphorylation for proton pumping across the membrane. A property of CL that has gone largely unrecognized is the ionization constants of its two phosphate diesters. Rather than displaying two pK values in the range of 2–4, pK_2 of CL is >8.5 [8] indicating that CL is protonated at physiological pH (Fig. 8). This property may make CL a proton sink or a conduit for protons in transfer processes. Although PG appears to substitute for CL in many processes in both bacteria and yeast, lack of CL results in a reduction in cell growth dependent on oxidative processes. Therefore, CL is not absolutely essential, but it appears to be required for optimal cell function.

3.4. What does the membrane bilayer look like?

The functional properties of natural fluid bilayers not only include the hydrophobic core and the hydrophilic surface but the interfacial region containing bound water and ions. Fig. 9A shows the distribution of the component parts of dioleoylphosphatidylcholine across the bilayer [9] and illustrates the dynamic rather than static nature of the membrane. The bilayer thickness of 30 Å is defined by the length of the fatty acid chains. However, the thickness is not a static number as indicated by the probability of finding CH_2 residues outside of this limit. Bilayer thickness can vary over the surface of a membrane if microdomains of lipids are formed with different alkyl chain lengths. What is generally not appreciated is the width (15 Å on either side of the bilayer) of the interface region between the hydrocarbon core and the free water phase of the cytosol. This region contains a complex mixture of chemical species defined by the ester linked glycerophosphate moiety, the variable headgroups, and bound water and ions. Many biological processes occur within this interface region and are dependent on its unique



CARDIOLIPIN ACID-ANION STRUCTURE



DEOXYCARDIOLIPIN ANION STRUCTURE

Fig. 8. Ionization state of CL and deoxy-CL at physiological pH. CL is only partially ionized under these conditions ($pK_2 > 8.5$) and therefore can trap a proton by hydrogen bonding with the *sn*-2 hydroxyl of glycerol that joins the two phosphatidic acids in the CL structure. Deoxy-CL lacking the *sn*-2 hydroxyl is fully ionized and cannot trap a proton.

properties including the steep polarity gradient (Fig. 9B) within which surface bound cellular processes occur.

4. Engineering of membrane lipid composition

Given the diversity in both lipid structure and function, how can the role of a given lipid be defined at the molecular level? Unlike proteins, lipids have neither inherent catalytic activity nor obvious functions in isolation (except for their physical organization). Many functions of lipids have been uncovered serendipitously based on their effect on catalytic processes or biological functions studied *in vitro*. Although considerable information has accumulated with this approach, such studies are highly prone to artifacts. The physical properties of lipids are as important as their chemical properties in determining function. Yet there is little understanding of how the physical properties of lipids measured *in vitro* relate to their *in vivo* function. In addition, the physical properties of lipids have been ignored in many *in vitro* studies. Genetic approaches are generally the most useful in studying *in vivo* function, but this approach has considerable limitations when applied to lipids. First, genes do not encode lipids, and in order to make mutants with altered lipid composition, the genes encoding enzymes along a biosynthetic pathway must be targeted. Therefore, the results of genetic mutation are indirect and many times far removed from the primary lesion. Second, a primary

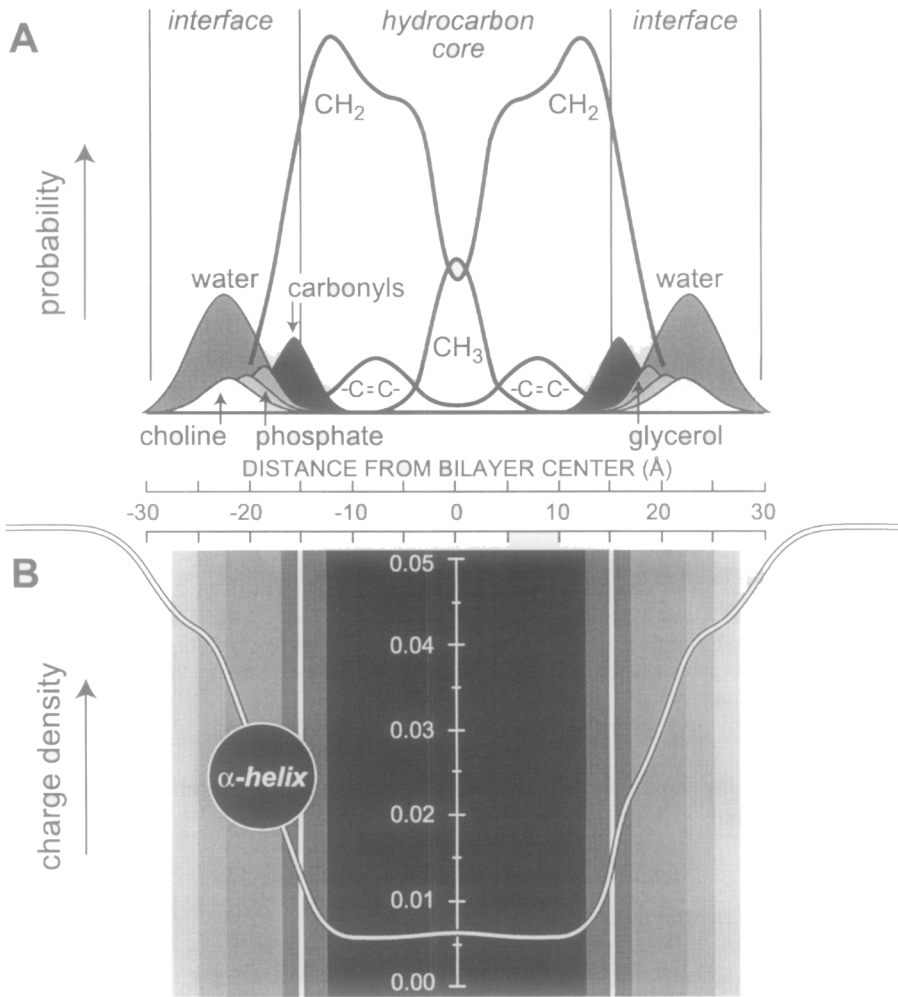


Fig. 9. The probability distribution for chemical constituents across a bilayer of PC. (A) The diagram was generated from X-ray and neutron diffraction data. The interface region between the hydrocarbon core and the free solvent region extends for approximately 15 Å on either side of the 30-Å-thick bilayer. The width of each peak defines the mobility of each constituent of PC. (B) As an α -helical peptide moves from either side of the bilayer towards the center, the charge density of the environment steeply declines as indicated by the line. Figure adapted with permission from White et al. [9]. Copyright 2001 American Society for Biochemistry and Molecular Biology.

function of most membrane lipids is to provide the permeability barrier of the cell. Therefore, alterations in lipid composition may compromise cell permeability before other functions of a particular lipid are uncovered. One may learn from genetics that a lipid is essential for cell viability but never learn the molecular bases for other requirements. Over the past 20 years, genetic approaches have been successfully used to establish the biosynthetic pathways of most of the common lipids. The challenge is to use this genetic information to manipulate the lipid composition of cells without

severely compromising cell viability. In those cases where this has been possible, the combination of the genetic approach to uncover phenotypes of cells with altered lipid composition and the dissection in vitro of the molecular basis for the phenotype has proven to be a powerful approach to defining lipid function. The more complex the organelle content and accompanying membrane structure of a cell the more difficult is the application of the genetic approach. Therefore, the most useful information to date has come from genetic manipulation of prokaryotic and eukaryotic microorganisms. However, the basic molecular principles underlying lipid function will be generally applicable to more complex mammalian systems.

4.1. Alteration of lipid composition in bacteria

The pathways for formation of the major phospholipids (PE, PG, and CL) of *E. coli* were biochemically established mainly by Eugene Kennedy and coworkers and subsequently verified using genetic approaches as described in Chapter 3. The design of strains in which lipid composition can be genetically altered in a systematic manner has been very important in defining new roles for lipids in cell function [2]. Unlike many other mutations affecting the metabolic pathways in *E. coli*, mutants in phospholipid biosynthesis cannot be bypassed by supplementation of the growth media with phospholipids due to the barrier function of the outer membrane. Therefore, the isolation and study of *E. coli* phospholipid auxotrophs has not been possible.

With the exception of the synthesis of CL, mutants in all steps of phospholipid biosynthesis were thought to be lethal even under laboratory conditions. To date, no growth conditions have been established for cells unable to synthesize CDP-diacylglycerol. Null mutants in the *pgsA* gene (encodes phosphatidylglycerophosphate synthase) that cannot synthesize PG and CL are lethal, but a suppressor of this mutation has been identified [10]. In such mutants, the major outer membrane lipoprotein precursor (see Fig. 4), that depends on PG for its lipid modification, accumulates in the inner membrane and apparently kills the cell. Cells unable to make this lipoprotein are viable but are temperature sensitive for growth indicating that PG and CL are not absolutely required for viability, only for optimal growth. However, the anionic nature of these lipids (apparently substituted by increased levels of PA) is necessary for the proper membrane association and function of peripheral membrane proteins as discussed in Sections 5.4 and 5.5.

The amine-containing lipids, PS and PE, were also thought to be essential based on initial mutants carrying temperature sensitive alleles of the genes (*pssA* and *psd*) encoding their respective biosynthetic enzymes. However, the growth phenotype of these mutants (as well as *pssA* null strains) with reduced amine-containing lipids could be suppressed by adding Ca^{2+} , Mg^{2+} , and Sr^{2+} in millimolar concentrations to the growth medium. These mutants, although viable, have a complex mixture of defects in cell division, reduced growth rate, loss of outer membrane barrier function, defects in energy metabolism, mis-assembly of membrane proteins, and defects in sugar and amino acid transport.

The key to defining new functions for the anionic and zwitterionic phospholipids of *E. coli* was the design of strains in which the content of PG/CL and PE could be

regulated in a systematic manner in viable cells. The *pgsA* gene (encoding the phosphatidylglycerophosphate synthase) was placed under the control of the exogenously regulated promoter *lacOP* (promoter of the *lac* operon) that is controlled by isopropyl- β -thiogalactoside levels in the growth media. Variation in PG plus CL levels were correlated in a dose-response manner with the functioning of specific cellular processes both in vivo and in vitro to determine lipid function. Similarly, the involvement of PE in function was uncovered by comparing phenotypes of cells with and without PE or by placing the *pssA* gene (encoding PS synthase) under exogenous regulation. Therefore, these genetically altered strains have been used as reagents to define potential lipid involvement in cellular process in vivo that can be verified by biochemical studies in vitro.

4.2. Alteration of lipid composition in yeast

The pathways of phospholipid synthesis and the genetics of lipid metabolism in yeast *Saccharomyces cerevisiae* [11] are as well understood as in *E. coli*. Yeast have pathways (see Chapters 3 and 8) similar to those in the *E. coli* for PE and PG synthesis. CL synthesis in all eukaryotes involves transfer of a phosphatidyl moiety from CDP-diacylglycerol to PG rather than from one PG to another PG as in bacteria. In addition, yeast utilize the mammalian pathways for synthesis of PI, PE, and PC including the methylation of PE to form PC (Chapter 8).

All gene products necessary for the synthesis of diacylglycerol, CDP-diacylglycerol, and PI in yeast are essential for viability. PS synthesis is not essential if growth medium is supplemented with ethanolamine in order to make PE and PC. However, PE is definitely required since *pssI* (encodes PS synthase) mutants also lacking a sphingolipid degradative enzyme that generates ethanolamine internally, require ethanolamine supplementation [12].

No gene products involved in lipid metabolism are encoded by the mtDNA which in *Saccharomyces cerevisiae* encodes eight proteins (subunits I, II, and III of cytochrome *c* oxidase, cytochrome *b*, the 3 subunits that make up the F_0 component of ATP synthase, and the *VARI* gene product which is part of the mitochondrial ribosome). The enzymes necessary for synthesis of PE from PS, and for PG and CL, are all encoded by nuclear genes and imported into the mitochondria. Null mutants of *crdI* (encodes CL synthase) grow normally on glucose for which mitochondrial function is not required. However, on non-fermentable carbon sources such as glycerol or lactate, they grow slower. Therefore, CL appears to be required for optimal mitochondrial function but is not essential for viability. However, lack of PG and CL synthesis due to a null mutation in the *PGSI* gene (encodes phosphatidylglycerophosphate synthase) results in the inability to utilize non-fermentable carbon sources for growth. Mitochondrial membrane potential is reduced to near undetectable levels although remains sufficient to support the import of all nuclear encoded proteins thus far investigated. Similar effects are seen in mammalian cells with a mutation in the homologous *PGSI* gene. The surprising consequence of lack of PG and CL in yeast is the lack of translation of mRNAs of four mitochondria-encoded proteins (cytochrome *b* and cytochrome *c* oxidase subunits I–III) as well as cytochrome *c* oxidase subunit IV [13] that is nuclear encoded. These results would indicate that

either some aspects of translation of a subset of mitochondrial proteins (those associated with electron transport complexes in the inner membrane but not ATP metabolism) require PG and/or CL or the lack of these lipids indirectly affects both mitochondrial and cytoplasmic mRNA translation.

5. Role of lipids in cell function

There are at least two ways by which lipids can affect protein structure and function and thereby cell function. Protein function is influenced by specific protein–lipid interactions that are dependent on the chemical and structural anatomy of lipids (headgroup, backbone, alkyl chain length, degree of unsaturation, chirality, ionization and chelating properties). However, protein function is also influenced by the unique self-association properties of lipids that result from the collective properties (fluidity, thickness, shape, packing properties) of the lipids organized into membrane structures.

5.1. The bilayer as a supramolecular lipid matrix

Biophysical studies on membrane lipids coupled with biochemical and genetic manipulation of membrane lipid composition have established that the L_α state of the membrane bilayer is essential for cell viability. However, membranes are made up of a vast array of lipids that have different physical properties, can assume individually different physical arrangements, and contribute collectively to the final physical properties of the membrane. Animal cell membranes are exposed to a rather constant temperature, pressure, and solvent environment and therefore do not change their lipid make up dramatically. The complex membrane lipid composition including cholesterol stabilizes mammalian cell membranes in the L_α phase over the variation in conditions they encounter. Microorganisms are exposed to a broad range of environmental conditions so have developed systems for changing membrane lipid composition in order to exist in the L_α phase. Yet all biological membranes contain significant amounts of non-bilayer-forming lipids.

5.1.1. Physical organization of the bilayer

As the growth temperature of *E. coli* is lowered, the content of unsaturated fatty acids in phospholipids increases. Genetic manipulation of phospholipid fatty acid composition in *E. coli* is possible by introducing mutations in genes required for the synthesis of unsaturated fatty acids (Chapter 3). The mutants require supplementation of unsaturated fatty acids from the growth medium and incorporate these fatty acids to adjust membrane fluidity in response to growth temperature. Mutants grown at low temperature with high unsaturated fatty acid content lyse when raised rapidly to high temperature probably due to the increased membrane permeability of fluid membranes and a transition from the L_α to H_{II} phase of the lipid bilayer. Conversely, growth at high temperatures with high saturated fatty acid content results in growth arrest after a shift to low temperature due to the reduced fluidity of the membrane. Wild-type cells, that do not normally contain such extremes in fatty acid content as can be generated with mutants, arrest growth

after a temperature shift until fatty acid composition is adjusted to provide a favorable membrane fluidity.

Bacterial cells also regulate the ratio of bilayer to non-bilayer-forming lipids in response to growth conditions [2]. Bacterial non-bilayer-forming lipids are PE with unsaturated alkyl chains, CL in the presence of divalent cations, and monoglucosyl diacylglycerol (MGDG). Extensive studies of lipid polymorphism have been carried out on *Acholeplasma laidlawii* because this organism alters its ratio of MGDG (capable of assuming the H_{II} phase) to DGDG (diglucosyl diacylglycerol, which only assumes the L_{α} or L_{β} phase) in response to growth conditions. High temperature and unsaturation in the fatty acids favor the H_{II} phase for MGDG. At a given growth temperature, the MGDG to DGDG ratio is inversely proportional to the unsaturated fatty acid content of MGDG. As growth temperature is lowered, *A. laidlawii* either increases the incorporation into MGDG of unsaturated fatty acids from the medium or increases the ratio of MGDG to DGDG to adjust the H_{II} phase potential of its lipids to remain just below the transition from bilayer to non-bilayer. Therefore, the cell maintains the physical properties of the membrane well within that of the L_{α} phase but with a constant potential to undergo transition to the H_{II} phase.

Contrary to *A. laidlawii*, *E. coli* maintains its non-bilayer lipids, CL (in the presence of divalent cations) and PE, within a narrow range and in wild-type cells adjusts the fatty acid content of PE to increase or decrease its non-bilayer potential. The unsaturated fatty acid content of inner membrane PE is higher than that of the PE on the inner leaflet of the outer membrane (which is 90% PE). The result is that the L_{α} to H_{II} transition for the inner membrane pool is only 10–15°C above the normal growth temperature of 37°C while this transition for the outer membrane phospholipids is 10°C higher than the inner membrane phospholipids. This increased potential for the inner membrane lipids to form non-bilayer structures is believed to be biologically significant to the function of the inner membrane. In mutants completely lacking PE, the role of non-bilayer lipid appears to be filled by CL. The growth defect of mutants lacking PE is suppressed by divalent cations in the same order of effectiveness ($Ca^{2+} < Mg^{2+} < Sr^{2+}$) as these ions induce the formation of non-bilayer phases of CL. Neither growth of the mutant nor the H_{II} phase for CL is supported by Ba^{2+} . The CL content of these mutants varies with the divalent cation used during growth. However, the L_{α} to H_{II} transition for the extracted lipids (in the presence of the divalent cation) is always the same as that of lipids from wild-type cells (containing PE) grown in the absence of divalent cations. Therefore, even though *E. coli* normally does not alter its PE or CL content to adjust the physical properties of the membrane, these mutants are able to adjust CL levels to maintain the optimal physical properties of the membrane bilayer.

5.1.2. Biological importance of non-bilayer lipids

It is obvious why prevention of formation of large amounts of non-bilayer phase would be important to maintaining cell integrity. However, why is there a need for non-bilayer-forming lipids? There are numerous biological processes that can be envisioned to require discontinuity in the membrane bilayer. Integration of proteins into the bilayer might require 'annular lipids' (those in close proximity to the protein) to interface with the more regular structure of the bilayer. Movement of proteins or other macromolecules

through the bilayer might also require such discontinuity. The process of membrane vesicle fusion and fission requires a transition state that is not bilayer in nature. Finally, the tension resulting from the pulling apart of the two halves of the bilayer induced by either one or both monolayers containing non-bilayer lipids may be of biological importance.

Since cells homeostatically adjust their mixture of bilayer and non-bilayer-forming lipids, some proteins must be sensitive to the intrinsic curvature of the composite membrane lipids. There is a correlation between the spontaneous curvature of the membrane and the performance of embedded proteins [14]. Protein kinase C is a peripheral membrane protein that binds to the membrane and is activated by a complex of PS (probably at least six molecules), one molecule of diacylglycerol, and one molecule of Ca^{+2} . In the presence of diacylglycerol protein kinase C is highly specific for PS but in the absence of diacylglycerol the kinase will bind to any anionic lipid. Stereoselectivity for the 1,2-diacyl-*sn*-glycerol is not absolute, but protein kinase C is stereospecific for the natural L-serine isomer of PS independent of whether or not other non-bilayer-forming lipids are present. However, this stereoselectivity appears to be related to the fact that in the presence of Ca^{2+} the natural isomer of PS undergoes the L_α to H_{II} transition at a lower temperature than the D-serine isomer [15]. Diacylglycerol is highly non-bilayer promoting and might selectively partition to a non-bilayer domain formed by the natural isomer of PS. The specific interaction of these two lipids may provide the unique allosteric switch regulating protein kinase C activity.

Phospholipase C activity is not directly influenced by the formation of non-bilayer structures. However, the presence of lipids (e.g. PE) with a tendency to form such structures, stimulates the enzyme even under conditions at which purely bilayer phases exist. Conversely sphingomyelin, a well-known stabilizer of the bilayer phase, inhibits the enzyme. Thus phospholipase C appears to be regulated by the overall geometry and composition of the bilayer [16] supporting the hypothesis that the collective physical properties of the lipid bilayer can modulate the activities of membrane-associated proteins.

Therefore, it is not always clear from initial studies which property of lipids, i.e. chemical or physical, is required for optimum function. The complex interplay between chemical and physical properties of lipids exemplifies the difficulty in understanding how lipids affect biological processes at the molecular level.

5.2. Selectivity of protein–lipid interactions

A specific phospholipid requirement has been determined for optimal reconstitution of function *in vitro* for more than 50 membrane proteins. If one considers specific lipid requirements for membrane association and activation of peripheral membrane proteins, the number is in the hundreds. Integral membrane proteins fold and exist in a very complex environment and have three modes of interaction with their environment. The extramembrane parts are exposed to the water milieu, where they interact with water, solutes, ions and water-soluble proteins. Part of the protein is exposed to the hydrophobic–aqueous interface region (see Fig. 9). The remainder of the protein is buried within the approximately 30 Å thick hydrophobic interior of the membrane.

Peripheral membrane proteins may spend part of their time completely in the cytosol and are recruited to the membrane surface, or even partially inserted into the membrane, in response to various signals.

Much of what is known about these protein–lipid interactions has come from protein purification and reconstitution of function dependent on lipids. Genetic approaches coupled with in vitro verification of function has uncovered new roles for lipids. Most exciting have been recent results from X-ray crystallographic analysis of membrane proteins which have revealed lipids in specific and tight association with proteins [17]. The predominant structural motif for the membrane spanning domain of membrane proteins is an α -helix of 20–25 amino acids which is sufficient to span the 30 Å core of the bilayer. A β -barrel motif is also found to a lesser extent.

5.2.1. Lipid association with α -helical proteins

CL is found aligned with a high degree of structural complementarity within a high-resolution structure of the light harvesting photosynthetic reaction center from *Rhodobacter sphaeroides*. The head group of CL is located on the surface of the reaction center, is in close contact with residues from all three of the reaction center subunits, and is engaged in hydrogen bond interactions with polar residues in the membrane interfacial region (at the cytoplasmic side of the membrane) (Fig. 10A). Bonding interactions between CL and the protein involve either direct contacts of the phosphate oxygens of the lipid headgroup with basic amino acids and backbone amide groups exposed at the protein surface, or indirect contacts with amino acid side chains that are mediated by bound water molecules. A striking observation was that the acyl chains of CL lie along grooves in the α -helices that form the hydrophobic surface of the protein and are restricted in movement by van der Waals interactions. A PE molecule was resolved in the X-ray structure of the photosynthetic reaction center from *Thermochromatium tepidum*. The phosphate group of PE is bound to Arg and Lys by electrostatic interaction, and to Tyr and Gly by hydrogen bonds. PE acyl chains fit into the hydrophobic clefts formed between α -helices of three different subunits of the complex.

Bacteriorhodopsin is a light-driven ion pump that is found in the purple membrane of the archaeobacterium *Halobacterium salinarum*. Bacteriorhodopsin monomers consist of a bundle of seven transmembrane α -helices that are connected by short inter-helical loops, and enclose a molecule of retinal that is buried in the protein interior, approximately half way across the membrane. Proton pumping by bacteriorhodopsin is linked to photoisomerization of the retinal and conformational changes in the protein, in a series of changes called a photochemical cycle. Specific lipids can influence the steps in this cycle. A combination of squalene (a reduced isoprenoid) and the methyl ester of phosphatidylglycerophosphate is required to maintain normal photochemical cycle behavior. In a high-resolution (1.55 Å) structure of bacteriorhodopsin, 18 full or partial lipid acyl chains per monomer were resolved (Fig. 10C), four pairs of which were linked with a glycerol backbone to form diether lipids identified as native archaeol-based lipids. One of the lipid alkyl chains buried in the center of the membrane appears to be squalene. Lipid chains were also observed in the hydrophobic crevices between the ends of the monomers in the trimeric structure and probable hold the complex together.

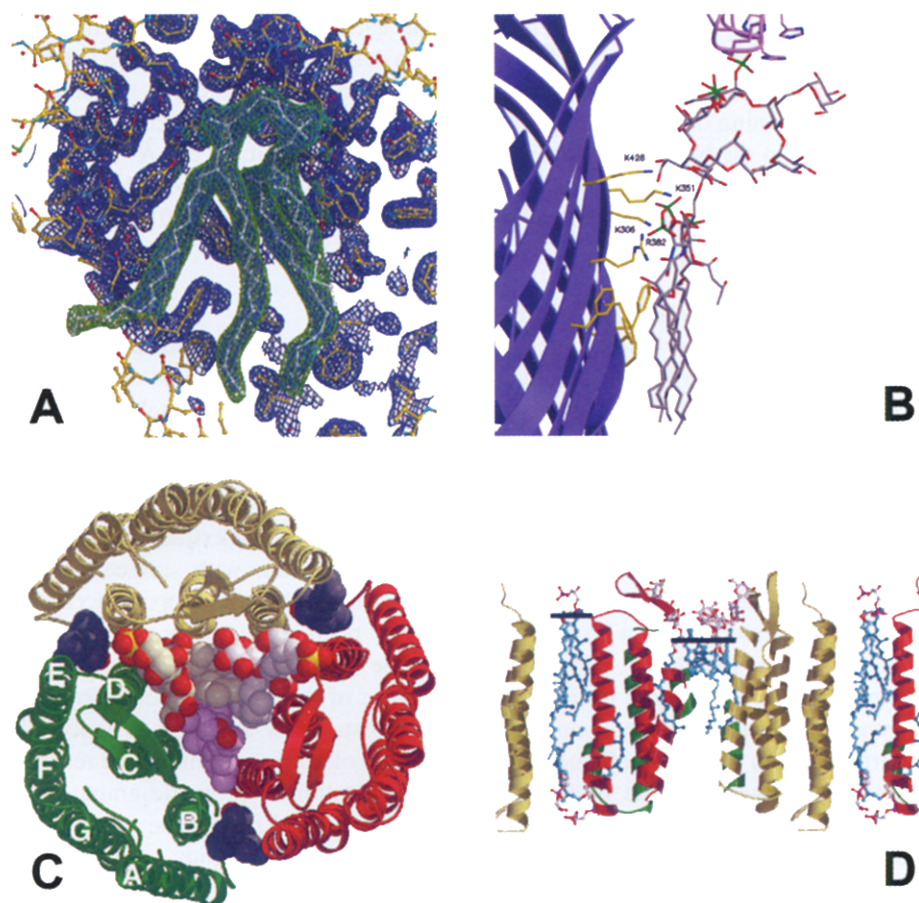


Fig. 10. Atomic structure of protein bound lipids. (A) Model of CL (green) tightly bound to the surface of photosynthetic reaction center (blue) from *R. sphaeroides*. The space-fill model was derived from X-ray crystallographic data that resolved between 9 and 15 carbons of the acyl chains of CL. *R. sphaeroides* contains mainly 18:0 and 18:1 fatty acids. Figure adapted with permission from McAuley et al. [18]. Copyright 1999 National Academy of Sciences, USA. (B) Crystal structure of FhuA complexed with lipopolysaccharide. The ribbon structure (blue) represents the outside surface of the β barrel of FhuA with extended chains (yellow) of amino acids. The amino acids of the aromatic belt interact with the acyl chains (gray) and the basic amino acids interact with the phosphate (green and red) groups of Lipid A. The remainder of the lipopolysaccharide structure extends upward into the periplasm. Figure adapted with permission from Ferguson et al. [19]. Copyright 1999 Elsevier Science Ltd. (C) Lipid packing in crystals of bacteriorhodopsin. Top view of the trimer in three different colors (domains A–E noted in one of the trimers) in complex with lipid (space-fill models) as viewed from the extracellular side. Three phytanyl chains of lipid (gray) lie in the crevices formed between the A and B domain of one monomer and the D and E domain of the adjacent monomer. The central core of the trimer is filled with a sulfated triglycoside attached to archaeol (glycolipid S-TGA-1). Red denotes the oxygen molecules of the sugars in white. (D) Cross section of the lipid bilayer showing phytanyl archaeol molecules (blue hydrophobic domain with red and white for the headgroups) extending from both sides of the bilayer and interacting with protein monomers (red and yellow helices) of bacteriorhodopsin. In the central core of the protein are two glycolipid S-TGA-1 molecules on the extracellular side of the membrane with a the gap on the cytosolic side of the membrane. The black lines indicate the boundary of the interface region. Figure adapted with permission from Essen et al. [20]. Copyright 1998 National Academy of Sciences, USA.

This organization explains the requirement for the natural archaeol lipids to maintain structure and function of the protein. The X-ray data demonstrate a good match between the hydrophobic face of membrane proteins and selective highly ordered surrounding lipids. The positioning of two glycolipid S-TGA-1 molecules to the extracellular side of the central hole in the bacteriorhodopsin trimer (Fig. 10D) results in a 5 Å ‘membrane thinning’ relative to the surrounding bilayer. This may cause a steeper electric field gradient across the central core than in the bulk lipid phase that might affect the proton pathway in bacteriorhodopsin.

5.2.2. Lipid association with β -barrel proteins

The pore-forming proteins of the outer membrane of *E. coli* are organized as antiparallel β chains forming a barrel structure with an aqueous pore on one side and an interface with the membrane bilayer on the other (Fig. 10B). X-ray crystal structure of *E. coli* outer membrane ferric hydroxamate uptake receptor (FhuA) contains a bound lipopolysaccharide in 1 : 1 stoichiometric amounts. The acyl chains of the lipopolysaccharide are ordered on the protein surface approximately parallel to the axis of the β -barrel along the half of the hydrophobic belt oriented toward the extracellular surface of the outer membrane. Numerous van der Waals interactions with surface-exposed hydrophobic residues are observed. The large polar headgroup of lipopolysaccharide makes extensive interactions with charged and polar residues of the protein near the outer surface of the membrane. Most of the favorable interactions are contributed by a cluster of 8 positively charged residues on the surface of the barrel, which interact by hydrogen bonding at distances around 3 Å or electrostatically at longer distances. In the interface region of the membrane there are clusters of aromatic amino acid residues positioned as belts around the protein. Similar organization of aromatic amino acids near the membrane interface region has been observed in other membrane proteins and may be involved in π -bonding interactions with the headgroups of lipids.

5.2.3. Organization of protein complexes

Rather than being associated with the exterior surface of membrane proteins, many phospholipids, particularly anionic ones like PG and CL, are found wedged between the subunits of monotropic and heterotropic oligomeric complexes. Anionic phospholipids have a particularly important function in energy-transducing membranes such as the bacterial cytoplasmic membrane and the inner mitochondrial membrane. In particular, CL has been shown to be a key factor in the maintenance of the optimal activity of the major integral proteins of the inner mitochondrial membrane, including NADH dehydrogenase, the cytochrome *bc*₁ complex, ATP synthase, cytochrome *c* oxidase, and the ATP/ADP translocase.

The crystal structure of bovine cytochrome *c* oxidase has been determined to a resolution of 2.3 Å. This integral membrane protein (a homodimer with 13 different subunits in each monomer) is responsible for the reduction of molecular oxygen to water during aerobic respiration, with concomitant proton pumping across the mitochondrial inner membrane. Several lipids were resolved in this structure: three PEs, seven PGs, one PC and two CLs. One of the CLs is located at the interface between the two monomers of the dimer and the remaining phospholipids are located between subunits

within each monomer. The resolution of CL in the structure of the bovine cytochrome *c* oxidase is particularly intriguing because it is well documented that CL cannot be removed from cytochrome *c* oxidase without a loss of enzyme activity.

In the above case, specific lipids mediate protein–protein contacts within a multimeric complex and may be very important for structural and functional integrity of complex membrane proteins. The advantage of using lipid molecules to form a significant part of the contact surface between adjacent protein subunits is that they have a high degree of conformational flexibility, and are usually available in a range of molecular shapes and sizes. Using lipids as interface material reduces the need for highly complementary protein–protein interactions and provides for flexible interactions between subunits.

5.2.4. *Binding sites for peripheral membrane proteins*

A common mechanism of cellular regulation is to organize functional complexes on demand from existing components. In many cases, the components are initially distributed between membranes and the cytosol. Post-translational modification of a protein, or the appearance of a cytoplasmic or membrane component, signals the organization of a functional complex on the membrane surface. There are many examples of primarily anionic phospholipids being either the signals or the organization site for such protein complexes at the membrane surface. Three structure-specific domains have been recognized, mostly in eukaryotes, that serve as specific lipid binding domains.

The C1 lipid clamp is a conserved cysteine-rich protein domain that binds lipids and is found in protein kinases C and other enzymes regulated by the second messenger diacylglycerol. This receptor domain interacts with one molecule of diacylglycerol and recruits the protein kinase C to specific membrane sites [21]. The C1 domain adopts a β -sheet structure with an open cavity.

The C2 domain generally binds anionic phospholipids such as PS in a Ca^{2+} -dependent manner and is conserved among phospholipase C, phospholipase A₂, PI-3-phosphate kinase and calcium-dependent protein kinase C [22]. The crystal structure of C2 domain of protein kinase C α in complex with PS reveals that the recognition of PS involves a direct interaction with two Ca^{2+} ions.

The PH, or pleckstrin homology, domain is shared by protein kinase C β and some phospholipases C. This domain is responsible for associating peripheral membrane proteins with the membrane via the phosphoinositide head group of polyphosphorylated PIs in an enantiomer-specific manner [23]. PH domains consist of 7-stranded β -sheets with positively charge pockets that attract the negatively charged PI head group.

In prokaryotic cells, the protein structural features defining lipid-binding domains is less well conserved than in eukaryotes, and the membrane ligand appears to be an anionic lipid-rich domain with little selectivity for the chemical species of lipid. DnaA protein and SecA protein (see Section 5.3) are peripheral membrane proteins in *E. coli* that carry out different functions but become membrane associated and activated by similar mechanisms. The involvement of anionic lipids in the function of these proteins was discovered through the use of *E. coli* mutants in which anionic lipid content could be controlled [2]. DnaA protein is required for initiation of DNA replication and is active in its ATP- but not ADP-bound form. In vitro, the exchange of ATP for ADP in the complex is greatly stimulated by almost any anionic phospholipid including non-*E.*

coli lipids like PI. In vivo DnaA function is compromised in mutants with reduced PG/CL levels. This phenotype can be suppressed by mutations that bypass the need for DnaA. An anionic specific membrane binding domain has been identified that appears to direct initial membrane association followed by partial insertion of the protein into the bilayer. The resulting conformational changes alter the ATP/ADP binding properties.

What was once thought to be a specific interaction of SecA and DnaA with either PG or CL, is actually an interaction with an anionic surface charge on the membrane. Mutants completely lacking PG and CL but with highly elevated levels of phosphatidic acid still initiate DNA replication (DnaA) and export protein (SecA) [10]. Both of these proteins can be activated in in vitro reconstituted systems with a wide range of anionic lipids including those not found in *E. coli*. It appears that both of these proteins recognize, via positively charged amphipathic helices, clusters or domains of negative charge rather than specific lipids on the membrane surface.

CTP: phosphocholine cytidylyltransferase is responsible for the synthesis of CDP-choline, an early precursor to PC synthesis in mammalian cells (Chapter 8). The enzyme has affinity for membranes depleted of PC that leads to activation of the enzyme and increased synthesis of PC. A complex mixture of factors including anionic lipids and non-bilayer-forming lipids stimulates membrane association. When binding occurs via two positively charged amphipathic helices, a large structural change occurs leading to enzyme activation. Affinity for an anionic membrane surface is understandable, but the role of other lipids such as PE and diacylglycerol in activation and the negative effect of fatty acids have only been recently clarified. The former two non-bilayer-forming lipids induce a negative curvature of the two halves of the bilayer toward the aqueous domain and away from the hydrophobic domain. Incorporation of free fatty acids (micelle forming) has the opposite effect. Surface association of the amphipathic helices would be favored by the decrease in hydration of the interfacial region due to the induced negative curvature, and penetration into the surface of the bilayer would reduce the stress imposed by negative curvature [24].

5.3. Translocation of proteins across membranes

Movement of proteins across membranes is required to transfer a protein from its site of synthesis to its site of function. The process involves the transfer of hydrophobic and hydrophilic segments of proteins through the hydrophobic core of the membrane. The role of lipids in this process has only recently received considerable attention. The in vivo evidence for the participation of anionic phospholipids in protein translocation was obtained from experiments with *E. coli* mutant strains defective in the biosynthesis of PG and CL [2]. The in vivo translocation of the outer membrane precursor proteins, prePhoE and proOmpA, was severely hampered in these cells. When the expression of the *pgsA* gene (encodes phosphatidylglycerophosphate synthase) placed under control of the *lac* promoter/operator was used to fine-tune the level of PG in the membrane, the translocation rate of the proteins was directly proportional to the amount of PG. The molecular basis for this anionic lipid requirement is for the function of SecA. SecA is a peripheral membrane protein that acts as a translocation ATP-driven motor which moves secreted proteins through the membrane translocation pore composed

of SecY and two other membrane proteins, SecE and SecG. SecA requires both anionic phospholipids and pore component SecY for high affinity binding to the membrane, for membrane penetration, and for high level ATPase-dependent function. Functional reconstitution of purified and delipidated SecYEG complex from *E. coli* and *Bacillus subtilis* into liposomes of defined lipid composition revealed an absolute requirement for PG [25]. Translocation activity was proportional to the amount of PG in reconstituted proteoliposomes and optimum activity was obtained only with the specific lipid composition of each organism.

The N-terminal signal peptides of *E. coli* secreted proteins possess at least one positively charged amino acid. Protein translocation efficiency is dependent on both the number of positive charges and the anionic phospholipid content of the membrane. Photocrosslinking of the secreted proteins with the fatty acid chains of membrane lipids demonstrated direct contact between the signal peptide and lipids during early stages of protein translocation.

Non-bilayer-forming lipids are also required for protein translocation of proteins across the membrane of *E. coli*. The only non-bilayer-forming lipid in *E. coli* mutants lacking PE is CL in the presence of Mg^{2+} or Ca^{2+} . Protein translocation into inverted membrane vesicles prepared from PE-deficient cells is reduced with divalent cation-depletion but can be enhanced by inclusion of Mg^{2+} or Ca^{2+} [2]. Translocation in the absence of divalent cations is restored by incorporation of non-bilayer PE (18:1 acyl chains) but not by bilayer-prone PE (14:0 acyl chains). These results would indicate that lipids with a tendency to form non-bilayer structures provide a necessary environment for the translocase of proteins across the membrane.

5.4. Assembly of integral membrane proteins

Much less is known about the role of phospholipids in insertion and organization of integral membrane proteins. Most such proteins are organized with several α -helical transmembrane domains spanning the membrane bilayer. These helices are connected by extramembrane loops alternately exposed on either side of the membrane. In bacteria these proteins generally obey the 'positive inside' rule in which those extramembrane loops on the cytoplasmic side of the membrane have a net positive charge and the loops exposed to the exterior are either neutral or negatively charged. Variation in the positive charge density of these loops as well as the anionic phospholipid content of the membrane indicates that these positive loops are anchored to the inside of the cell by interaction with anionic phospholipids.

5.4.1. Lipid-assisted folding of membrane proteins

The membrane clearly serves as the solvent within which integral membrane proteins fold and function [9]. However, do lipids act in more specific ways to guide and determine final membrane protein structure and organization? Recent evidence supports a role for lipids analogous to that of protein molecular chaperones in the folding of membrane proteins. Molecular chaperones facilitate the folding of substrate proteins by interacting with non-native folding intermediates but do not interact with native or totally unfolded proteins, and are not required to maintain native conformation. Lipids

that fulfill these requirements in assisting the folding of specific membrane proteins have been termed 'lipochaperones' to distinguish their function from simply providing a solvent for the folding process.

The major evidence for the existence of lipochaperones comes from studies on the requirement for PE in the assembly and function of lactose permease (LacY) of *E. coli* [26]. LacY is a polytopic membrane protein with 12 transmembrane-spanning domains connected by alternating cytoplasmic and periplasmic loops (Fig. 11). LacY carries out transport of lactose either in an energy-independent mode to equilibrate lactose across the membrane (facilitated transport) or by coupling uphill movement of lactose against a concentration gradient with downhill movement of a proton coupled to the proton electrochemical gradient across the membrane (active transport). Uncovering a lipochaperone role for PE in the assembly of LacY came about by the fortuitous availability of reagents and techniques. A large number of biochemical and molecular genetic resources are available for studying LacY including antibodies and molecularly engineered derivatives of LacY. The availability of viable *E. coli* strains that either lack PE, or in which the level of PE can be regulated, provided reagents to study the requirement for PE in the assembly of LacY in vivo and in isolated membranes. The development of a blotting technique termed an 'Eastern-Western' made possible the screening for lipids affecting the refolding of LacY in vitro or the conformation of LacY made in vivo.

In the Eastern-Western procedure, lipids are first applied to a solid support such as nitrocellulose. Next, proteins subjected to sodium dodecyl sulfate polyacrylamide gel electrophoresis are transferred by standard Western blotting techniques to the solid support in such a manner that the protein of interest is transferred to the lipid patch. During electrotransfer of protein to the solid support, protein, lipid, and sodium dodecyl sulfate mix and as transfer continues the sodium dodecyl sulfate is removed leaving behind the protein to refold in the presence of lipid. Attachment of the refolded protein to a solid support allows one to probe protein structure using conformation-sensitive antibodies or protein function by direct assay. This combined blotting technique allows the detection of membrane protein conformational changes as influenced by individual lipids during refolding. Since many membrane proteins including LacY retain significant amounts of secondary structure even in sodium dodecyl sulfate, refolding from sodium dodecyl sulfate detects intermediate and late steps of folding.

The initial observation that PE was required for LacY function was concluded from studies of reconstitution of transport function in sealed vesicles made of purified LacY and lipid. When reconstituted in lipid mixtures containing PE, both active and facilitated transport was restored. In mixtures containing only PG and/or CL or even PC only facilitated transport occurred. The physiological importance of PE for LacY function was established using mutants lacking PE. LacY expressed in PE-containing cells had full transport function, but cells lacking PE only displayed facilitated transport even though bioenergetic parameters of the membrane were normal. Using Western and Eastern-Western blotting techniques and a conformation sensitive antibody, it was established that LacY assembled in the presence, but not in the absence, of PE displays 'native' structure. LacY maintains its native structure even when PE is completely removed, and LacY originally assembled in the absence of PE is restored to native structure by partial denaturation in sodium dodecyl sulfate followed by renaturation in

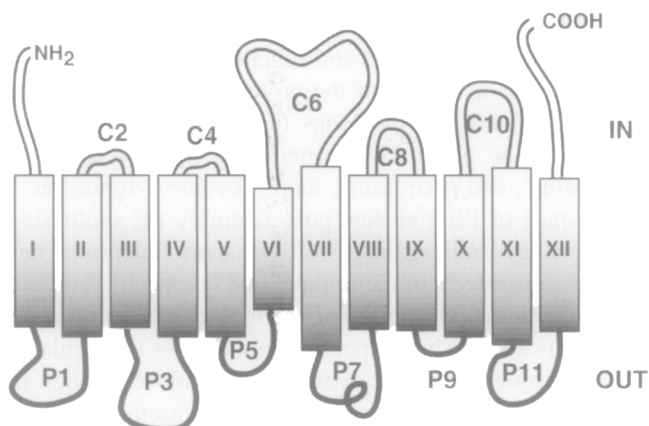
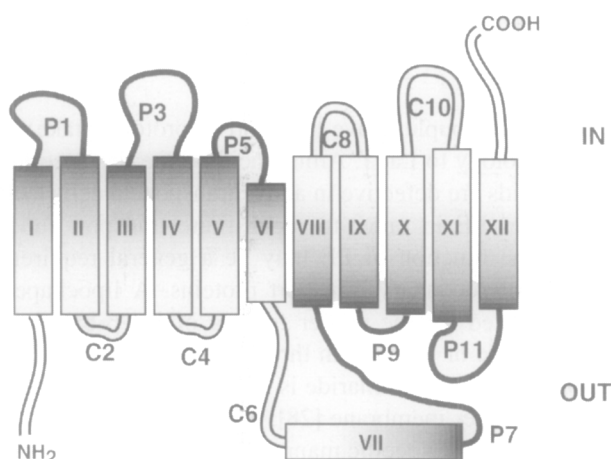
A**B**

Fig. 11. Topological organization of LacY in the *E. coli* inner membrane. (A) The topological organization of LacY when assembled in membranes containing PE (normal configuration). The 12 hydrophobic membrane spanning α -helices are numbered in Roman numerals from the amino (NH_2) to the carboxyl (COOH) terminus. The even-numbered hydrophilic loops ('C' and open) connect the transmembrane domains on the cytoplasmic side (IN) of the membrane. The odd-numbered hydrophilic loops ('P' and filled) connect the transmembrane domains on the periplasmic side (OUT) of the membrane. (B) The topological organization of lactose permease when assembled in the absence of PE. Note the first six transmembrane domains along with their connected hydrophilic loops are inverted with respect to the plane of the membrane when compared to A. Transmembrane domain VII is very hydrophilic and is postulated to reside outside of the membrane in the altered structure.

the presence of specifically PE (or other primary amine-containing phospholipids such as PS). LacY assembled (either *in vivo* or *in vitro*) in membranes lacking PE is restored to native structure by post-assembly addition of PE to the membranes. Furthermore, LacY extensively denatured in urea-sodium dodecyl sulfate (which eliminates most secondary structure) cannot be renatured by simple exposure to PE. Taken together

these data strongly suggest that PE assists in the folding of LacY by a transient non-covalent interaction with a late-folding and non-native intermediate thereby fulfilling the minimum requirements of a molecular chaperone [26].

The molecular basis for the loss of native structure and function of LacY assembled in the absence of PE is a topological mis-assembly of the protein [27]. In the absence of PE the first six transmembrane domains and the loops that connect them assume an inverted topology with respect to the plane of the membrane bilayer (Fig. 11). Cytoplasmic loops become periplasmic, and periplasmic loops become cytoplasmic. If LacY is first assembled in the absence of PE and then post-assembly PE synthesis is initiated, there is correction of topology and regain of full transport function. These results dramatically illustrate the specific effects of membrane lipid composition on both structure and function of membrane proteins. The ability of changes in lipid composition to effect such large changes in protein structure has important implications for regulatory roles of lipids in cell processes. For example, as eukaryotic proteins move through the secretory pathway, they encounter different membrane lipid compositions that might affect protein structure in dramatic ways to turn on or turn off function. Local changes in lipid composition may also result in similar changes in structure and function.

5.4.2. Scope of lipochaperone function

Secondary transport proteins (those that couple active transport to proton uptake) in *E. coli* show high overall structural homology to LacY. Amino acid permeases specific for Phe, Lys, Pro, and aromatic amino acids are defective in active transport in cells lacking PE. The melibiose permease requires PE in order to reconstitute transport function in vitro. Therefore, the lipochaperone function of PE may be a general requirement for proper assembly of this family of secondary transport proteins. A lipochaperone requirement in *E. coli* has been reported for the proper folding of the β -barrel outer membrane protein PhoE. During passage of the protein through the inner membrane of *E. coli*, interaction with precursors to lipopolysaccharide is required to obtain assembly competent protein upon arrival at the outer membrane [28]. Therefore, a role for lipids to act in membrane protein folding in a more specific manner than providing the solvent has been established.

5.5. Lipid domains

Compartmentalization of many biological processes such as biosynthesis, degradation, energy production, and metabolic signaling plays an important role in cell function. Subcellular organelles, multiple membrane structures, cytosol versus membrane are all utilized to compartmentalize functions. The original fluid mosaic model envisioned the membrane bilayer as a homogenous sea of lipid into which proteins are dispersed (Fig. 1). The current view of biological membranes is that they contain microdomains of different lipid and protein composition than is reflected by the whole membrane and that these domains serve to further compartmentalize cellular processes.

Lipid mixtures made up of defined lipids undergo phase separations due to lipid polymorphism and differences in steric packing of the acyl chains. Mixtures of bilayer and non-bilayer lipids undergo multiple phase transitions supporting the existence

of segregated domains within the bilayer. In model systems amphipathic polar lipid analogues self associate into domains if their hydrophobic domains are the same even if their polar domains carry the same net charge. Therefore, headgroup repulsive forces can be overcome by orderly packing of the hydrophobic domains. There is considerable acyl chain mismatch between phospholipids and sphingolipids (see Chapter 14 for structures), i.e., phospholipids tend to have shorter acyl chains (16–18) with higher degrees of unsaturation as compared to the longer (20–24 for the acyl group) saturated chains of sphingolipids. Naturally occurring sphingolipids undergo the L_{β} to L_{α} transition near the physiological temperature of 37°C while this transition for naturally occurring phospholipids is near or below 0°C. Therefore, the more laterally compact hydrophobic domains of sphingolipids can readily segregate from the more disordered and expanded domains of unsaturated acyl chains of phospholipids. Lipid segregation can also be facilitated by specific polar headgroup interactions, particularly intermolecular hydrogen bonding to other lipids and to protein networks involving hydroxyls, phosphates, amines, carbohydrates, and alcohols. The hydrogen bonding properties of CL due to its high pK_2 , as noted earlier, may be the basis for the formation of clusters of CL in natural and artificial membranes [29].

5.5.1. Lipid rafts

Lipid rafts are liquid ordered phases of cholesterol, glycosphingolipids (gangliosides), sphingomyelin, and proteins that exist as microdomains within the more dispersed liquid crystalline bilayer. Lipid rafts are operationally defined as the membrane fraction of eukaryotic cells that is resistant to solubilization in the cold by the detergent Triton X-100 (detergent-resistant membrane fraction). This fraction is greatly enriched in cholesterol, glycosphingolipids, sphingomyelin and a subset of membrane proteins. The proteins co-clustered in lipid rafts are soluble globular proteins tethered to the raft lipids via covalent linkage to fatty acids, cholesterol, or phosphatidylinositol (Chapter 2) [30]. The latter glycosylphosphatidylinositol-linked proteins are attached directly to the amino group of ethanolamine phosphate which in turn is linked to a trisaccharide and then to the inositol of PI (Fig. 12). The sphingolipids and glycosylphosphatidylinositol-linked proteins occupy the outer surface monolayer of the plasma membrane bilayer, and the acyl chains of these lipids are generally more saturated and longer than those of the plasma membrane phospholipids. The similarity in the structure of the more ordered hydrophobic domains of the raft lipids and their dissimilarity with the surrounding more fluid phospholipids favor a self-association of the raft lipids and the glycosylphosphatidylinositol-linked proteins. The hydrogen-bonding properties of the glycosphingolipids with themselves and with the constituents of the glycosylphosphatidylinositol-linked proteins stabilizes the complexes. Finally, the planar shape of cholesterol favors its intercalation parallel to the ordered acyl chains of the raft lipids with its single hydroxyl group facing the surface. The stability of this structure appears to be why it is not dissipated by detergent extraction.

Lipid rafts appear to be a mechanism to compartmentalize various processes on the cell surface by bringing together various receptor-mediated and signal transduction processes. A general phenomenon is that when glycosylphosphatidylinositol-linked proteins aggregate on the surface, they also become enriched in the detergent-resistant

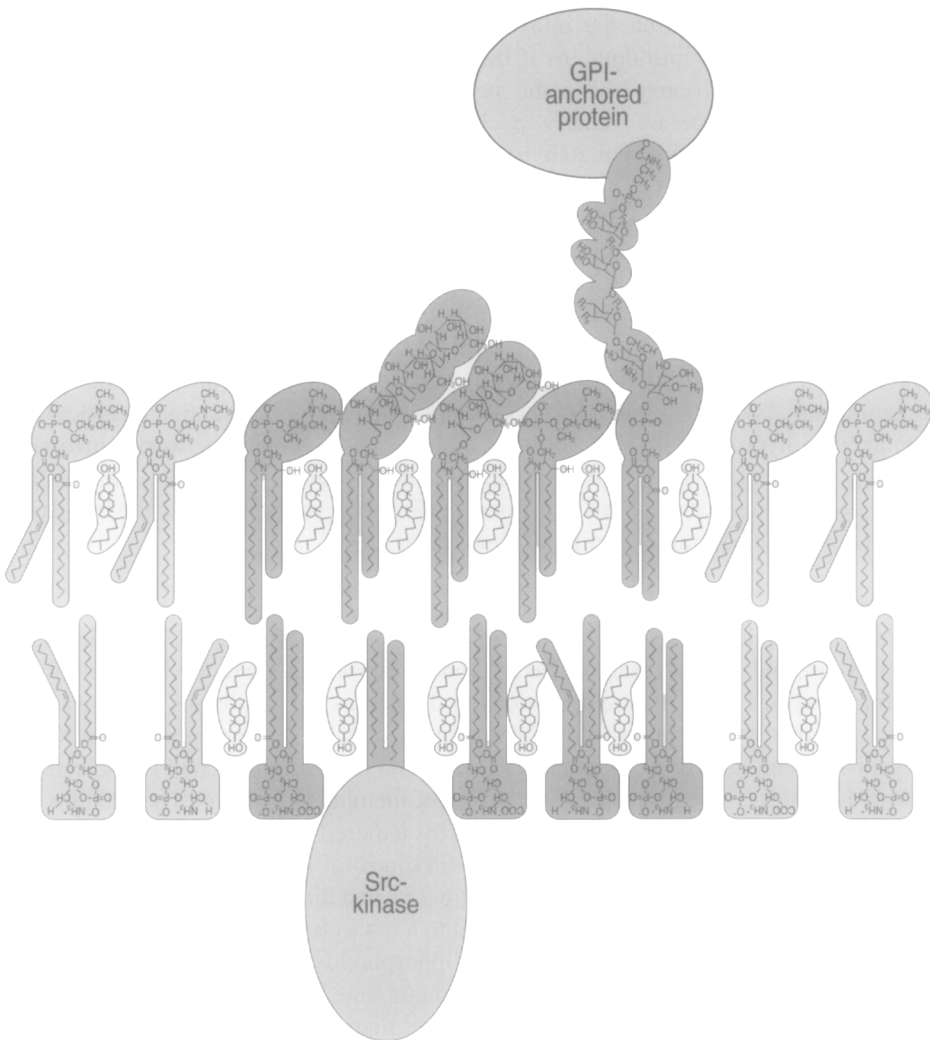


Fig. 12. Model of lipid raft. A glycosylphosphatidylinositol-linked protein is attached to the exterior monolayer of the membrane and a Src-kinase to the interior monolayer of the membrane by their respective covalently attached lipids. The mechanism for clustering and coupling Src-kinase with a glycosylphosphatidylinositol-linked protein is hypothetical. Clustered (dark gray) around the glycosylphosphatidylinositol are ordered (straight alkyl chains) glycosphingolipids, sphingomyelin, and PC with intercalated cholesterol. The phospholipids with kinked (unsaturated) chains indicate the more disordered liquid crystalline state of the surrounding bilayer. Reprinted (abstracted/excerpted) with permission from Simons and Ikonen [30]. Copyright 2000 American Association for the Advancement of Science.

membrane fraction as well as phosphorylated. For example, clustering of the IgE receptor, FcεRI, by binding its ligand on the cell surface results in its phosphorylation by a Src-family kinase that activates a signaling cascade. Only the receptor associated with the detergent-resistant membrane fraction is phosphorylated. The Src-family of

kinases is believed to localize to lipid rafts from the cytosolic side of the membrane via covalently attached fatty acids that insert into the membrane (Fig. 12).

A class of lipid domains related to rafts are caveolae which are invaginations on the surface of the plasma membrane of eukaryotic cells (Chapter 20). They contain the protein caveolin which when complexed with cholesterol forms a hairpin structure believed to induce curvature in the bilayer to form the caveolae. Caveolin is an integral part of the intracellular cholesterol transport machinery and caveolae are involved in regulating influx and efflux of cellular cholesterol. Caveolae are also resistant to detergent solubilization and also contain several proteins responsible for signal transduction suggesting a regulatory role for these structures in response to cholesterol.

The existence of lipid rafts and their function is still an evolving area. Isolation and characterization of detergent-resistant membrane fractions, studies in model systems, and studies with whole cells all support the concept of lipid rafts. However, they have never been observed in the native state, presumably due to their small size and possibly their loose association and dynamic properties. The raft components may normally have low affinity for each other, but increase their affinity when clustering is induced by other processes. It still is not clear how glycosylphosphatidylinositol-linked proteins on the exterior of the cell interact with acylated proteins that appear to be associated with rafts on the interior surface of the membrane.

5.5.2. Lipid domains in bacteria

One of the few examples of lipid domains being observed in living cells comes from the staining of CL in the membrane of *E. coli* by the fluorescent dye 10-*N*-nonyl acridine orange [31]. This reagent is highly specific for CL and has little or no interaction with other lipids or cell components. In wild-type cells, fluorescence is observed at the poles of the cell and at the division septum. In filamentous cells of *E. coli* with multiple genomes distributed along their length, the fluorescence is localized between the genomes. These fluorescence-enriched domains may be due either to regions enriched in CL (or possibly anionic lipids including CL) or areas enriched in lipid relative to protein. Initiation of DNA replication and the formation of the septal domain of *E. coli* have been postulated to be dependent on anionic phospholipids that would be consistent with the localization of CL domains.

5.6. Cytokinesis

The function of cytokinesis is to divide one cell into two by building a membrane barrier between the two daughter cells. In eukaryotic cells, the interaction of actin filaments with myosin filaments applies tension to the membrane to form a cleavage furrow, which gradually deepens until it encounters the narrow remains of the mitotic spindle between the two nuclei. Phospholipids play an essential role in the division processes in eukaryotic cells.

Phospholipids in biological membranes are distributed asymmetrically between the inner and outer leaflets of the lipid bilayer [32]. In the plasma membrane of eukaryotic cells PE and PS are localized to the inner leaflet and PC and sphingomyelin are enriched in the outer leaflet. Using a cyclic peptide highly specific for binding to PE, it was

demonstrated that PE is exposed on the cell surface of the cleavage furrow of eukaryotic cells at the final stage of cytokinesis. Immobilization of cell surface PE by the PE-binding peptide inhibited disassembly of the contractile ring resulting in formation of a long cytoplasmic bridge between the daughter cells. Removal of the peptide from the surface of arrested cells allowed cell division to proceed with disappearance of exposed PE. Furthermore, a mutant cell line defective in PE biosynthesis was isolated as a variant resistant to the cytotoxicity of the PE-binding peptide [33]. This cell line required either PE or ethanolamine for normal growth and cell division. In ethanolamine-deficient medium these mutant cells arrested with a cytoplasmic bridge between the two daughter cells. Addition of PE or ethanolamine restored normal cytokinesis. These findings provide the evidence that transbilayer movement of PE at the cleavage furrow contributes to regulation of cytokinesis.

In *E. coli*, cell division is initiated after genome duplication by organization of FtsZ protein monomers midway between the poles of the cell. This protein ring is the scaffold that recruits a series of proteins to the division site that brings about constriction and eventually cell division. An *E. coli* mutant completely lacking PE propagates as long filamentous cells. The FtsZ protein complex is recruited to the division site, but the FtsZ ring fails to constrict [34]. This phenotype is not observed in strains with specific defects in other steps of phospholipid biosynthesis. Although not firmly documented, prokaryotic membranes also appear to have an asymmetric enrichment of PE on the inner leaflet of the cytoplasmic membrane [35]. It is likely that PE is essential for cytoskeletal organization in the completion of cytokinesis in prokaryotic cells as well as mammalian cells.

6. Future directions

The roles lipids play in cellular processes is as diverse as the chemical structures of lipids found throughout nature. Although a single phospholipid can form a sealed bilayer vesicle in solution, a diversity of lipid structure and physical properties is necessary to fill the broad range of roles that lipids play in cells. Lipid structures vary greatly from the archaeobacteria, with their often-harsh environments, to the eubacteria that also must carry out a diversity of processes in one or two membrane structures, and to eukaryotic cells that have specialized organelles with different lipid compositions tailored to their function.

Defining lipid function is a challenging undertaking because of the diversity of chemical and physical properties of lipids and the fact that each lipid type potentially is involved at various levels of cellular function. Biological membranes are flexible self-sealing boundaries that form the permeability barrier for cells and organelles and provide the means to compartmentalize functions, but at the same time they perform many other duties. As a support for both integral and peripheral membrane processes, their physical properties directly affect these processes in ways that are often difficult to assess. Each specialized membrane has a unique structure, composition and function. Also within each membrane exist subdomains such as lipid rafts, lipid domains, and organizations of membrane associated complexes with their own unique composition. These complexes

can be made up of specific lipids as the organization site for integral and peripheral membrane proteins and many times are transient responding to cellular signals that can themselves be lipids. Lipids are integral components of stable complexes and serve specific structural roles by affecting protein conformation, by serving as the 'glue' that holds complexes together, or by providing the flexible interface between protein subunits. Lipids provide the complex hydrophobic–hydrophilic solvent within which membrane proteins fold and function, but they can also act in a more specific manner as molecular chaperones directing the attainment of final membrane protein organization. These diverse functions of lipids are made possible by a family of low molecular weight molecules that are physically fluid and deformable to enable interaction in a flexible and specific manner with macromolecules. At the same time they can organize into the very stable but highly dynamic supramolecular structures we know as membranes.

The challenge for the future will be to determine the function at the molecular level for the many lipid species already discovered. Coupling genetic and biochemical approaches has been historically a very powerful approach to defining structure–function relationships of physiological importance. Using this approach in microorganisms has proven to be very fruitful. As the sophistication of mammalian cell and whole animal genetics evolves, genetic manipulation coupled with biochemical characterization will begin to yield new and useful information on the function of lipids in more complex organisms. The interest in understanding biodiversity through the detailed characterization of the vast number of microorganisms will yield additional novel lipids that must be characterized structurally and functionally. Finally, as we discover more about the role of lipids in normal cell function, the role lipids play in disease will become more evident.

Abbreviations

CL	cardiolipin
DGDG	diglucosyl or digalactosyl diacylglycerol
MGDG	monoglucosyl or monogalactosyl diacylglycerol
PA	phosphatidic acid
PC	phosphatidylcholine
PE	phosphatidylethanolamine
PG	phosphatidylglycerol
PI	phosphatidylinositol
PS	phosphatidylserine

References

1. Singer, S.J. and Nicolson, G.L. (1972) The fluid mosaic model of the structure of cell membranes. *Science* 175, 720–731.
2. Dowhan, W. (1997) Molecular basis for membrane phospholipid diversity: why are there so many phospholipids. *Annu. Rev. Biochem.* 66, 199–232.

3. Kates, M. (1993) Biology of halophilic bacteria. Part II. Membrane lipids of extreme halophiles: biosynthesis, function and evolutionary significance. *Experientia* 49, 1027–1036.
4. Raetz, C.R. (2001) Regulated covalent modifications of lipid A. *J. Endotoxin Res.* 7, 73–78.
5. Tanford, C. (1980) *The Hydrophobic Effect: Formation of Micelles and Biological Membranes*. Wiley, New York.
6. Garavito, R.M. and Ferguson-Miller, S. (2001) Detergents as tools in membrane biochemistry. *J. Biol. Chem.* 276, 32403–32406.
7. Seddon, J.M., Cevc, G. and Marsh, D. (1983) Calorimetric studies of the gel-fluid ($L\beta$ - $L\alpha$) and lamellar-inverted hexagonal ($L\alpha$ - H_{II}) phase transitions in dialkyl- and diacylphosphatidylethanolamines. *Biochemistry* 22, 1280–1289.
8. Kates, M., Syz, J.Y., Gosser, D. and Haines, T.H. (1993) pH-dissociation characteristics of cardiolipin and its 2'-deoxy analogue. *Lipids* 28, 877–882.
9. White, S.H., Ladokhin, A.S., Jayasinghe, S. and Hristova, K. (2001) How membranes shape protein structure. *J. Biol. Chem.* 276, 32395–32398.
10. Matsumoto, K. (2001) Dispensable nature of phosphatidylglycerol in *Escherichia coli*: dual roles of anionic phospholipids. *Mol. Microbiol.* 39, 1427–1433.
11. Carman, G.M. and Henry, S.A. (1999) Phospholipid biosynthesis in the yeast *Saccharomyces cerevisiae* and interrelationship with other metabolic processes. *Prog. Lipid Res.* 38, 361–399.
12. Birner, R., Burgermeister, M., Schneider, R. and Daum, G. (2001) Roles of phosphatidylethanolamine and its several biosynthetic pathways in *Saccharomyces cerevisiae*. *Mol. Biol. Cell* 12, 997–1007.
13. Ostrander, D.B., Zhang, M., Mileykovskaya, E., Rho, M. and Dowhan, W. (2001) Lack of mitochondrial anionic phospholipids causes an inhibition of translation of protein components of the electron transport chain. A yeast genetic model system for the study of anionic phospholipid function in mitochondria. *J. Biol. Chem.* 276, 25262–25272.
14. Bezrukov, S.M. (2000) Functional consequences of lipid packing stress. *Curr. Opin. Coll. Interface Sci.* 5, 237–243.
15. Epand, R.M., Stevenson, C., Bruins, R., Schram, V. and Glaser, M. (1998) The chirality of phosphatidylserine and the activation of protein kinase C. *Biochemistry* 37, 12068–12073.
16. Ruiz-Arguello, M.B., Goni, F.M. and Alonso, A. (1998) Phospholipase C hydrolysis of phospholipids in bilayers of mixed lipid compositions. *Biochemistry* 37, 11621–11628.
17. Fyfe, P.K., McAuley, K.E., Roszak, A.W., Isaacs, N.W., Cogdell, R.J. and Jones, M.R. (2001) Probing the interface between membrane proteins and membrane lipids by X-ray crystallography. *Trends Biochem. Sci.* 26, 106–112.
18. McAuley, K.E., Fyfe, P.K., Ridge, J.P., Isaacs, N.W., Cogdell, R.J. and Jones, M.R. (1999) Structural details of an interaction between cardiolipin and an integral membrane protein. *Proc. Natl. Acad. Sci. USA* 96, 14706–14711.
19. Ferguson, A.D., Welte, W., Hofmann, E., Lindner, B., Holst, O., Coulton, J.W. and Diederichs, K. (2000) A conserved structural motif for lipopolysaccharide recognition by procaryotic and eucaryotic proteins. *Struct. Fold. Des.* 8, 585–592.
20. Essen, L., Siegert, R., Lehmann, W.D. and Oesterhelt, D. (1998) Lipid patches in membrane protein oligomers: crystal structure of the bacteriorhodopsin–lipid complex. *Proc. Natl. Acad. Sci. USA* 95, 11673–11678.
21. Hurley, J.H., Newton, A.C., Parker, P.J., Blumberg, P.M. and Nishizuka, Y. (1997) Taxonomy and function of C1 protein kinase C homology domains. *Protein Sci.* 6, 477–480.
22. Rizo, J. and Sudhof, T.C. (1998) Mechanics of membrane fusion. *Nat. Struct. Biol.* 5, 839–842.
23. Rebecchi, M.J. and Scarlata, S. (1998) Pleckstrin homology domains: a common fold with diverse functions. *Annu. Rev. Biophys. Biomol. Struct.* 27, 503–528.
24. Davies, S.M., Epand, R.M., Kraayenhof, R. and Cornell, R.B. (2001) Regulation of CTP: phosphocholine cytidyltransferase activity by the physical properties of lipid membranes: an important role for stored curvature strain energy. *Biochemistry* 40, 10522–10531.
25. van der Does, C., Swaving, J., van Klompenburg, W. and Driessen, A.J. (2000) Non-bilayer lipids stimulate the activity of the reconstituted bacterial protein translocase. *J. Biol. Chem.* 275, 2472–2478.
26. Bogdanov, M. and Dowhan, W. (1999) Lipid-assisted protein folding. *J. Biol. Chem.* 274, 36827–36830.

27. Bogdanov, M., Heacock, P. and Dowhan, W. (2002) A polytopic membrane protein displays a reversible topology dependent on membrane lipid composition. *EMBO J.* 21, 2107–2116.
28. De Cock, H., Brandenburg, K., Wiese, A., Holst, O. and Seydel, U. (1999) Non-lamellar structure and negative charges of lipopolysaccharides required for efficient folding of outer membrane protein PhoE of *Escherichia coli*. *J. Biol. Chem.* 274, 5114–5119.
29. Mileykovskaya, E., Dowhan, W., Birke, R.L., Zheng, D., Lutterodt, L. and Haines, T.H. (2001) Cardiolipin binds nonyl acridine orange by aggregating the dye at exposed hydrophobic domains on bilayer surfaces. *FEBS Lett.* 507, 187–190.
30. Simons, K. and Ikonen, E. (2000) How cells handle cholesterol. *Science* 290, 1721–1726.
31. Mileykovskaya, E. and Dowhan, W. (2000) Visualization of phospholipid domains in *Escherichia coli* by using the cardiolipin-specific fluorescent dye 10-*N*-nonyl acridine orange. *J. Bacteriol.* 182, 1172–1175.
32. Sprong, H., van der Sluijs, P. and van Meer, G. (2001) How proteins move lipids and lipids move proteins. *Nat. Rev. Mol. Cell. Biol.* 2, 504–513.
33. Emoto, K. and Umeda, M. (2000) An essential role for a membrane lipid in cytokinesis. Regulation of contractile ring disassembly by redistribution of phosphatidylethanolamine. *J. Cell. Biol.* 149, 1215–1224.
34. Mileykovskaya, E., Sun, Q., Margolin, W. and Dowhan, W. (1998) Localization and function of early cell division proteins in filamentous *Escherichia coli* cells lacking phosphatidylethanolamine. *J. Bacteriol.* 180, 4252–4257.
35. Rothman, J.E. and Kennedy, E.P. (1977) Asymmetrical distribution of phospholipids in the membrane of *Bacillus megaterium*. *J. Mol. Biol.* 110, 603–618.

Lipid modifications of proteins

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1. Preamble

Lipid modifications of proteins (Fig. 1) are widespread and functionally important in eukaryotic cells. For example, many intracellular proteins such as the signal-transducing heterotrimeric GTP-binding proteins (G proteins) and the Ras superfamily of G proteins are modified by 14- or 16-carbon fatty acids and/or 15- or 20-carbon isoprenoids. Also, a variety of cell surface proteins such as acetylcholinesterase, the T lymphocyte surface antigen Thy-1, and members of the cell adhesion protein family are modified by glycosylphosphatidylinositol (GPI) anchors. In most cases, the lipid moiety is crucial to protein function as it allows an otherwise water-soluble protein to interact strongly with membranes. The lipid moiety may also aid in the sorting of the protein to membrane domains that promote lateral and transbilayer protein-protein interactions that are critical for cell function. In some instances, the covalent lipid acts as a functional switch resulting in functional membrane association of certain protein conformations but not of others.

The covalent attachment of lipid to protein was first described in a study of myelin protein in 1951, but only clearly documented as important for protein biosynthesis and function in a study of the outer membrane murein lipoprotein of *Escherichia coli* by Braun and Rehn in 1969. These early discoveries were followed, in the 1970s, by the identification of fatty acids linked to viral glycoproteins and of isoprenoids covalently attached to fungal mating factors and to GTP-binding proteins. The 1980s saw the identification and characterization of *N*-myristoylated proteins and GPI-anchored proteins, and work on tissue patterning factors in the 1990s revealed a new class of autoprocessed proteins modified by cholesterol. Our purpose in this chapter is to document the structure of these various lipid modifications, describe their biosynthesis, and survey their functional significance. The chapter does not cover the structure and biosynthesis of diacylglycerol-modified proteins found in *E. coli* and other bacteria; information on this may be found in articles by Wu and colleagues (Wu, 1993).

2. Protein prenylation

Prenylated proteins constitute approximately 0.5–2% of all proteins in mammalian cells. They contain a farnesyl (15-carbon) or geranylgeranyl (20-carbon) isoprenoid attached via a thioether linkage to a cysteine residue at or near the carboxy-terminus of the protein. Protein prenylation was originally discovered when certain fungal

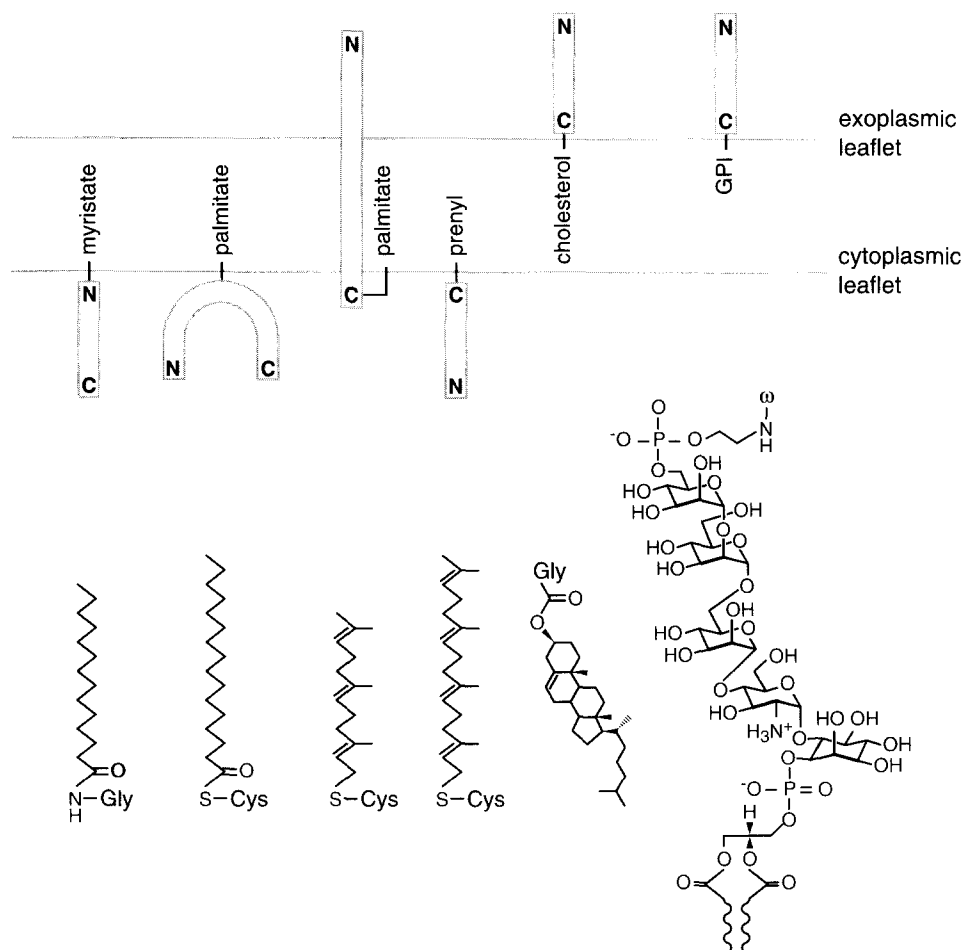


Fig. 1. Membrane topology of lipid modified proteins and structures of the covalently attached lipids. The structures (left to right) represent *N*-myristoyl glycine, palmitate thioether-linked to cysteine, farnesyl thioether-linked to cysteine, geranylgeranyl thioether-linked to cysteine, cholesterol ester-linked to glycine, and a minimal GPI anchor linked to the ω amino acid in a GPI-anchored protein. The GPI structure is shown with a diacylglycerol moiety containing two, ester-linked fatty acids. Other GPI anchors are based on ceramide, while yet others have monoacylglycerol, a fatty acid in ether-linkage to glycerol, and/or a fatty acid ester-linked to inositol.

peptide mating factors were shown to contain a carboxy-terminal cysteine modified by a thioether-linked farnesyl group. However, it was not until Glomset and colleagues (Schmidt, 1984) subsequently showed that animal cell proteins could be metabolically radiolabeled with radioactive mevalonate, an isoprenyl group precursor, that protein prenylation was more widely appreciated. Glomset and colleagues initially observed that the growth arrest of mammalian cells induced by compactin, an inhibitor of isoprenoid biosynthesis, could not be reversed by exogenously added sterols including cholesterol, the major product of the isoprenoid pathway (Brown, 1980). The compactin-

induced growth arrest could, however, be alleviated by small amounts of mevalonate, suggesting that mevalonate itself or a non-sterol metabolite of mevalonate played an important role in the growth cycle of cells. This result prompted studies in which cells were metabolically labeled with radioactive mevalonate and led to the discovery that almost 50% of the cell-associated radioactive mevalonate could not be extracted into lipid solvents as a result of post-translational (cycloheximide-insensitive) covalent association with proteins.

Protein prenylation is catalyzed by one of three different multisubunit prenyltransferases located in the cytoplasm of cells [1]. The majority of prenylated proteins, including most members of the Ras family of G proteins [2], contain a carboxy-terminal CaaX motif (CaaX box) composed of a conserved cysteine residue, two aliphatic amino acids (a) and a variable carboxy-terminal residue (X). The CaaX box is recognized by CaaX prenyltransferases that catalyze the attachment of a farnesyl or geranylgeranyl group from the corresponding isoprenyl pyrophosphates to the cysteine residue (Fig. 2). The CaaX prenyltransferases involved in these reactions are protein farnesyltransferase (FTase) and protein geranylgeranyltransferase type I (GGTase-I). FTase recognizes CaaX boxes where X = M, S, Q, A, or C, whereas GGTase-I recognizes CaaX boxes with X = L or F. Other prenylated proteins, such as the Rab proteins involved in vesicular transport, terminate in a CC or CXC motif; these proteins are substrates for protein geranylgeranyltransferase type II (GGTase-II) [3].

Subsequent to prenyl modification, Ras and most other CaaX proteins are further processed by two ER-localized, membrane-bound enzymes. The first prenyl-dependent processing step is the proteolytic removal of the -aaX tripeptide by the CaaX protease

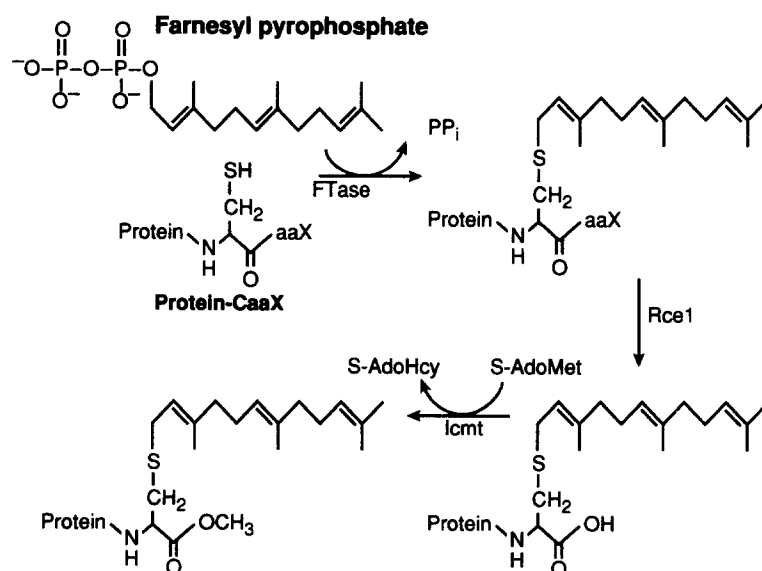


Fig. 2. Farnesylation, proteolysis and carboxymethylation of a CaaX protein. The farnesyl donor is farnesyl pyrophosphate.

Rce1; this is followed by carboxymethylation of the now C-terminal prenylcysteine residue by the methyltransferase Icmt (Fig. 2). The result of these modifications is to produce a protein that exhibits a high affinity for cellular membranes and also to impart a unique structure at the C-terminus that can serve as a specific recognition motif in certain protein–protein interactions [4].

The importance of prenylation in CaaX protein function, most notably as a regulator of the oncogenic potential of the Ras proteins [2], has led to considerable efforts to identify inhibitors of the prenyltransferases involved for evaluation as therapeutic agents [5,6]. The majority of these studies have focused on FTase, since this enzyme modifies Ras proteins, and early preclinical studies indicated significant anticancer potential for FTase inhibitors. A wide variety of FTase inhibitors have been developed, including some very promising ones that possess antitumor activity in animal models and are now in clinical development [6]. In addition, the success of FTase inhibitors in preclinical models of tumorigenesis, the increasing realization that proteins modified by GGTase-I play important roles in oncogenesis, and the finding that post-prenylation processing by Rce1 is important in the function of Ras and other CaaX proteins has led to the current situation in which all of the enzymes involved in CaaX protein processing are viewed as potential therapeutic targets.

2.1. The CaaX prenyltransferases FTase and GGTase-I

FTase is a heterodimer consisting of 48-kDa (α) and 46-kDa (β) subunit polypeptides. GGTase-I also consists of two subunits, a 48-kDa α subunit shared with FTase, and a 43-kDa β subunit. The isoprenoid substrates for the two enzymes are farnesyl pyrophosphate and geranylgeranyl pyrophosphate. Protein substrates for FTase in mammalian cells include Ras GTPases, lamin B, several proteins involved in visual signal transduction and at least three protein kinases and phosphatases. Known targets of GGTase-I include most γ subunits of heterotrimeric G proteins and Ras-related GTPases such as members of the Ras and Rac/Rho families. Both FTase and GGTase-I recognize short peptides containing appropriate CaaX motifs, and tetrapeptide substrates were instrumental in purifying the enzymes to homogeneity.

Both FTase and GGTase-I are zinc metalloenzymes in which the single bound zinc atom participates directly in catalysis. FTase additionally requires high concentrations (>1 mM) of magnesium for catalysis. FTase proceeds via a functionally ordered kinetic mechanism, with farnesyl pyrophosphate binding first to create an FTase–farnesyl pyrophosphate binary complex that then reacts rapidly with a CaaX substrate to form a prenylated product. In the absence of excess substrate, the dissociation rate is so slow that FTase–product complexes can be isolated. A wealth of structural information has emerged for FTase in the past several years beginning with the first X-ray crystal structure of unliganded FTase solved at 2.2 Å resolution (Park, 1997), greatly enhancing the ability of investigators to conduct structure–function analyses on the enzyme and investigating the roles of specific residues in substrate binding and catalysis.

3. Myristoylation

N-Myristoylated proteins comprise a large family of functionally diverse eukaryotic and viral proteins. Myristate, a relatively rare 14-carbon, saturated fatty acid, is transferred from myristoyl-CoA and linked via an amide bond to the N-terminal glycine residue of the target protein. The reaction is catalyzed by the enzyme myristoyl-CoA: protein *N*-myristoyltransferase (NMT) which recognizes and modifies N-terminal glycine residues presented in a particular sequence context. Although myristate can be attached post-translationally to N-terminal glycine in synthetic peptides of the appropriate sequence, in vivo myristoylation is an early co-translational event occurring in the cytoplasm as soon as ~60 amino acids of the nascent peptide emerge from the ribosomal tunnel and after the N-terminal glycine residue is made available by cellular methionyl-aminopeptidases that remove the initiator methionine residue. The myristoyl-CoA pools used by NMT appear to be supplied by de novo synthesis and by activation of exogenous myristate. Of 6220 open reading frames surveyed in the *Saccharomyces cerevisiae* genome data base for an appropriately positioned glycine residue, 70 (1.1%) are known to be or predicted to be *N*-myristoylated proteins. Some of these proteins play critical roles in cell survival since NMT is an essential enzyme in *S. cerevisiae*. NMT is also essential in *Candida albicans*, and *Cryptococcus neoformans*, the most common causes of systemic fungal infections in immunocompromised individuals.

3.1. *N*-Myristoyltransferase (NMT)

NMT was first purified from *S. cerevisiae* and characterized as a ~55-kDa monomeric protein with no apparent cofactor requirements. The crystal structures of the *S. cerevisiae* and *C. albicans* NMTs have been determined, the former as a co-crystal with a non-hydrolyzable myristoyl-CoA derivative and a dipeptide inhibitor of the enzyme [7].

NMT uses an ordered Bi–Bi reaction mechanism: myristoyl-CoA binds first, followed by the peptide substrate. After catalysis, CoA is discharged first, followed by the *N*-myristoyl peptide. The enzyme is highly selective for myristoyl-CoA and for polypeptide substrates with an N-terminal glycine, an uncharged residue at position 2, and neutral residues at positions 3 and 4. Serine is found at position 5 of all known yeast *N*-myristoyl proteins, while lysine is commonly found at position 6. In vitro analyses of human and fungal NMTs indicate that while their reaction mechanism and acyl-CoA substrate specificities are the same, their peptide specificities are different — this difference has been exploited to develop species-selective NMT inhibitors that act as fungicidal agents.

3.2. Myristoyl switches to regulate protein function

The myristoyl group in *N*-myristoylated proteins frequently acts as a key regulator of protein function. In some cases, the myristate residue provides a constitutive source of membrane affinity that needs to be supplemented by a second interaction between the protein and the membrane in order for the protein to stay membrane associated. For the MARCKS protein (myristoylated alanine-rich C-kinase substrate) as well as the tyrosine kinase Src, this second interaction is provided by electrostatic affinity

between a polybasic region of the protein and the negatively charged headgroups of phospholipids in the cytoplasmic leaflet of cell membranes. When serine residues in the polybasic region of MARCKS or Src are phosphorylated, the electrostatic contribution to membrane binding is reduced, and the protein moves off the membrane into the cytoplasm.

Myristate can also provide a regulated source of membrane affinity. Some proteins such as ARF (ADP ribosylation factor) and recoverin exist in alternate conformations in which the myristoyl group is exposed and available for membrane binding, or sequestered within a hydrophobic pocket in the protein. On ligand binding (GTP for ARF, and Ca^{2+} for recoverin), the myristoyl group is exposed and becomes available to promote interactions with target membranes and or protein partners.

4. Protein thioacylation

Thioacylated proteins contain fatty acids in thioester linkage to cysteine residues [8,9]. This class of lipid modification of proteins was first identified in studies of brain myelin protein, but only firmly established in the late 1970s when Schlesinger, Schmidt and co-workers reported the palmitoylation of Sindbis virus and vesicular stomatitis virus glycoproteins. Protein thioacylation is frequently referred to as palmitoylation, although fatty acids other than palmitate may be found on thioacylated proteins. Membrane proteins as well as hydrophilic proteins are thioacylated, the latter, in many cases, acquiring the modification when they become associated with a membrane compartment as a result of *N*-myristoylation or prenylation. Thioacylated cysteine residues are found in a variety of sequence contexts and are invariably located in portions of the protein that are cytoplasmic or within a predicted transmembrane domain. Unlike the other known lipid modifications of proteins, thioacylation is reversible: the protein undergoes cycles of acylation and deacylation, and as a result, the half-life of the acyl group is much shorter than that of the polypeptide (~ 20 min for the acyl group versus ~ 1 day for the polypeptide in the case of N-Ras (Magee, 1987)). Several protein acyltransferases have been isolated, but it is unclear whether they are required for thioacylation in living cells: non-enzymatic thioacylation has been observed *in vitro* suggesting that acyl transfer may occur through an autocatalytic process. In contrast, a number of thioacyl protein thioesterases have been identified and these appear to be responsible for the deacylation of thioacylated proteins.

4.1. Examples of thioacylated proteins

There are three classes of thioacylated proteins [9]: polytopic membrane proteins such as some G-protein coupled receptors (β -adrenergic receptor, rhodopsin), monotopic membrane proteins including viral glycoproteins, the transferrin receptor, and the cation-dependent mannose-6-phosphate receptor, and hydrophilic proteins such as members of the Src family of protein tyrosine kinases (e.g., p59^{lyn} and p56^{lck}) [10], as well as H-Ras, N-Ras, and the synaptic vesicle protein SNAP-25. The functional significance of thioacylation of polytopic and monotopic membrane proteins is unclear, but the acyl

modification may dictate how the protein is trafficked between membranes and whether it partitions into sterol/sphingolipid-rich membrane domains (lipid rafts) (Chapter 1) (Melkonian, 1999). For intrinsically hydrophilic proteins such as p59^{lyn} and SNAP-25, the function of thioacylation is clear — the acyl group functions together with other acyl groups, or other lipid modifications, to provide membrane anchoring in the cytoplasmic leaflet of the membrane bilayer for an otherwise soluble protein.

4.2. Membrane anchoring of thioacylated proteins: the need for multiple lipid modifications and the role of dynamic thioacylation

Thioacylated proteins that lack transmembrane spans must have more than one covalently bound lipid chain in order to be stably associated with membranes. This is also true for *N*-myristoylated proteins and prenylated CaaX proteins that, when newly synthesized, are modified by only a single lipid moiety. The association of monoacylated or monoprenylated proteins with the lipid bilayer appears to be rapidly reversible and thus, for stable membrane association, lipid modified proteins require at least two lipid chains or must rely on some other interaction with membranes such as recognition by a membrane-bound receptor or electrostatic interaction between charged amino acids in the protein and charged phospholipids in the membrane [10].

The *N*-myristoylated Src family protein tyrosine kinases are frequently thioacylated at one or more cysteine residues near the myristoylated glycine, and it is these doubly or triply lipid modified proteins that are found associated with the cytoplasmic face of the plasma membrane [10] (the N-terminal sequences of p59^{lyn} and p56^{lck} are **¹MGCVC**– and **¹MGCVQC**–, respectively, where the *N*-myristoylated glycine is shown in bold and the thioacylated cysteine residues are underlined). A similar situation is seen for the prenylated Ras proteins which must be thioacylated before they associate firmly with membranes (the C-terminal sequences of H-Ras and N-Ras are –G**C**MS**C**KCVLS-COOH and –G**C**MGLPCVVM-COOH, respectively (the farnesylated cysteine is shown in bold and the thioacylated cysteines are underlined)). A third example is provided by proteins such as SNAP-25 which are exclusively thioacylated, but display at least four thioacyl chains through which they become stably associated with the synaptic vesicle membrane.

An interesting and persuasive model for the targeting of lipid modified proteins to particular membranes was suggested by Shahinian and Silvius [11] (Fig. 3), based on the notion that single lipid modifications allow the protein to undergo transient interactions with a variety of intracellular membranes whereas tandem modifications promote stable membrane association. Thus, a protein with a single lipid modification such as the cytoplasmically synthesized *N*-myristoylated p59^{lyn} becomes stably associated with the cytoplasmic face of the plasma membrane only when it becomes thioacylated at this location. In this scenario, thioacylation would not only provide for stable membrane association of a protein with a single lipid modification, but it would also ensure targeting of that protein to the membrane where thioacylation occurs. Thus, within 5 min of the completion of peptide synthesis, p59^{lyn} becomes *N*-myristoylated, thioacylated and located to the cytoplasmic face of the plasma membrane (van 't Hof, 1997). Removal of the palmitoylation sites slows the kinetics of membrane association,

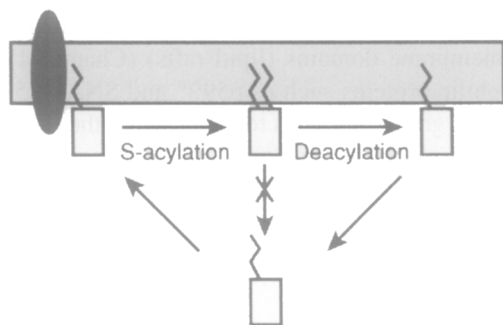


Fig. 3. Bilayer trapping mechanism for membrane targeting of lipid-anchored proteins lacking transmembrane spans (redrawn from [11]). A singly lipid-modified protein associates transiently with a membrane containing a membrane targeting receptor, possibly a thioacyltransferase. Thioacylation at the membrane yields a dual-anchored species that is stably associated with the membrane until it is deacylated.

and reduces the proportion of $p59^{\text{fyn}}$ that is membrane associated. The dynamic nature of thioacylation would suggest that the duration of association of $p59^{\text{fyn}}$ with the plasma membrane would be dictated by the half-life of the thioacyl chain. However, since $p59^{\text{fyn}}$ is doubly thioacylated, it is unlikely that it would revert to its solely *N*-myristoylated state and re-enter the cytoplasm.

Thioacylation frequently dictates plasma membrane targeting of proteins lacking transmembrane spans. In the case of $p59^{\text{fyn}}$, targeting occurs directly, with the *N*-myristoylated protein becoming thioacylated and plasma membrane associated rapidly upon completion of synthesis. In contrast, $p56^{\text{lc}}$ appears to be thioacylated on intracellular membranes and arrive at the plasma membrane via vesicular transport (bound to the cytoplasmic face of secretory vesicles) (Bijlmakers, 1999). In yet another targeting variation, newly synthesized *N*-myristoylated G_{α} , a dually acylated trimeric G protein α subunit, associates with all cellular membranes but accumulates eventually at the plasma membrane: the plasma membrane form is the only one that is both *N*-myristoylated and thioacylated.

4.3. Thioesterases

Three thioacyl protein thioesterases have been identified [9]. Two of these, PPT1 (Camp, 1994) and PPT2 (Soyombo, 1997) (palmitoyl-protein thioesterases 1 and 2) are localized to the lysosomes and are thus likely to be involved in the catabolism of thioacylated proteins or peptides. The thioacylated molecules are presumed to gain access to the lysosomal lumen by an autophagic pathway in which membrane fragments are captured into a vacuole that subsequently fuses with lysosomes. A defect in PPT1 leads to a severe neurodegenerative disorder termed infantile neuronal ceroid lipofuscinosis characterized by the accumulation of autofluorescent material (including lipid thioesters) in all tissues. No diseases have been linked to PPT2.

A third thioesterase was purified from rat liver cytosol using palmitoylated G-protein α subunit as a substrate (Duncan, 1998). This thioesterase, a 25-kDa monomeric protein, is likely to be the one involved in turnover of thioacyl groups on proteins. It

displays both acylprotein thioesterase activity as well as lysophospholipase activity, but thioacylproteins are by far the preferred substrate.

5. Cholesterol modification

In addition to its numerous roles in membrane architecture and steroid and bile acid synthesis (Chapters 15 and 16), cholesterol has recently been discovered as a post-translational protein modification on a family of signaling proteins referred to as hedgehog (Hh) proteins [12]. This modification was identified in studies on the processing of the *Drosophila* Hh protein (Porter, 1996). During biosynthesis, the Hh protein undergoes an autocatalytic cleavage event during which the carboxy-terminal domain is removed and cholesterol is covalently attached to the amino-terminal domain yielding an active signaling molecule (Fig. 4). The Hh proteins, found in insects, vertebrates, and other multicellular organisms, are part of a family of secreted signaling molecules involved in the patterning of diverse tissues during development. These proteins function

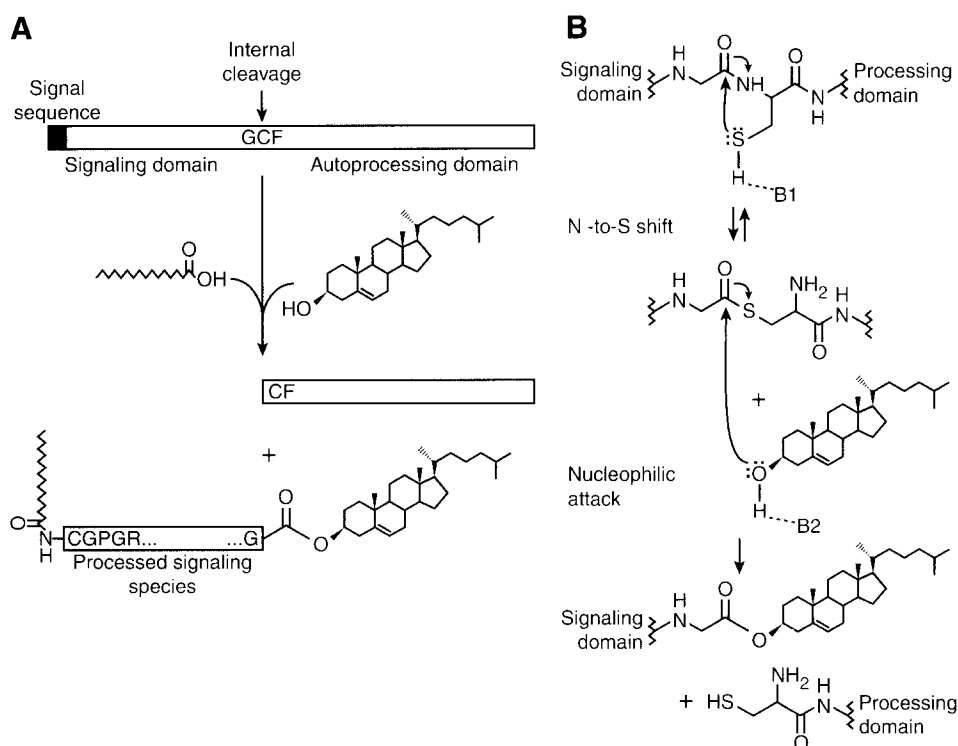


Fig. 4. Modification of hedgehog with ester-linked cholesterol and an N-terminal amide-linked fatty acid (redrawn from [12]). (A) Overall processing steps leading to lipid-modified Hh. (B) Mechanism of cholesterol addition. Residues indicated as B1 and B2 in panel B are catalytic bases presumably contained in the carboxy-terminal domain of the Hh precursor.

in a graded manner, emanating from cells in which the protein is produced and acting upon target cells several cell diameters away, thus specifying different cell fates by altering programs of gene expression. Cholesterylation of the Hh signaling protein is crucial for its proper targeting to specific tissues during development [13,14].

5.1. Addition of cholesterol to hedgehog proteins

Newly synthesized Hh precursor proteins of approximately 45 kDa contain an N-terminal signal sequence which targets the protein to the ER and is subsequently trimmed, followed by an N-terminal signaling domain (~19 kDa), an absolutely conserved Gly–Cys–Phe tripeptide motif, and a C-terminal processing domain which is removed to yield the active signaling molecule (Fig. 4). Addition of cholesterol to Hh proteins proceeds via an autoproteolytic internal cleavage reaction at the Gly–Cys–Phe tripeptide sequence. The cleavage is initiated through a nucleophilic attack by the thiol side chain of Cys on the Gly carbonyl, replacing the Gly–Cys peptide bond with a thioester linkage. A second nucleophilic attack on the same carbonyl by the hydroxyl group of cholesterol results in removal of the C-terminal processing domain and produces the active Hh signaling molecule containing a C-terminal cholesterol modification (Fig. 4). Although other sterols have been shown to be able to substitute for cholesterol in *in vitro* Hh processing assays, the free 3 β -hydroxyl group is essential for processing [12].

The autocleavage reaction is mediated by the C-terminal processing domain of the Hh precursor polypeptide. The first 145 amino-terminal residues of C-terminal processing domain are sufficient for thioester formation as indicated by the ability of Hh proteins truncated after this point to undergo cleavage in the presence of dithiothreitol. At least some part of the remaining carboxy-terminal residues is required for cholesterol addition and contains a sterol recognition region (Hall, 1997).

In addition to the cholesterol modification, Hh proteins can also be palmitoylated at the N-terminal Cys residue (part of a CGPGR motif) after cholesterol is added. An acyltransferase, named Skinny Hh, was discovered in *Drosophila*. Skinny Hh mutant cells express Hh proteins containing the cholesterol modification but lacking aminoterminal palmitate (Chamoun, 2001). Available information indicates that the palmitate residue is amide-linked to the N-terminus of the protein and not thioester-linked to the cysteine side chain.

5.2. Biological significance of cholesterol modification

The role of cholesterol-anchoring of the hedgehog signaling molecule is not fully understood. Cholesterylated Hh must travel to other cells to elicit its effect, even though the hydrophobic cholesterol molecule is presumed to act as a membrane anchor. Hh proteins truncated at the cleavage site are not modified by cholesterol, but nevertheless produce active signaling molecules; however, studies in *Drosophila* have shown that these proteins are not targeted properly and cause mispatterning and lethality in embryos. The subcellular location of the autocatalytic cleavage and processing events is not clear. Cholesterol-modified Hh in *Drosophila* partitions into

detergent insoluble membranes, or rafts, which are enriched in specific molecules such as sterols, sphingolipids, thioacylated proteins and GPI-anchored proteins and usually considered predominantly plasma membrane domains (Chapter 1) (Rietveld, 1999). Several proteins, each containing a sterol-sensing domain, have been identified that function in the release of Hh from Hh-producing cells and proper sequestering of Hh in target cells. For example, the proteins Patched and Dispatched are involved in sequestration of the Hh signal within tissues and release of functional signal from Hh-producing cells, respectively. A soluble form of mammalian Hh (sonic Hh) that is cholesterol-modified, multimeric and biologically potent has been isolated from developing chick limb (Zeng, 2001). The protein Patched is required for release of this soluble complex. A proposed model is that cholesterol-modified Hh protein is recruited to lipid rafts (exoplasmic leaflet of the plasma membrane) where multimerization occurs and the soluble complex can then travel to target cells and elicit its effect.

6. GPI anchoring of proteins

Glycosylphosphatidylinositols (GPIs) are a ubiquitous family of eukaryotic glycolipids. GPIs were originally discovered covalently linked to cell surface glycoproteins and recognized to be an important alternative mechanism for anchoring proteins to the cell surface [15]. After synthesis of GPI anchors and attachment to protein in the endoplasmic reticulum (ER), GPI-anchored proteins are transported via the secretory pathway to the cell surface. GPI anchors are found on a variety of functionally diverse proteins and glycoconjugates including cell surface receptors (folate receptor, CD14), cell adhesion molecules (NCAM isoforms, carcinoembryonic antigen variants), cell surface hydrolases (5'-nucleotidase, acetylcholinesterase), complement regulatory proteins (decay accelerating factor), and protozoal surface molecules (*Trypanosoma brucei* variant surface glycoprotein, *Leishmania* lipophosphoglycan). Along with serving to attach proteins to the cell surface, GPI-anchored proteins appear to be markers and major constituents of 'detergent-resistant' lipid rafts, the sphingolipid- and sterol-rich domains in membranes that are postulated to play an important role in the activation of signaling cascades [16].

6.1. Biosynthesis of GPI

GPIs are assembled via a biosynthetically and topologically complex metabolic pathway that is comprised of at least 10 steps and requires the participation of at least 20 distinct gene products (reviewed in [17–19]). In basic terms, synthesis occurs in the ER and involves the sequential addition of monosaccharides to phosphatidylinositol (PI) yielding the core GPI structure of ethanolamine-*P*-6Man α 1–2Man α 1–6Man α 1–4GlcN α 1–6*myo*-inositol-*P*-lipid which is then added to proteins. Many of the genes encoding enzymes involved in synthesis have been identified. Fig. 5 illustrates the sequence of reactions involved in the assembly of a GPI protein anchor precursor in mammalian cells; variations of this general sequence are found in all organisms studied thus far.

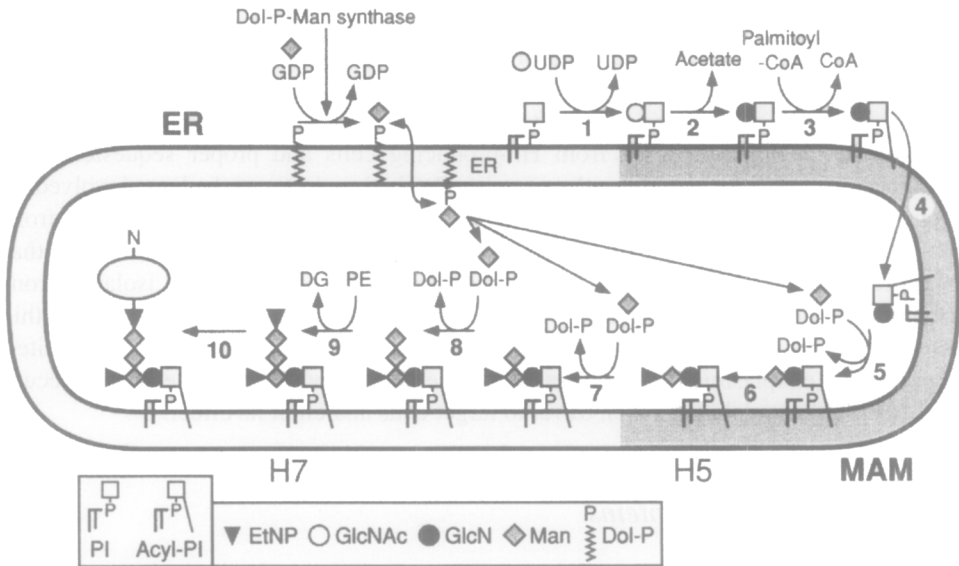


Fig. 5. GPI biosynthesis in the ER (redrawn from [17]). The pathway is initiated on the cytoplasmic face of the ER and proceeds clockwise in the figure to yield the lumenally oriented mature GPI structure H7 (mammalian cells synthesize another mature GPI termed H8 (not depicted) which is identical to H7 except that it contains an additional EtNP residue linked to the second mannose). H7 is attached to ER-translocated proteins bearing a C-terminal, GPI-directing signal sequence. MAM, mitochondria-associated membrane. See text for other abbreviations.

Initiation of GPI biosynthesis occurs on the cytoplasmic leaflet of the ER via the addition of *N*-acetylglucosamine (GlcNAc) from UDP-GlcNAc to PI yielding GlcNAc-PI (Fig. 5, step 1). The enzyme which mediates this first step is a complex consisting of at least six proteins in mammalian cells: PIG-A, PIG-C, PIG-H, GPI1, PIG-P, and DPM2; however, the subunit stoichiometry and functions of most of the subunits are not yet known. PIG-A which has homology to many glycosyltransferases is most likely the catalytic component. PIG-C, PIG-H, and PIG-P are essential components of the complex but their precise functions are unknown. GPI1 is important but not essential for activity and DPM2 appears to be a positive regulator stimulating GlcNAc transferase activity approximately 3-fold.

GlcNAc-PI is subsequently de-*N*-acetylated to generate GlcN-PI by the GlcNAc-PI de-*N*-acetylase, PIG-L (Fig. 5, step 2). The third step in biosynthesis is the acylation, predominantly palmitoylation, of the inositol residue of GlcN-PI at the 2 position (Fig. 5, step 3). The acylation step is required prior to the addition of three mannose residues (four in yeast) to GlcN-(acyl)PI and modification of the mannose residues with phosphoethanolamine (EtNP) side chains. In contrast to the first two steps of synthesis which occur on the cytoplasmic leaflet of the ER, the mannosylation reactions are thought to occur lumenally based on the predicted topology of the mannosyl transferases. This indicates that the substrate (GlcN-PI or GlcN-(acyl)PI) must be flipped across the ER membrane bilayer (Fig. 5, step 4).

Dolichol-P-mannose serves as the mannose donor for all three mannosylation steps. The first mannosylation is catalyzed by PIG-M, the GPI α 1–4-mannosyltransferase (Fig. 5, step 5). Phosphoethanolamine is then added to the 2 position of the first mannose residue (Fig. 5, step 6); PIG-N is involved in this reaction and is probably the EtNP transferase. A second mannose residue is then added by an α 1–6-mannosyltransferase that has not yet been identified (Fig. 5, step 7). An α 1–2-mannosyltransferase, most likely the protein PIG-B, mediates transfer of the third mannose residue to the GPI moiety (Fig. 5, step 8). EtNP residues can be added to both the second and third mannose residues as well (Fig. 5, steps 9 and 10). PIG-O and PIG-F are the gene products responsible for transferring EtNP from phosphatidylethanolamine (PE) to the 6 position on the third mannose (Fig. 5, step 10). These proteins form a complex, with PIG-O most likely serving the catalytic role. This modification is crucial as it is this EtNP residue on the third, or terminal mannose that is involved in the linkage of GPI to the carboxy-terminus of proteins destined to be GPI-anchored (Fig. 5, step 11). Fully assembled GPI structures, or the GPI moiety in GPI-anchored proteins, are frequently subject to lipid re-modeling reactions in which fatty acids or the entire lipid structure are replaced with different fatty acids or lipids.

6.2. Subcellular location and membrane topology of GPI biosynthesis

GPI biosynthesis occurs in the ER, however the ER is an extremely heterogeneous organelle and GPI biosynthetic activities are not uniformly distributed throughout the bulk ER [19]. The capacity to synthesize GlcNAc-PI appears to be uniformly distributed in the ER in mammalian cells while subsequent biosynthetic steps leading to the formation of (Etn-P)Man-GlcN-PI are highly enriched in an ER domain that is associated with mitochondria (mitochondria-associated ER membrane domain) (Fig. 5). The reason for segregation of some GPI biosynthetic reactions to the mitochondria-associated membrane is not known.

In addition to spatial segregation of GPI biosynthetic activities within subdomains of the ER, the biosynthetic pathway is also topologically complex with the first two steps occurring in the cytoplasmic leaflet of the membrane and later mannosylation steps and attachment of GPI to protein occurring in the luminal leaflet (Vidugiriene, 1993, 1994). Thus, for GPI synthesis to go to completion GPI precursors must be flipped across the ER bilayer to access the enzymes involved in the late steps of synthesis [19]. The point at which this flipping occurs as well as the mechanism for flipping GPI precursors remain to be found.

6.3. Attachment of GPIs to proteins

Newly synthesized proteins are attached to pre-existing GPIs in the luminal leaflet of the ER by a GPI: protein transamidase complex [20,21]. Proteins destined to be GPI-anchored contain an N-terminal signal sequence for targeting to the ER and a C-terminal signal sequence that directs GPI-anchoring. The N-terminal signal sequence is cleaved by signal peptidase during or after translocation of the nascent polypeptide into the ER lumen. The C-terminal GPI signal sequence is then

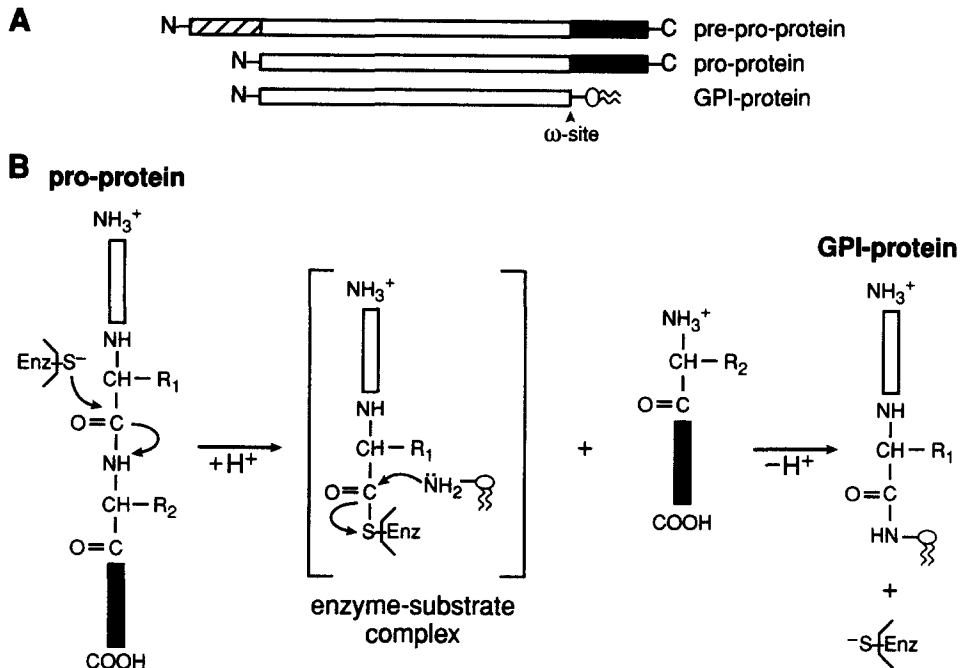


Fig. 6. GPI anchoring of proteins. (A) The processing steps involved in the conversion of a pre-pro-protein to a GPI-anchored protein. (B) The transamidation reaction in which a GPI anchor is attached to a pro-protein. The transamidase enzyme, which contains an enzymatically critical cysteine residue, is depicted as Enz-S^- . The ω amino acid is depicted with a side chain labeled R_1 ; the $\omega + 1$ amino acid has a side chain labeled R_2 .

cleaved and replaced with a GPI anchor through the action of GPI transamidase (Fig. 6).

The C-terminal GPI-signal peptide, which is cleaved at what is referred to as the ω site, is necessary and sufficient for designating that a protein becomes GPI-anchored. The sequence motif for GPI-anchoring includes four sequence elements: (1) an unstructured linker region of about 11 residues ($\omega - 11$ to $\omega - 1$); (2) four preferably small amino acids ($\omega - 1$ to $\omega + 2$) including the cleavage site; (3) a moderately polar spacer region ($\omega + 3$ to $\omega + 9$); and (4) a hydrophobic tail from $\omega + 9$ or $\omega + 10$ to the C-terminal end [21].

The GPI transamidase complex responsible for attaching the protein to the GPI-anchor consists of at least five subunits: Gpi8p, Gaa1p, PIG-S, PIG-T and PIG-U (Vidugiriene, 2001; Ohishi, 2001; Fraering, 2001; Vainauskas, 2002). The enzyme recognizes a pro-protein containing a GPI-anchoring signal sequence, cleaves the signal sequence and attaches the new carboxy terminus to the terminal EtNP of a pre-existing GPI. Gpi8p is a membrane protein with a large luminal domain and has homology to a family of cysteine proteases, one of which is a plant endopeptidase with transamidase activity. Cysteine and histidine residues conserved between the cysteine proteases and Gpi8p are essential for GPI transamidase activity, thus it is proposed that Gpi8p cleaves

the signal sequence and functions in the transamidation reaction. The functions of the remaining subunits are unknown.

6.4. GPI-anchoring in mammals, parasitic protozoa and yeast

GPI-deficient mammalian cells are viable in tissue culture, but a GPI defect has clear consequences for multicellular organisms. Transgenic mouse embryos lacking the ability to initiate GPI biosynthesis (defective in PIG-A) do not develop beyond the ninth day of gestation [18]. The inability of certain blood cells to express GPI-anchored proteins results in the rare human disease, paroxysmal nocturnal hemoglobinuria (PNH), characterized by intravascular hemolysis, thrombosis and bone marrow failure. The disease is caused by a somatic mutation of the X-linked PIG-A gene (encoding the catalytic component of the enzyme responsible for the first step in GPI biosynthesis, see above) in hematopoietic stem cells. Cells defective in PIG-A are either unable to synthesize GPIs or exhibit a significant decrease in the synthesis of GPIs and thus decreased expression of GPI-anchored proteins. Red blood cells no longer expressing GPI-anchored complement regulatory proteins (e.g. CD55 and CD59) become susceptible to complement-mediated lysis, resulting in the release of heme and hemoglobin into the blood, filtering by the kidney, and excretion in the urine. Interestingly, while the GPI-deficient phenotype of hematopoietic stem cells makes the cells more sensitive to complement-mediated lysis, the defective PNH clone still persists. This implies that there must be other factors which select for clonal expansion and maintenance of GPI-deficient blood cells. One proposal is that PNH patients possess autoreactive T cells that target GPI on the surface of hematopoietic stem cells, thus PNH cells would evade damage because they lack surface GPI molecules [22].

GPI anchoring is the most prominent mode of attachment for cell surface proteins and glycans in parasitic protozoa [23]. Pathogenic protozoa, including species of the genera *Trypanosoma*, *Leishmania*, and *Plasmodium*, display abundant GPI-anchored cell surface macromolecules that play crucial roles in parasite infectivity and survival. An example is the GPI-anchored variant surface glycoprotein (VSG) of bloodstream forms of *Trypanosoma brucei*, the causative agent of African sleeping sickness.

GPIs serve a unique function in yeast (*S. cerevisiae*). In addition to anchoring secretory proteins to the cell surface, GPIs play an additional role in yeast cell wall biosynthesis [24]. The yeast cell wall consists of a fibrous lattice of mannoproteins, β 1,3-glucan, β 1,6-glucan and chitin. During cell wall biosynthesis GPI-anchored mannoproteins are transported through the secretory pathway to the cell surface. After arrival at the plasma membrane, a transglycosylation reaction (catalyzed by an unknown enzyme) results in cleavage of the GPI moiety between GlcN and the first mannose residue and formation of a glycosidic linkage between the mannoprotein-GPI-remnant and β 1,6-glucan. Mutations in GPI biosynthesis are lethal in yeast and decreased levels of GPI biosynthesis cause growth defects and aberrant cell wall biogenesis. However, not all GPI-anchored proteins become crosslinked to β 1,6-glucan; the amino acids within four or five residues upstream of the ω -site determine whether the protein becomes incorporated into the cell wall or remains anchored to the plasma membrane. A unique characteristic of yeast GPIs is the addition of a fourth mannose residue to the core GPI structure (via an α 1-2

linkage to the third mannose). Most GPI-anchored proteins in yeast also undergo lipid remodeling replacing the glycerolipid backbone with ceramide. The remodeling occurs after the attachment of protein to the GPI and can occur in both the ER and the Golgi.

6.5. Functions of GPI anchors

GPIs function as membrane anchors for secretory proteins and are believed to provide targeting signals that influence the intracellular trafficking of these proteins. Recent results indicate that GPI-proteins are packaged into unique transport vesicles, distinct from those carrying other secretory proteins, for export from the ER (Muñiz, 2001). GPI-anchored proteins have been shown to coalesce with sphingolipids and cholesterol into detergent insoluble membrane domains, or lipid rafts. Association of molecules in rafts at the plasma membrane, including GPI-anchored proteins (in the exoplasmic leaflet) and acylated signaling molecules (in the cytoplasmic leaflet), is postulated to play an important role in the activation of signaling cascades [16]. In some polarized epithelial cells, GPI-anchored proteins and many glycosphingolipids are sorted in the *trans* Golgi network and specifically targeted to the apical membrane. Once at the plasma membrane GPI-anchored proteins can undergo endocytosis and recycling back to the cell surface. However, uptake of GPI-anchored proteins is approximately five-times slower than that of receptor-mediated endocytosis and recycling to the plasma membrane is about three-times slower than that of recycling receptors (Sabharanjak, 2002). The current view is that GPI-anchoring of proteins directs their segregation into lipid rafts and affects their sorting in both the exocytic and endocytic pathways.

A GPI-anchor may also allow a protein to be selectively released from the cell surface upon hydrolysis by a GPI-specific phospholipase (e.g., PI-PLC or GPI-PLD). This has been shown to occur for certain GPI-anchored proteins in mammalian cell culture. One example is GPI-anchored membrane dipeptidase which is released from the adipocyte cell surface by a phospholipase C in response to insulin (Movahedi, 2000). Interestingly, other GPI-anchored proteins are not released indicating a level of regulation in insulin-stimulated hydrolysis of GPI-anchored proteins. GPI-anchored molecules have also been shown to transfer between cells and stably insert in the external leaflet of the acceptor cell's plasma membrane (Low, 1998). The biological significance of this is unclear; however, the ability of GPI-anchored proteins to transfer between cells has implications for the expression of foreign proteins on the cell surface.

7. Future directions

This chapter describes post-translational lipid modifications of proteins that represent functionally critical elaborations of protein structure. These modifications, with the possible exception of thioacylation, can be anticipated at the level of primary protein sequence, enabling predictions about the localization and likely behavior of the modified proteins within cells. Although the functional diversity of lipid-modified proteins makes it difficult to arrive at generalizations about the evolutionary impetus for lipid anchoring compared to the use of 'conventional' protein transmembrane domains, the observation

that many lipid modified proteins are associated with cholesterol and sphingolipid-rich membrane domains (rafts) is suggestive. Analyses of the biophysical characteristics of membrane association of lipid modified proteins, the role of raft-associated proteins in cell signaling, and the regulation of protein function by lipid modification are likely to remain fruitful areas of investigation.

Although many biochemical and cell biological aspects of the various lipid modifications remain to be elucidated, it is worth highlighting the scope for investigation offered by the GPI biosynthetic pathway. GPI anchoring is arguably the most biosynthetically and topologically complex of the lipid modifications described in this chapter. Despite the identification of numerous genes and gene products associated with the GPI biosynthetic pathway, the enzymology of GPI biosynthesis, including analyses of enzyme structure, basis for enzyme localization to the ER and ER domains, and transbilayer distribution of the pathway remains open to new investigation.

Abbreviations

ER	endoplasmic reticulum
EtN	ethanolamine
FTase	protein farnesyltransferase
FTI	FTase inhibitor
GGTase	protein geranylgeranyltransferase
GlcN	glucosamine
GlcNAc	<i>N</i> -acetylglucosamine
GPI	glycosylphosphatidylinositol
G proteins	GTP-binding proteins
Hh	hedgehog protein
Man	mannose
NMT	<i>N</i> -myristoyltransferase
PE	phosphatidylethanolamine
PI	phosphatidylinositol
PNH	paroxysmal nocturnal hemoglobinuria

References

1. Zhang, F.L. and Casey, P.J. (1996) Protein prenylation: molecular mechanisms and functional consequences. *Annu. Rev. Biochem.* 65, 241–269.
2. Magee, T. and Marshall, C. (1999) New insights into the interaction of ras with the plasma membrane. *Cell* 98, 9–12.
3. Pereira-Leal, J.B., Hume, A.N. and Seabra, M.C. (2001) Prenylation of rab GTPases: molecular mechanisms and involvement in genetic disease. *FEBS Lett.* 498, 197–200.
4. Gelb, M.H. (1997) Protein prenylation — signal transduction in two dimensions. *Science* 275, 1750–1751.
5. Gelb, M.H., Scholten, J.D. and Sebolt-Leopold, J.S. (1998) Protein prenylation: from discovery to prospects for cancer treatment. *Curr. Opin. Chem. Biol.* 2, 40–48.

6. Johnston, S.R. (2001) Farnesyltransferase inhibitors: a novel targeted therapy for cancer. *Lancet Oncol.* 2, 18–26.
7. Farazi, T.A., Waksman, G. and Gordon, J.I. (2001) The biology and enzymology of protein N-myristoylation. *J. Biol. Chem.* 276, 39501–39504.
8. Mumby, S.M. (1997) Reversible palmitoylation of signaling proteins. *Curr. Opin. Cell Biol.* 9, 148–154.
9. Linder, M.E. (2000) Reversible modification of proteins with thioester-linked fatty acids. In: F. Tamanoi and D.S. Sigman (Eds.), *The Enzymes*, Volume XXI 'Protein Lipidation'. Academic Press, New York, pp. 215–240.
10. Resh, M.D. (1999) Fatty acylation of proteins: new insights into membrane targeting of myristoylated and palmitoylated proteins. *Biochim. Biophys. Acta* 1451, 1–16.
11. Shahinian, S. and Silvius, J.R. (1995) Doubly-lipid-modified protein sequence motifs exhibit long-lived anchorage to lipid bilayer membranes. *Biochemistry* 34, 3813–3822.
12. Mann, R.K. and Beachy, P.A. (2000) Cholesterol modification of proteins. *Biochim. Biophys. Acta* 1529, 188–202.
13. Incardona, J.P. and Eaton, S. (2000) Cholesterol in signal transduction. *Curr. Opin. Cell Biol.* 12, 193–203.
14. Ingham, P.W. (2000) Hedgehog signaling: how cholesterol modulates the signal. *Curr. Biol.* 10, R180–R183.
15. Low, M.G., Ferguson, M.A.J., Futerman, A.H. and Silman, I. (1986) Covalently attached phosphatidylinositol as a hydrophobic anchor for membrane proteins. *Trends Biochem. Sci.* 11, 212–215.
16. Simons, K. and Toomre, D. (2000) Lipid rafts and signal transduction. *Nat. Rev. (Mol. Cell Biol.)* 1, 31–39.
17. Kinoshita, T. and Inoue, N. (2000) Dissecting and manipulating the pathway for glycosylphosphatidylinositol-anchor biosynthesis. *Curr. Opin. Chem. Biol.* 4, 632–638.
18. Tiede, A., Bastisch, I., Schubert, J., Orlean, P. and Schmidt, R.E. (1999) Biosynthesis of glycosylphosphatidylinositols in mammals and unicellular microbes. *Biol. Chem.* 380, 503–523.
19. McConville, M.J. and Menon, A.K. (2000) Recent developments in the cell biology and biochemistry of glycosylphosphatidylinositol lipids. *Mol. Membr. Biol.* 17, 1–16.
20. Udenfriend, S. and Kodukula, K. (1995) How glycosyl-phosphatidylinositol-anchored proteins are made. *Annu. Rev. Biochem.* 64, 563–591.
21. Eisenhaber, B., Bork, P. and Eisenhaber, F. (2001) Post-translational GPI lipid anchor modification of proteins in kingdoms of life: analysis of protein sequence data from complete genomes. *Protein Eng.* 14, 17–25.
22. Karadimitris, A. and Luzatto, L. (2001) The cellular pathogenesis of paroxysmal nocturnal hemoglobinuria. *Leukemia* 15, 1148–1152.
23. Ferguson, M.A.J. (2000) Glycosylphosphatidylinositol biosynthesis validated as a drug target for African sleeping sickness. *Proc. Natl. Acad. Sci. USA* 97, 10673–10675.
24. Lipke, P.N. and Ovalle, R. (1998) Cell wall architecture in yeast: new structure and new challenges. *J. Bacteriol.* 180, 3735–3740.

Fatty acid and phospholipid metabolism in prokaryotes

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1. Bacterial lipid metabolism

Bacteria are a versatile tool for the study of metabolic pathways. This is especially true for the Gram-negative *Escherichia coli*. These bacteria are easy to grow, and the growth conditions can be controlled and manipulated by the investigator. Most importantly, they are suitable for genetic manipulation and their genome sequence is available [1]. *E. coli* have been studied extensively, and the genes and enzymes of phospholipid metabolism were first delineated using this organism [2]. However, *E. coli* is not typical of all Gram-negative bacteria, let alone all eubacteria. Certain key differences between this model system and other organisms will be highlighted in the relevant sections.

Phospholipids in bacteria comprise about 10% of the dry weight of the cell, and each mole of lipid requires about 32 mole of ATP for its synthesis. Thus, phospholipid synthesis requires significant investment by the cell, and the advantages of maintaining fine control over the pathway are obvious. The pathway in most bacteria is catalyzed by a series of discrete proteins: the enzymes of fatty acid synthesis are cytosolic, while those of membrane lipid synthesis are mainly integral inner membrane proteins. The differences between the bacterial and mammalian enzymes offer attractive targets for novel antimicrobial drugs, and this has been a driving force behind much of the recent research.

The study of phospholipid enzymology in *E. coli* dates back to the early 1960s, when work in the laboratory of Vagelos discovered that the intermediates in fatty acid synthesis are bound to a heat stable cofactor termed acyl carrier protein (ACP) [3]. The enzymes of fatty acid synthesis are soluble proteins whose individual activities can be assayed in crude cell extracts or purified preparations. This is markedly different from the mammalian fatty acid synthase, a large multi-functional polypeptide with intermediates covalently attached (Chapter 6). Thus, the bacterial enzymes became a focus of intensive study, especially in the laboratories of Vagelos, Bloch and Wakil. Soon, the structures of all of the intermediates were known, and the basic chemical reactions required had been described [4]. During the late 1960s, work on the enzymes of phospholipid synthesis in bacteria flourished, and, mainly through classical identification experiments in the Kennedy laboratory, the intermediates in that pathway were established [5].

A second phase of bacterial phospholipid research, during the 1970s and early 1980s,

was the identification of mutants in the pathway [6]. The genetic manipulation of *E. coli* is relatively facile. Mutants in many specific enzymes were generated by employing mutagens in combination with a battery of clever selection and screening techniques [7]. Such mutations generally fall into one of two classes. Firstly, they may confer an auxotrophy on a strain, such as a requirement for unsaturated fatty acids or glycerol phosphate. Such mutants have generally lost the ability to produce a key biosynthetic enzyme (e.g. *fabA* mutants, which require supplementation with unsaturated fatty acids), or they may be more complex (e.g. *plsB* mutants require high glycerol phosphate concentrations due to a K_m defect in the enzyme and a second site mutation). Secondly, they may be conditionally defective, usually at elevated temperatures. For example, strains with the *fabI*(Ts) mutation grow at 30°C, but are not viable at 42°C. This type of defect is usually ascribed to an amino acid change in an essential enzyme that renders it unstable at higher temperatures. Techniques using the bacteriophage P1 are available for the movement of these alleles into other host strains, thus allowing for the mapping of the genes to specific regions of the chromosome, or the generation of strains with particular combinations of mutations. Regulatory mutants were identified, affecting multiple enzymes with a single mutation, that allowed for regulatory networks to be investigated. The membrane-bound enzymes of phospholipid synthesis were not amenable to analysis using standard biochemical approaches, and the genetic approach allowed these enzymes to be identified.

Next came the cloning and detailed study of the enzymes of lipid metabolism during the late 1980s and 1990s [2]. Plasmid-based expression systems were used to examine overexpression of enzymes on pathway regulation, and purified enzymes could be more easily obtained for biochemical analysis. With the availability of the sequence data generated by these clones, more precise methods for the construction of specific mutations also became available. Specific genes, or portions of genes, could be 'knocked-out' by targeted replacement based on sequence information, as opposed to random insertion of phage DNA. In the late 1990s, the genomic sequences of a broad spectrum of bacteria started to become available [8]. These data fuel the current phase of bacterial lipid metabolic research and comparative enzymology.

With the advent of genome information, many of the genes and gene products first identified and characterized in *E. coli* (Table 1) were identified in other bacteria, thus these genes can easily be cloned and their properties compared to those of *E. coli*. The pathways, especially fatty acid biosynthesis, are generally highly conserved amongst bacteria. Coupled with recent findings about the efficacy of certain inhibitors of the pathway, considerable attention is now being focused on fatty acid synthesis as a target for novel drug design [9]. The availability of pure enzymes has also stimulated progress in structural biology, and many new 3-dimensional structures of enzymes related to lipid metabolism have been recently solved.

2. Membrane systems of bacteria

Phospholipids in *E. coli* and other Gram-negative bacteria are used in the construction of the inner and outer membranes (Fig. 4 in Chapter 1). The inner membrane is im-

permeable to solutes unless specific transport systems are present. The outer membrane contains pores that allow the passage of molecules having a molecular weight less than 600, and is rich in structural lipoproteins and proteins involved in the transport of high molecular weight compounds. The outer layer of the outer membrane is composed primarily of lipopolysaccharides rather than phospholipid. Between the inner and outer membranes is an osmotically active compartment called the periplasmic space. Membrane-derived oligosaccharides, peptidoglycan, and binding proteins involved with metabolite transport are found in this compartment. Gram-positive bacteria do not possess an outer membrane. Instead, they have a membrane bilayer surrounded by a thick layer of peptidoglycan decorated with proteins, carbohydrates and, often, teichoic and lipoteichoic acid.

3. *Bacterial fatty acid biosynthesis*

3.1. *Acyl carrier protein*

A unique feature of fatty acid synthesis in bacteria is the presence of the small (8.86 kDa), acidic and highly soluble ACP, the product of the *acpP* gene. ACP is one of the most abundant proteins in *E. coli*, constituting about 0.25% of the total soluble protein ($\sim 6 \times 10^4$ molecules/cell). The acyl intermediates of fatty acid biosynthesis are bound to the protein through a thioester linkage to the terminal sulfhydryl of the 4'-phosphopantetheine prosthetic group. The prosthetic group sulfhydryl is the only thiol group of ACP and is attached to the protein via a phosphodiester linkage to Ser-36. ACP must interact specifically and transiently with all of the enzymes of fatty acid biosynthesis (except acetyl-CoA carboxylase), and does so through interactions with exposed negative residues on ACP with a patch of positive residues on the surfaces of the *fab* enzymes.

The ACP pool in normally growing cells is approximately one-eighth the coenzyme A (CoA) pool, the other acyl group carrier in cells. The prosthetic group of ACP is produced from CoA, and a common feature of both is the pantetheine arm for thioester formation. Virtually all of the ACP is maintained in the active, holo-form in vivo indicating that the supply of prosthetic group does not limit fatty acid biosynthesis. During logarithmic growth, a significant pool of ACP is unacylated. The ACP pool must be severely depleted before an effect on fatty acid and phospholipid synthesis can be detected. Overproduction of ACP generally yields high levels of apo-ACP, which is toxic to the cell by inhibition of the glycerol phosphate acyltransferase. The 4'-phosphopantetheine prosthetic group is transferred from CoA to apo-ACP by the 14-kDa monomeric [ACP]synthase. The [ACP]synthase from *Bacillus subtilis* has been crystallized in complex with ACP to give the first detailed look at ACP-protein interactions. ACP plays other roles in cell physiology, donating acyl chains to membrane-derived oligosaccharides, lipoic acid and quorum sensors.

Table 1
Genes of lipid metabolism

Gene	Protein
<i>aas</i>	2-acyl-GPE acyltransferase
<i>accA</i>	Carboxyltransferase subunit
<i>accB</i>	Biotin carboxy carrier protein
<i>accC</i>	Biotin carboxylase
<i>accD</i>	Carboxyl transferase subunit
<i>acpP</i>	Acyl carrier protein
<i>acpD</i>	Azoreductase
<i>acpS</i>	Acyl carrier protein synthase
<i>cdh</i>	CDP-diacylglycerol hydrolase
<i>cdsA</i>	CDP-diacylglycerol synthase
<i>cdsS</i>	Stabilizes mutant CDP-diacylglycerol synthase
<i>cfa</i>	Cyclopropane fatty acid synthase
<i>cls</i>	Cardiolipin synthase
<i>desA</i>	Desaturase
<i>dgk</i>	Diacylglycerol kinase
<i>fabA</i>	β -Hydroxydecanoyl-ACP dehydrase
<i>fabB</i>	β -Ketoacyl-ACP synthase I
<i>fabD</i>	Malonyl-CoA : ACP transacylase
<i>fabF</i>	β -Ketoacyl-ACP synthase II
<i>fabG</i>	β -Ketoacyl-ACP reductase
<i>fabH</i>	β -Ketoacyl-ACP-synthase III
<i>fabI</i>	Enoyl-ACP reductase I
<i>fabK</i>	Enoyl-ACP reductase II
<i>fabL</i>	Enoyl-ACP reductase III
<i>fabZ</i>	β -Hydroxyacyl-ACP dehydrase
<i>fadA</i>	β -Ketoacyl-CoA thiolase
<i>fadB</i>	4-Function enzyme of β -oxidation: β -hydroxyacyl-CoA dehydrogenase and epimerase; <i>cis</i> - β - <i>trans</i> -2-enoyl-CoA isomerase and enoyl-CoA hydratase
<i>fadD</i>	Acyl-CoA synthetase
<i>fadE</i>	Electron transferring flavoprotein
<i>fadF</i>	Acyl-CoA dehydrogenase
<i>fadG</i>	Acyl-CoA dehydrogenase?
<i>fadH</i>	2,4-dienoyl-CoA reductase
<i>fadL</i>	Long-chain fatty acid transport protein precursor
<i>fadR</i>	Transcriptional regulator
<i>fatA</i>	Unknown, possible transcription factor
<i>gpsA</i>	Glycerol phosphate synthase
<i>htrB</i>	KDO ₂ -lipid IV _A acyloxy lauroyltransferase
<i>kdtA</i>	KDO transferase
<i>lpxA</i>	UDP-GlcNAc β -hydroxymyristoyl-ACP acyltransferase
<i>lpxB</i>	Disaccharide-1-P synthase
<i>lpxC</i>	UDP- β -O-hydroxymyristoyl-GlcNAc deacetylase
<i>lpxD</i>	UDP- β -O-hydroxymyristoyl-GlcNAc <i>N</i> -acyltransferase
<i>lpxK</i>	Disaccharide-1-P 4'-kinase
<i>lpxP</i>	KDO ₂ -lipid IV _A acyloxy palmitoyltransferase
<i>mdoB</i>	Phosphatidylglycerol : membrane-oligosaccharide glycerophosphotransferase
<i>msbA</i>	Lipid flippase
<i>msbB</i>	KDO ₂ -lipid IV _A acyloxy myristoyltransferase

Table 1 (continued)

Gene	Protein
<i>pgpA</i>	PGP phosphatase
<i>pgpB</i>	PGP phosphatase
<i>pgsA</i>	PGP synthase
<i>pldA</i>	Detergent-resistant phospholipase A
<i>pldB</i>	Inner membrane lysophospholipase
<i>plsB</i>	Glycerol phosphate acyltransferase
<i>plsC</i>	1-Acylglycerol phosphate acyltransferase
<i>plsX</i>	Unknown
<i>plsD</i>	Glycerol phosphate acyltransferase
<i>psd</i>	PS decarboxylase
<i>pssA</i>	PS synthase
<i>tesA</i>	Thioesterase I
<i>tesB</i>	Thioesterase II

3.2. Acetyl-CoA carboxylase

Acetyl-CoA carboxylase catalyzes the first committed step of fatty acid synthesis, the conversion of acetyl-CoA to malonyl-CoA. Acetyl-CoA is a key intermediate in many pathways, and forms the majority of the CoA species within the cell at concentrations of about 0.5–1.0 mM during logarithmic growth on glucose [10]. Malonyl-CoA is normally present at 0.5% of this level, and is used exclusively for fatty acid biosynthesis. The overall carboxylation reaction is composed of two distinct half reactions: the ATP-dependent carboxylation of biotin with bicarbonate to form carboxybiotin; and transfer of the carboxyl group from carboxybiotin to acetyl-CoA, forming malonyl-CoA (Fig. 1).

Each acetyl-CoA carboxylase half reaction is catalyzed by a different protein subcomplex. The vitamin biotin is covalently coupled through an amide bond to a lysine residue on biotin carboxyl carrier protein (BCCP; a homodimer of 16.7-kDa

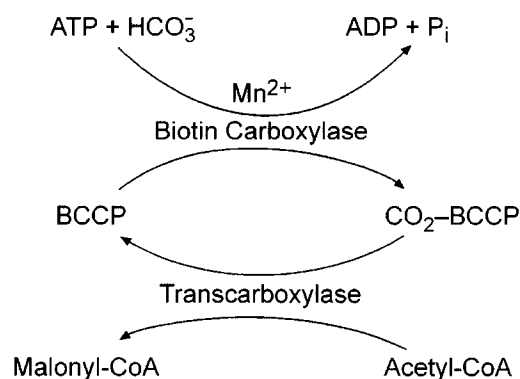


Fig. 1. The acetyl-CoA carboxylase reaction is performed in two steps. Biotin, covalently attached to BCCP (biotin carboxyl carrier protein, *accB*), is carboxylated by the carboxylase subunit (*accC*). The heterodimeric transcarboxylase (*accA* and *accD*) then transfers the CO₂ to acetyl-CoA, forming malonyl-CoA.

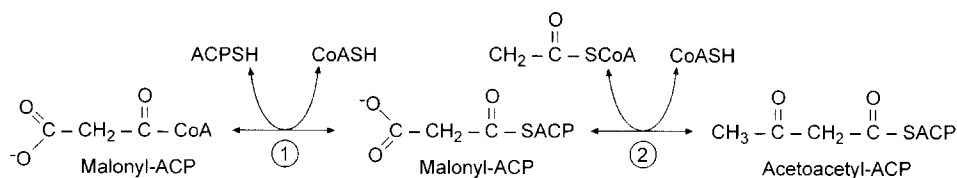


Fig. 2. Initiation of fatty acid synthesis. (1) Malonyl-CoA : ACP transacylase (FabD) transfers the malonyl group from CoA to ACP and then (2) β -ketoacyl-ACP synthase III (FabH) catalyzes the initial irreversible condensation of malonyl-ACP with acetyl-CoA to form acetoacetyl-ACP.

monomers encoded by *accB*) by a specific enzyme, biotin-apoprotein ligase (encoded by *birA*), and is essential to activity. The crystal and solution structures of the biotinyl domain of BCCP have been determined, and reveal a unique ‘thumb’ required for activity. Carboxylation of biotin is catalyzed by biotin carboxylase (encoded by *accC*), a homodimeric enzyme composed of 55-kDa subunits that is copurified complexed with BCCP. The *accB* and *accC* genes form an operon. The 3-dimensional structure of the biotin carboxylase subunit has been solved by X-ray diffraction revealing an ‘ATP-grasp’ motif for nucleotide binding. The mechanism of biotin carboxylation involves the reaction of ATP and CO_2 to form the short-lived carboxyphosphate, which then interacts with biotin on BCCP for CO_2 transfer to the 1'-nitrogen.

The carboxyltransferase enzyme that transfers the carboxy group from the biotin moiety of BCCP to acetyl-CoA is a heterotetramer composed of two copies of two dissimilar subunits, α (35 kDa) and β (33 kDa) (encoded by *accA* and *accD*, respectively). Sequence analysis suggests that the acetyl-CoA binding site lies within the AccA subunit. Strains with mutations in *accB* and *accD* have been obtained that are temperature sensitive for growth, indicating that this reaction is essential. It is thought that the enzyme present *in vivo* is composed of one copy of each subcomplex, with a combined molecular weight of 280 kDa.

3.3. Initiation of fatty acid biosynthesis

For the malonate group to be used for fatty acid synthesis, it must first be transferred from malonyl-CoA to malonyl-ACP by the 32.4-kDa monomeric malonyl-CoA : ACP transacylase, the product of the *fabD* gene (Fig. 2). A stable malonyl-serine enzyme intermediate is formed during the course of the FabD reaction, and subsequent nucleophilic attack on this ester by the sulfhydryl of ACP yields malonyl-ACP. The high reactivity of the serine in malonyl-ACP transacylase is due to the active site being composed of a nucleophilic elbow as observed in α/β hydrolases. The serine is hydrogen bonded to His-201 in a fashion similar to serine hydrolases.

The last two carbons of the fatty acid chain (i.e. those most distal from the carboxylate group) are actually the first introduced into the nascent chain, and acetyl-CoA can be thought of as the ‘primer’ molecule of fatty acid synthesis in *E. coli*. The initial condensation reaction, catalyzed by β -ketoacyl-ACP synthase III (FabH), utilizes acetyl-CoA and malonyl-ACP to form the four carbon acetoacetyl-ACP with concomitant loss of CO_2 (Fig. 2). FabH also possesses acetyl-CoA : ACP transacylase

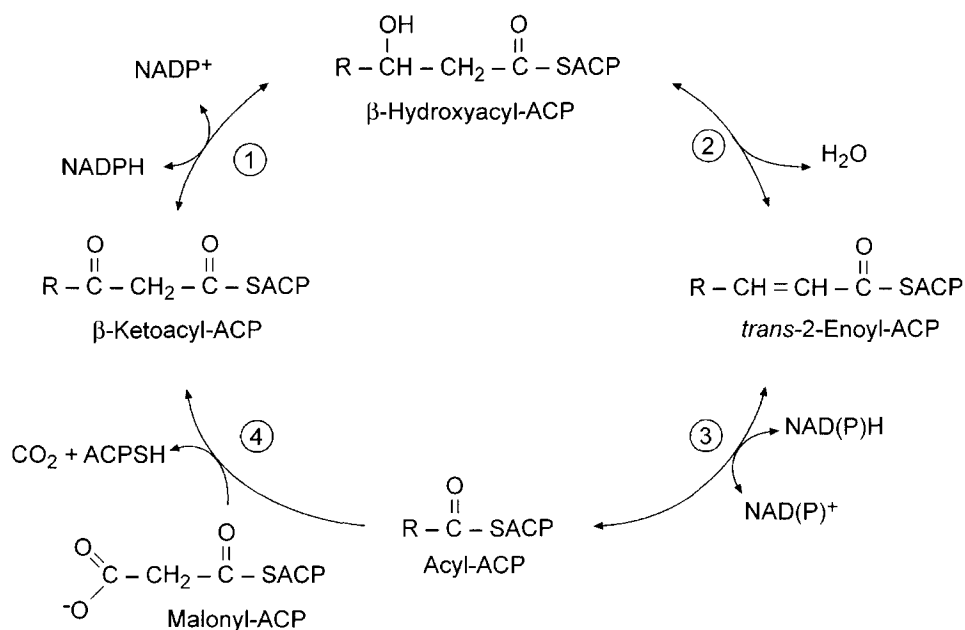


Fig. 3. Cycles of fatty acyl chain elongation. All intermediates in fatty acid synthesis are shuttled through the cytosol as thioesters of the acyl carrier protein (ACP). (1) β -Ketoacyl-ACP reductase (FabG). (2) β -Hydroxyacyl-ACP dehydrase (FabA or FabZ). (3) *trans*-2-Enoyl-ACP reductase I (FabI). (4) β -Ketoacyl-ACP synthase I or II (FabB or FabF).

activity, and for many years it was thought that acetyl-ACP was the actual primer. However, acetyl-ACP appears to be a product of a side reaction, and plays no direct role in fatty acid synthesis.

3.4. Elongation of acyl chains

Four enzymes participate in each iterative cycle of chain elongation (Fig. 3). First, β -ketoacyl-ACP synthase I or II (*fabB* or *fabF*) adds a two-carbon unit from malonyl-ACP to the growing acyl-ACP. The resulting ketoester is reduced by a NADPH-dependent β -ketoacyl-ACP reductase (*fabG*), and a water molecule is then removed by a β -hydroxyacyl-ACP dehydrase (*fabA* or *fabZ*). The last step is catalyzed by enoyl-ACP reductase (*fabI*) to form a saturated acyl-ACP, which in turn serves as the substrate for another condensation reaction.

3.4.1. The β -ketoacyl-ACP synthases

Three *E. coli* enzymes catalyze the Claisen condensation that is the β -ketoacyl-ACP synthase reaction. These enzymes are the products of the *fabB*, *fabF*, and *fabH* genes. β -Ketoacyl-ACP synthase I, or FabB, is composed of two identical 42.6-kDa subunits, and has both malonyl-ACP and fatty acyl-ACP binding sites. In the condensation reaction, the acyl group is covalently linked to the active site cysteine. The acyl-enzyme undergoes condensation with malonyl-ACP to form β -ketoacyl-ACP, CO_2 , and free enzyme.

Overproduction of FabB has two effects: an increased amount of *cis*-vaccenic acid in phospholipids; and resistance to the antibiotics thiolactomycin and cerulenin (Section 12.2). β -Ketoacyl-ACP synthase II (FabF) is very similar to FabB (38% identical at the amino acid level). Like FabB, FabF has a dimeric structure of 43-kDa subunits, and is inhibited by cerulenin and thiolactomycin. FabF is not essential to growth in *E. coli*, but is essential for the regulation of fatty acid composition in response to temperature fluctuations. Mutants lacking FabF activity, unlike wild-type *E. coli*, do not produce increased amounts of the long, unsaturated fatty acid *cis*-vaccenate at lower temperatures. β -Ketoacyl-ACP synthase III (FabH) is a dimeric protein of identical 33.5-kDa subunits first detected as a condensation activity resistant to cerulenin. The FabH reaction is characterized by the preference for a CoA-linked primer, rather than acyl-ACP.

The crystal structures of all three synthases have been determined, and all share a common thiolase fold. Structures of FabB and FabF are virtually identical. Both utilize a conserved catalytic triad of Cys–His–His. FabH has a much more closed active site, reflecting its use of less hydrophobic, shorter chain, substrates than FabB and FabF. FabH also differs in that it contains a catalytic triad composed of Cys–His–Asn. This difference renders FabH more resistant to thiolactomycin and cerulenin than FabB or FabF. The exact reason for this discrepancy is not clear at this time, since the key nitrogens of the His or Asn residues of the respective synthases occupy equivalent space in the respective structures.

3.4.2. β -Ketoacyl-ACP reductase

The β -ketoacyl-ACP reductase gene (*fabG*) is located within the *fab* gene cluster between the *fabD* and *acpP* genes and is cotranscribed with *acpP*. Insertional mutants that prevent *fabG* transcription while allowing ACP to be produced were generated in Cronan's laboratory and suggest that *fabG*-encoded reductase activity is essential in *E. coli*. The *fabG*-encoded NADPH-specific β -ketoacyl-ACP reductase is a homotetrameric protein of 25.6-kDa monomers. The protein functions with all chain lengths in vitro and exhibits cooperative binding of NADPH. A dramatic conformational change occurs on cofactor binding, as evidenced by the crystal structures of the free and NADPH-bound protein.

3.4.3. β -Hydroxyacyl-ACP dehydrase

E. coli possesses two β -hydroxyacyl-ACP dehydrases (more properly termed dehydratases). One is encoded by *fabZ*, and is active on all chain lengths of saturated and unsaturated intermediates. This enzyme is distinct from the dual-function β -hydroxydecanoyl-ACP dehydrase/isomerase (encoded by *fabA*) first described by Bloch and coworkers. The FabA enzyme dehydrates saturated, but not unsaturated, fatty acid intermediates and catalyzes a key isomerization reaction at the point where the biosynthesis of unsaturated fatty acids diverges from saturated fatty acids (Fig. 4). Both enzymes share weak overall homology (28% identity and 50% similarity at the amino acid level). The mono-functional FabZ protein (17 kDa) is somewhat smaller than the *fabA*-encoded bifunctional enzyme (19 kDa). The FabA dehydrase/isomerase has been crystallized, and its structure solved. The active site His is located in a long tunnel which acts a molecular ruler to ensure that only 10-carbon intermediates are isomerized.

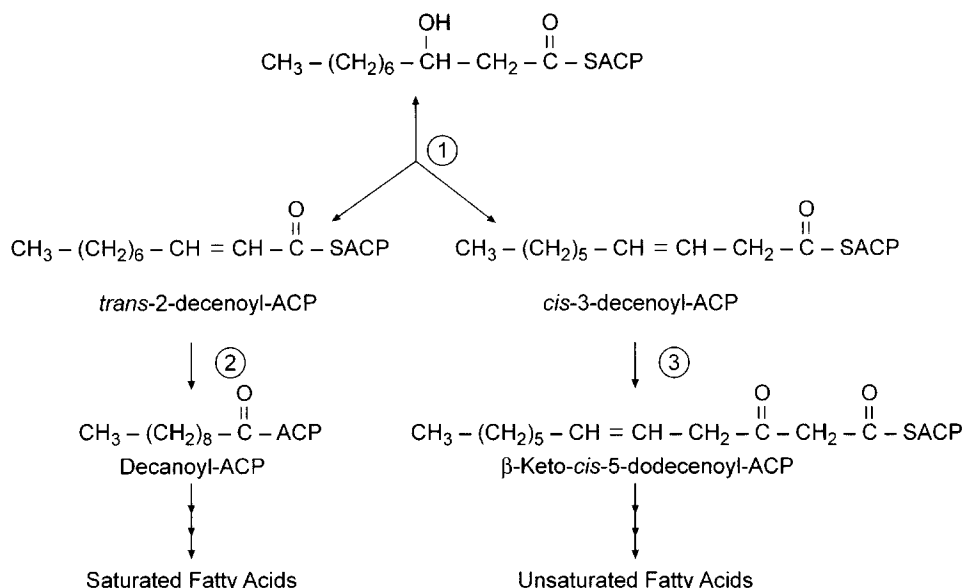


Fig. 4. Branch-point in unsaturated fatty acid synthesis. (1) FabA catalyzes the interconversion of β -hydroxydecanoyl-ACP, *trans*-2-decenoyl-ACP and *cis*-3-decenoyl-ACP. (2) *trans*-2-Decenoyl-ACP is a substrate for enoyl-ACP reductase (FabI), while (3) *cis*-3-decenoyl-ACP is elongated by β -ketoacyl-ACP synthase I (FabB). Competition between FabI and FabB is partly responsible for the ratio of unsaturated fatty acids made.

3.4.4. Enoyl-ACP reductase

The final step in each round of fatty acyl elongation in *E. coli* is the NADH-dependent reduction of the *trans* double bond, catalyzed by the homotetrameric (subunit mass of 29 kDa) NADH-dependent enoyl-ACP reductase I (encoded by *fabI*). The FabI amino acid sequence is similar (34% identical) to the product of a gene (called *inhA*) from *Mycobacteria*. InhA is involved in mycolic acid biosynthesis. The synthesis of these unusual 70–80 carbon mycobacterial acids requires a pathway composed of enzymes essentially identical to those of fatty acid synthesis. Missense mutations within the *inhA* gene result in resistance to the anti-tuberculosis drugs, isoniazid and ethionamide. The crystal structures of FabI and InhA have been solved, and are virtually superimposable for most of the protein. FabI has a flexible substrate binding loop that becomes ordered on binding of the specific inhibitors, diazaborine or triclosan (and presumably substrate), while InhA has two short helices in this region that move 4 Å on drug or substrate binding. Novel enoyl-ACP reductase isoforms II (FabK) and III (FabL) have been discovered recently in Gram-positive bacteria by bioinformatics (Section 11).

3.5. Synthesis of unsaturated fatty acids

The pathway described above suffices to produce the straight chain saturated fatty acids found in the membrane phospholipids, mainly palmitic acid (16:0) in *E. coli*.

Gram-negative bacteria also contain unsaturated fatty acids, and the ratio of saturated:unsaturated fatty acids in the membrane phospholipids is a key determinant in membrane fluidity, and changes according to temperature [11]. The *fabA*-encoded dehydrase/isomerase (Section 3.4.3) catalyzes a vital reaction at the branch point of the two pathways: the isomerization of *trans*-2-decenoyl-ACP to *cis*- β -decenoyl-ACP (Fig. 4). The *cis*-3 compound is not a substrate for the enoyl-ACP reductase, but instead is rapidly condensed in a reaction requiring FabB, but not FabF. Strains with mutations in either *fabA* or *fabB* require supplementation with unsaturated fatty acids for growth, showing the specific requirement for both enzymes. As their names suggest, these were the first two mutants in fatty acid biosynthesis identified by classical genetic techniques [12,13].

Both FabB and FabF are capable of participating in saturated and unsaturated fatty acid synthesis and the enzymes have been shown, *in vitro*, to function similarly with all long-chain acyl-ACPs except palmitoleoyl-ACP. Palmitoleoyl-ACP is an excellent substrate for FabF, but not for FabB. *In vivo*, the reactivity of FabF towards this substrate increases at lower temperatures, leading to increased amounts of the more fluid *cis*-vaccenate (18:1 Δ 11) in membrane phospholipids.

3.6. Afterword: dissociable or dissociated enzymes?

Historically, scientists in the field have debated whether the enzymes of bacterial fatty acid synthase system are ‘dissociable’ or ‘dissociated’. This seemingly minor semantic distinction has larger ramifications for the *in vivo* physiology of the cell. The implication of the use of the word *dissociable* is that the enzymes form a complex *in vivo*, and only became separated (or dissociated) on cell disruption; whereas in a *dissociated* system, the enzymes do not form a complex *in vivo*. The concept of a large complex mimicking the multifunctional type I enzyme would support the notion of substrate channeling between the active sites, increasing the catalytic efficiency of the pathway as a whole. However, there are no data to support the existence of either a large complex or substrate channeling. A minimal fatty acid synthetic unit must consist of at least six separate activities, encoded by the *acpP*, *fabD*, *fabH*, *fabG*, *fabI*, *fabZ* and *fabB* genes. *In vitro*, these enzymes appear as monomers (ACP, FadD), dimers (FabH, FabZ and FabB) or tetramers (FabG and FabI). Thus, a complex of these proteins *in vivo* would possess 16 subunits (assuming only one of each individual complex was present) with a combined mass of over 440 kDa. With our current knowledge of the 3-dimensional structures of these proteins (only FabZ has not been solved to date), it is hard to envision how the pieces of such a puzzle would fit together. Many of the enzymes have active sites located in narrow tunnels, leaving no opportunity for the prosthetic group to swing between them. The acyl-ACP intermediate must instead completely dissociate from the enzyme to interact with the next enzyme in the pathway. Studies with the yeast two-hybrid system have also failed to detect interactions between different enzymes. Thus, type II fatty acids synthases should be considered dissociated and not dissociable.

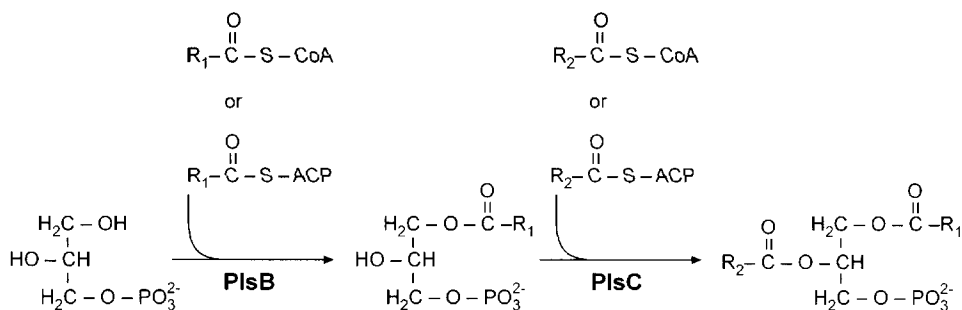


Fig. 5. Transfer of fatty acyl groups to the membrane. A (primarily saturated) fatty acid is transferred from acyl-ACP or acyl-CoA to the 1 position of *sn*-glycerol-3-phosphate by glycerol phosphate acyltransferase (PlsB). A second fatty acid is transferred to the 2-position by the 1-acylglycerol phosphate acyltransferase (PlsC).

4. Transfer to the membrane

Fatty acid biosynthesis in *E. coli* normally ends when the acyl chain is 16 or 18 carbons in length. These acyl-ACPs are now substrates for the acyltransferases that will transfer the fatty acyl chain into the membrane phospholipids (Fig. 5). Alternatively, *E. coli* can incorporate exogenous fatty acids, following esterification to CoA. The first enzyme (the *plsB* gene product) transfers fatty acids from either the soluble acyl-ACP or acyl-CoA to the 1-position of glycerol phosphate. The product of the reaction, 1-acylglycerol phosphate, partitions into the membrane. The PlsB protein is an integral inner membrane protein of 91 kDa, and has a preference for saturated fatty acids. The second acyltransferase (the *plsC* gene product), a membrane protein of 27 kDa, esterifies the 2-position of the glycerol backbone and prefers unsaturated acyl chains. Thus, bacterial phospholipids have an asymmetric distribution of fatty acids between the 1- and 2-positions of the glycerol phosphate backbone. The glycerol phosphate acyltransferase system influences the chain length of the fatty acids incorporated into the phospholipids by competition with the elongation condensing enzymes for acyl-ACP, and the rate of fatty acid biosynthesis via modulation of its activity by ppGpp (Section 10.5).

The isolation of *E. coli* mutants with defective acyltransferase activity (*plsB*) by Bell's laboratory heralded a major advance in the study of the acyltransferases. These mutants were glycerol phosphate auxotrophs and exhibited an increased Michaelis constant for glycerol phosphate in *in vitro* acyltransferase assays. The increased K_m was subsequently shown to arise from a single missense mutation in the open reading frame. Therefore, *plsB* mutants require an artificially high intracellular concentration of glycerol phosphate for activity. Complementation of these mutants facilitated the cloning of the glycerol phosphate acyltransferase. Plasmids that suppressed the glycerol phosphate requirement of *plsB* strains overexpressed glycerol phosphate acyltransferase activity 10-fold. The protein possesses a catalytic His-Asp dyad common to all glycerolipid acyltransferases. The glycerol phosphate acyltransferase is specifically activated by acidic phospholipids, phosphatidylglycerol and cardiolipin, as shown by

micelle assays containing detergents and phospholipid, and is active as a monomer. Glycerol phosphate acyltransferase also exhibits negative cooperativity with respect to glycerol phosphate binding, a property that may account in part for the finding that dramatic increases in the intracellular glycerol phosphate concentration do not increase the amount of phospholipid in *E. coli*.

The interpretation of the *plsB* mutants is complicated by the finding that the *plsB* growth phenotype depends on mutations in two unlinked genes. One mutation is in the *plsB* gene discussed above, and the second is in a gene called *plsX*. Both mutations are required for a strain to exhibit a requirement for glycerol phosphate since strains harboring either the *plsB* or *plsX* lesion do not have a defective growth phenotype. The *plsX* gene is located in the *fab* cluster next to *fabH* and predicted to encode a protein of 37.1 kDa. Despite extensive study, no enzymatic activity has been described for the PlsX protein. It is possible that the native PlsB protein exists in a complex with PlsX. Analysis of the genomic databases reveals many bacteria contain *plsX* homologues, but provides no further clues as to its biochemical function.

The next step in phospholipid biosynthesis is catalyzed by 1-acylglycerol phosphate acyltransferase (the *plsC* gene product) which acylates the product of the PlsB step to form phosphatidic acid (Fig. 5). Phosphatidic acid comprises only about 0.1% of the total phospholipid in *E. coli* and turns over rapidly, a property consistent with its role as an intermediate in phospholipid synthesis. The 1-acyl-glycerol phosphate acyltransferase is thought to transfer unsaturated fatty acids selectively to the 2-position.

Certain bacteria, such as Clostridia, do not possess a *plsB* homologue. Instead, the *plsD* gene was isolated from a *Clostridium difficile* genomic library by functional complementation of the *plsB* phenotype of *E. coli*. The protein encoded by *plsD* is a glycerol phosphate acyltransferase, but shares no homology with the *plsB*-encoded enzyme, except for a predicted His–Asp dyad active site. Overall, it appears more similar to the *plsC*-encoded 1-acylglycerol phosphate acyltransferase of *E. coli* in both amino acid composition (27% identity) and size (26.5 kDa). However, there are many bacterial species that lack both *plsB* and *plsD* genes, thus the enzyme(s) that acylates glycerol phosphate in these organisms remains unknown.

5. Phospholipid biosynthesis

E. coli possesses only three major phospholipid species in its membranes, making it one of the simplest organisms to study with regard to phospholipid biosynthesis. Phosphatidylethanolamine (PE) comprises the bulk of the phospholipids (75%), with phosphatidylglycerol (PG) and cardiolipin (CL) forming the remainder (15–20% and 5–10%, respectively). The scheme for the synthesis of membrane phospholipids follows the classic Kennedy pathway (Fig. 6).

5.1. Phosphatidate cytidyltransferase

The key activated intermediate in bacterial phospholipid synthesis, CDP-diacylglycerol, comprises only 0.05% of the total phospholipid pool. The 27.6-kDa enzyme phosphati-

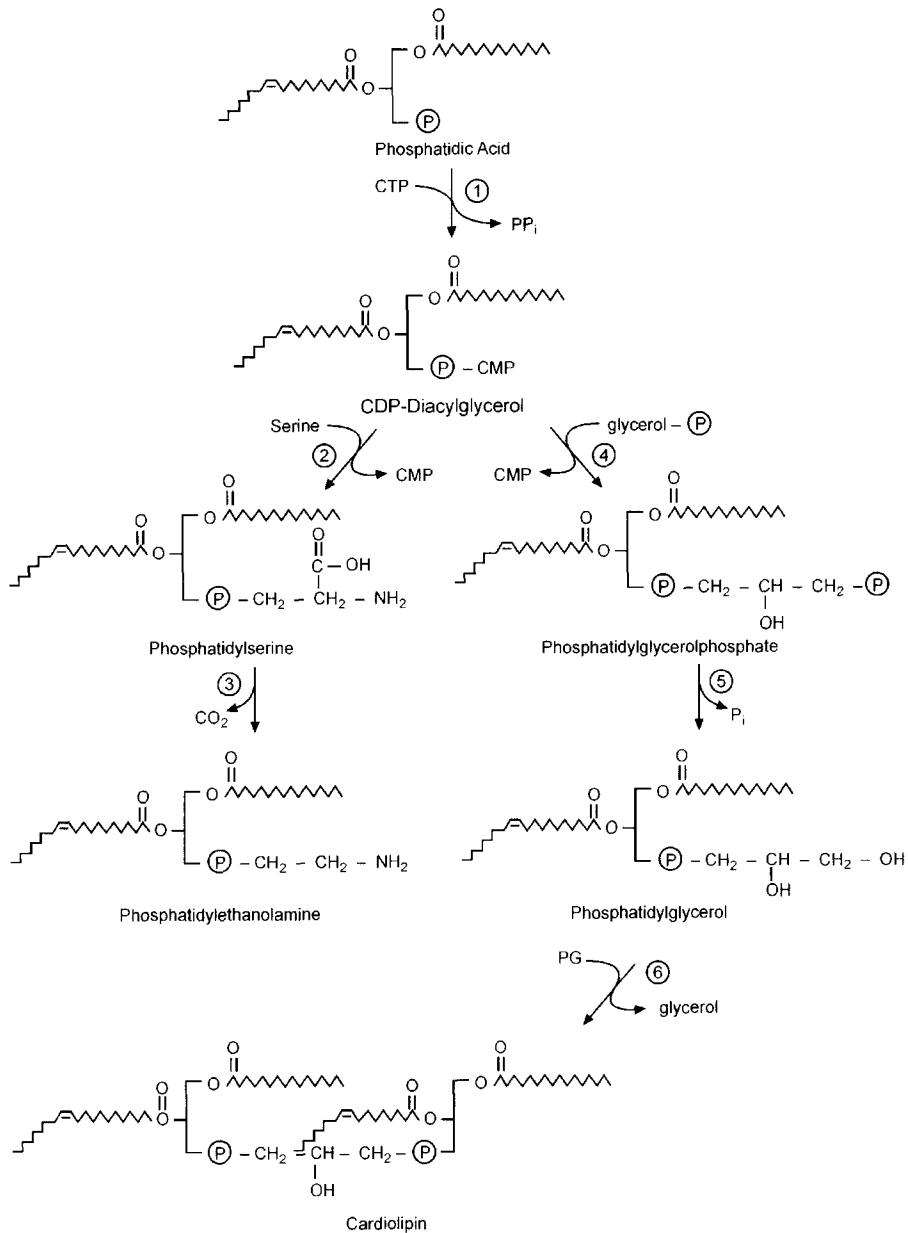


Fig. 6. Synthesis of phospholipid polar headgroups. The three major phospholipid species in *E. coli* are synthesized by a total of six different enzymatic activities: (1) phosphatidate cytidylyltransferase (Cds); (2) phosphatidylserine synthase (Pss); (3) phosphatidylserine decarboxylase (Psd); (4) phosphatidylglycerolphosphate synthase (PgsA); (5) phosphatidylglycerolphosphate phosphatase (PgpA or PgpB); and (6) cardiolipin synthase (Cls).

date cytidyltransferase (or CDP-diacylglycerol synthase) catalyzes the conversion of phosphatidic acid to a mixture of CDP-diacylglycerol and dCDP-diacylglycerol. Strains of *E. coli* with mutations in the *cds* gene retain 5% of the normal levels of CDP-diacylglycerol synthase and grow normally under standard laboratory conditions, although are hypersensitive to erythromycin and elevated pH. Thus, CDP-diacylglycerol synthase is present in large excess of the minimum amount of enzyme required to sustain phospholipid synthesis. These mutants accumulate substantial amounts of phosphatidic acid (up to 5% of the total phospholipid). Null mutations in *cds* have not been reported, and would presumably be non-viable due to the complete lack of phospholipid synthesis from this point.

CDP-diacylglycerol stands at the branch point between PE synthesis and PG and CL synthesis (Fig. 6). It has been hypothesized that the presence of both ribo and deoxyribo forms of the liponucleotide could play a role in determining the relative amount of intermediate entering these two arms of the pathway. For this to be true, the respective synthases that utilize this compound would have to be selective toward either dCDP- or CDP-diacylglycerol. In vivo the ratio of dCDP- to CDP-diacylglycerol is 0.88. A change in this ratio to 3.1 has no effect on the relative rates of PE and PG synthesis in vivo, arguing against selectivity of the subsequent enzymes for one form of the other. Further, both ribo- and deoxyribo-liponucleotides are substrates for PS synthase in vitro, and thus, the significance, if any, of the two forms of liponucleotide remains to be determined.

5.2. Phosphatidylethanolamine production

5.2.1. Phosphatidylserine synthase

The first step in the synthesis of PE is the condensation of CDP-diacylglycerol with serine catalyzed by PS synthase to form PS. During cell disruption, the 58-kDa PS synthase appears associated with ribosomes, but reattaches to the membrane vesicles once substrate is added. PS is a minor membrane constituent of *E. coli* since it is rapidly converted to PE by PS decarboxylase. Mutants in the *pss* gene encoding PS synthase are viable only when supplemented with divalent metal ions. PE is capable of forming the hexagonal (non-bilayer) H_{II} lipid phase, and Dowhan has demonstrated that the divalent cations interact with CL to replace the function of PE in the formation of an H_{II} phase (Chapter 1). The cells lack PS synthase activity, and thus contain no PS or PE in their membranes. There are also perturbations in the function of permeases, electron transport, motility and chemotaxis.

5.2.2. Phosphatidylserine decarboxylase

PS is decarboxylated by PS decarboxylase to yield the zwitterionic PE. This inner membrane enzyme has a subunit molecular mass of 36 kDa. PS decarboxylase has a pyruvate prosthetic group that participates in the reaction by forming a Schiff base with PS. Overproduction of the enzyme 30–50-fold by plasmid-borne copies of the *psd* gene has no effect on membrane phospholipid composition indicating that the level of this enzyme does not regulate the amount of PE in the membrane. The majority of the PE is found in the periplasmic leaflet of the inner membrane, and there is a rapid flipping from the inner to outer leaflet by the MsbA lipid flippase (Section 7).

Mutants with a temperature-sensitive decarboxylase accumulate PS at the non-permissive temperature. The mutants continue to grow for several hours after the shift to the non-permissive temperature, despite the reduced levels of PE and the concomitant increase in PS. Complete inactivation of *psd* by insertional mutagenesis has the same divalent cation-requiring phenotype as the *pss* mutants described above. The requirement for CL is consistent with the inability to introduce a null *cls* allele into *pss* strains. Thus, PE is essential for the polymorphic regulation of lipid structure. Evidence from the Dowhan laboratory demonstrates that PE is a molecular chaperone that is essential for the proper folding of integral membrane proteins (Chapter 1). All of the physiological processes dependent on the formation of local regions of non-bilayer structure or that specifically require PE remain to be elucidated, but the process of cell division, the formation of contacts between inner and outer membranes, and the translocation of molecules across the membrane are viable candidates.

5.3. Phosphatidylglycerol synthesis

5.3.1. Phosphatidylglycerolphosphate synthase

CDP-diacylglycerol is condensed with glycerol phosphate to form phosphatidylglycerolphosphate (PGP), an intermediate in the production of the acidic phospholipids PG and CL (Fig. 6). The reaction is analogous to the synthesis of PS, with the product CMP being released. Mutants (*pgsA*) defective in PGP synthesis contain less than 5% of normal PGP synthase activity in vitro, however, there is no growth phenotype associated with these mutants. The PgsA protein is predicted to be a 20.7-kDa integral membrane protein. It has long been thought that PGP synthase is essential, and that cells cannot survive without acidic phospholipids. There are many important cellular functions that are affected by reduced PG and/or CL content of the membrane. PG is required for protein translocation across the membrane, and acidic phospholipids are required for channel activity of bacterial colicins and the interaction of antibiotics with the membrane. Cell division proteins such as FtsY also apparently require acidic phospholipids for activity, as does the DnaA protein involved in chromosome segregation. However, Matsumoto has recently inactivated the *pgsA* gene with a kanamycin cassette. This *pgsA::kan* strain has no detectable PG or CL, is not viable above 40°C, and contains increased concentrations of PA, which may at least partially compensate for the absence of PG and CL.

5.3.2. PGP phosphatases

The second step in the synthesis of PG is the dephosphorylation of PGP (Fig. 6). Two independent genes have been identified, *pgpA* and *pgpB*, that encode PGP phosphatases based on an in vitro assay. Both proteins are small (19.4 and 29 kDa, respectively) but share no sequence homology. In vitro, the *pgpA*-encoded phosphatase specifically hydrolyzes PGP, whereas the PgpB phosphatase also hydrolyzes phosphatidic acid. However, disruption of both of these genes in a single strain did not impair PG synthesis, although the respective phosphatase activities were reduced. Thus, neither of these phosphatases is required for PG synthesis suggesting that another phosphatase capable of operating in the PG biosynthetic pathway remains to be discovered.

5.4. Cardiolipin biosynthesis

Unlike in mammalian mitochondria, where CL is synthesized by the reaction of CDP-diacylglycerol with PG (Chapter 8), CL is produced in bacteria by the condensation of two PG molecules (Fig. 6). CL accumulates as the cells enter the stationary phase of growth, and is required for prolonged survival of the bacteria. CL synthase is post-translationally processed from a 55-kDa precursor to a 45–46-kDa form. Mutants deficient in CL synthase (*cls*) possess very low levels of CL and lose viability in stationary phase. The mutants also grow at a slower rate and to a lower density than the corresponding wild-type cells. Low, residual concentrations of CL are present in the *cls* null mutants, hampering efforts to isolate the role of this lipid in cell physiology. A second gene has recently been described that can catalyze the formation of CL in vitro, but does not appear to do so in vivo. Amplification of CL synthase leads to the overproduction of CL, a decrease in membrane potential, and loss of viability. Therefore, *E. coli* can tolerate changes in the overall CL content but the elimination or significant overproduction of CL leads to significant physiological imbalance.

5.5. Cyclopropane fatty acids

Fatty acids attached to membrane phospholipids can be post-synthetically converted to their cyclopropane derivatives during the stationary phase of bacterial growth. Their biosynthesis and function have been elucidated by the Cronan laboratory. *E. coli* mutants that completely lack cyclopropane fatty acid synthase activity (owing to null mutations in the *cfa* gene) grow and survive normally under virtually all conditions, except that *cfa* mutant strains are more sensitive to freeze–thaw treatment and acid shock than are isogenic *cfa*⁺ strains. Thus, the stable cyclopropane derivative protects the reactive double bond from adverse reactions during stationary phase. Cyclopropanation involves a significant energy commitment by the cell: the reaction uses *S*-adenosylmethionine, which requires three molecules of ATP are required for regeneration. It is not known how the soluble enzyme and substrate gain access to the phospholipids of the inner and outer membranes.

The 44-kDa cyclopropane synthase protein is metabolically unstable, but protein levels peak sharply due to increased *cfa* transcription as cultures enter the stationary phase. Cfa levels drop in late stationary phase cultures as the enzyme is destroyed by proteolysis, probably by a protease of the heat shock response. The *cfa* gene possesses two promoters of approximately equal strengths, with the more distal promoter functioning throughout the growth cycle. The proximal promoter requires the specialized sigma factor, σ^S (encoded by *rpoS*), for transcription, and is thus active only as cultures enter stationary phase. Indeed, the cyclopropane fatty acid content of *rpoS* strains is low and transcription from the proximal promoter is absent in these strains. As cells remain in stationary phase and phospholipid biosynthesis ceases, the low levels of Cfa that do persist no longer encounters an expanding substrate pool. Thus, an increasing amount of the fatty acyl chains are converted to their cyclopropane derivatives over time. The instability of Cfa results in little carry-over of synthetic capacity when exponential growth resumes, and the existing cyclopropane fatty acids are quickly diluted by de novo phospholipid synthesis.

6. Lipid A biosynthesis

Lipopolysaccharides (LPS) form the majority of the outermost leaflet of the membrane in most Gram-negative bacteria (Fig. 4 in Chapter 1), and display a tremendous amount of structural variability [14]. LPS is essential to the growth of Gram-negative bacteria, and provides an effective hydrophobic barrier to toxic compounds. LPS are comprised of three components: the O-antigen, a core polysaccharide and lipid A. The O-antigen is a polysaccharide that extends from the cell surface. O-antigens are constructed from 10 to 30 repeats of specific β -6 sugar oligosaccharide units, and each is essentially unique to a given serotype of bacteria. The O-antigen is linked to the core polysaccharide region, which is common to groups of bacteria. The membrane associated portion of LPS is lipid A. The core polysaccharide is attached to lipid A by a 2-keto- β -deoxyoctonate (KDO) disaccharide. Lipid A anchors the LPS to the outer membrane and functions as an endotoxin and a mitogen during bacterial infections. The lipid A is synthesized and ligated to the oligosaccharide core on the cytoplasmic face of the inner membrane, while the O-antigen is added in the periplasm. O-antigen is not essential for the viability of *E. coli*, and is in fact missing from *E. coli* K12, making it safe for laboratory use. Details of the synthesis of the O-antigen and core region are outside the scope of this discussion and can be found in a review by Raetz [14].

The pathway of lipid A synthesis has been determined mainly in the laboratory of Raetz (Fig. 7). The first step in lipid A synthesis is the reversible transfer of the β -hydroxymyristoyl group from ACP to UDP-*N*-acetylglucosamine (UDP-GlcNAc) by the UDP-GlcNAc acyltransferase (*lpxA*). Competition between LpxA and the FabZ dehydratase of fatty acid synthesis help determine the rate of lipid A synthesis. The second step in the pathway is the deacetylation of UDP-3-Acyl-GlcNAc by the zinc-dependent 34-kDa UDP- β -*O*-acyl-GlcNAc deacetylase (*lpxC*). The *lpxC* gene was first described as *envA*, an essential gene involved in envelope production. Mutations in *lpxC* cause a plethora of effects, including increased sensitivity to antibiotics, increased dye permeability and defects in cell division. Null mutations in *lpxC* are lethal. The LpxA-catalyzed acyltransfer step is thermodynamically unfavorable, thus the irreversible deacetylase reaction is the first committed step in the pathway. The essential nature of LpxC has driven the development of a novel group of antimicrobial compounds active against this step (Section 12). These compounds are effective against a wide range of Gram-negative bacteria *in vivo*.

The third step is a second β -hydroxymyristoyl-ACP acyltransferase, catalyzed by UDP- β -*O*-[β -hydroxymyristoyl]GlcN acyltransferase (*lpxD*). Like the LpxA acyltransferase, the 36-kDa LpxD possesses repeating hexapeptide units and will presumably have a fold and trimeric structure similar to LpxA. The product of the LpxD reaction is UDP-2,3-diacyl-GlcN. UMP is removed from this compound to form 2,3-diacyl-GlcN-1-phosphate (lipid X), but the enzyme responsible for this reaction is not known. There is an approximately 10-fold excess of lipid X over UDP-2,3-diacyl-GlcN in wild-type cells. Mutations in *lpxB*, cause a 500-fold increase in the amount of lipid X in the membrane, although LpxB is not a pyrophosphatase. LpxB, a dimer of 42-kDa monomers, catalyzes the condensation of lipid X with UDP-2,3-diacyl-GlcN to form the lipid A disaccharide-1-phosphate. *lpxB* is present in a complex cluster with *lpxA*, *lpxD*

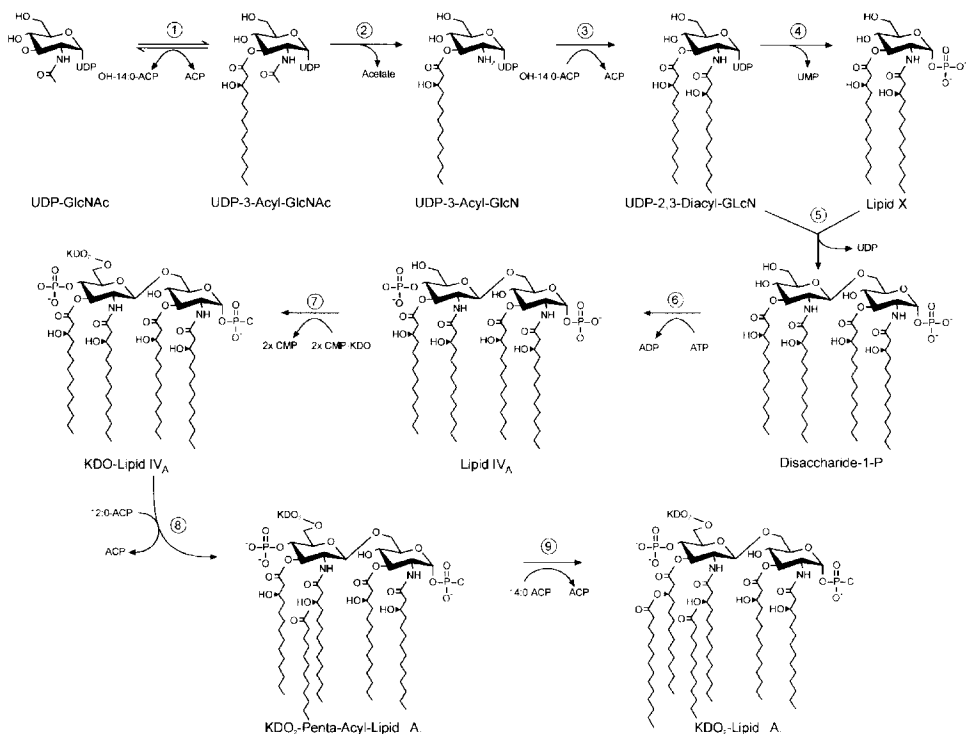


Fig. 7. Biosynthesis of endotoxin in *E. coli*. The first step (1) in the pathway is catalyzed by UDP-N-acetylglucosamine (UDP-GlcNAc) acyltransferase (LpxA). (2) The committed step is catalyzed by the LpxC deacetylase, followed by (3) a second acyltransferase (LpxD). (4) Lipid X is generated by the removal of UMP from UDP-2,3-diacyl-GlcNAc by an unknown enzyme. (5) Lipid X and UDP-2,3-diacyl-GlcNAc are then condensed together by LpxD to form lipid IV_A. (6) A 4'-kinase phosphorylates the disaccharide to produce lipid IV_A. (7) Two consecutive additions of KDO by KdtA, and two O-acylations by (8) HtrB and (9) MsbB yield KDO₂-lipid A. Subsequent addition of core sugars and O-antigen chains (not shown) yield the mature lipopolysaccharide.

and *fabZ*, and other genes encoding enzymes of phospholipid synthesis, outer membrane proteins, DNA synthesis and fatty acid synthesis.

Strains mutated in *lpxK* accumulate lipid A disaccharide-1-P, which led to the cloning and characterization of lipid A disaccharide kinase, a Mg^{2+} -dependent 4'-kinase activity stimulated by CL, that makes the key intermediate, lipid IV_A. The presence of the 4'-phosphate is essential for the recognition of lipid A by mammalian cells during endotoxin stimulation. Two subsequent additions of KDO are catalyzed by the KDO transferase, the 47-kDa KdtA protein. A single protein catalyzes both additions, since overexpression of *kdtA* causes a large increase in both transfers. KDO transferase activity is essential for growth, and conditional mutants accumulate massive amounts of lipid IV_A prior to cessation of growth at the non-permissive temperature.

In the final steps of lipid A synthesis, two fatty acids are transferred to the hydroxyl groups of the β -hydroxymyristate on the distal unit. The first of these is usually laurate (12 carbons) added from lauroyl-ACP by the enzyme HtrB, and then MsbB adds

myristate from myristoyl-ACP. During cold shock (on shifting *E. coli* from 30 to 12°C), palmitoleate appears in the lipid A at the expense of laurate. A novel gene, *lpxP*, is induced for this reaction, and null mutants in *HtrB* are not defective in this adaptation. Double null *htrB msbB* mutants can also be generated. Thus, the extensive acylation of the lipid A is not absolutely required for its proper insertion into the outer membrane or the formation of a hydrophobic barrier.

7. Phospholipid flippase

All of the enzymes of phospholipid and lipid A biosynthesis are either cytosolic or located on the inner aspect of the inner membrane. How then do the lipids get to the outer face of the inner membrane, or to the outer membrane? Recent work from Raetz's laboratory has shown that the product of the *msbA* gene is a lipid 'flippase' required for the trafficking of lipids across the inner membrane. A temperature-sensitive mutant, *msbA*(Ts), with a A270T substitution in the MsbA protein, shows a rapid and dramatic reduction in the export of all major lipid classes, including PE and core-lipid A, to the outer membrane. Lipid export is inhibited by >90% after 30 min at the non-permissive temperature, while protein transport is not affected. *E. coli* harboring this mutation possess duplicated inner membranes at the elevated temperature. Null mutations in *msbA* are lethal, and this is the only bacterial transporter that has been shown to be essential. The MsbA protein is a member of the ATP-binding cassette (ABC) family of transporters, and has most similarity to mammalian P-glycoprotein multiple-drug resistance ABC transporters (>30% identity). MsbA exists as a homodimer of 64.6-kDa monomers, each of which has a single membrane spanning region (comprised of six transmembrane helices) and a nucleotide binding domain. This arrangement is distinct from the mammalian multiple-drug resistance pumps, which have two membrane spanning regions and two nucleotide binding domains fused into a single polypeptide. Thus, MsbA is a 'half-transporter', and is actually homologous to both amino- and carboxy-terminal halves of the mammalian multiple-drug resistance protein.

The crystal structure of the integral membrane MsbA has been determined by Chang and Roth to 4.5 Å [15]. The protein traverses the entire inner membrane and is cone-shaped, with an opening of about 25 Å on the cytoplasmic side, leading into a chamber of sufficient size for core-lipid A. The nucleotide binding domains are on the cytoplasmic face, and share no contacts. Dimer contacts are made in the half of the protein in the outer membrane. A model for transport has been suggested in which the lipid A or phospholipid enters the chamber through the cytoplasmic membrane, the chamber then closes allowing for flip-flop of the molecule, and then the chamber re-opens, and the lipid is ejected. Presumably, the lipid is released into the outer leaflet of the inner membrane, since MsbA does not extend across the periplasm. Thus, lipids that are required in the outer membrane must also traverse the periplasm. LPS molecules must also be flipped across the outer membrane to be displayed on the surface of the bacteria. Currently, nothing is known about these processes.

8. Degradation of fatty acids and phospholipids

8.1. β -Oxidation of fatty acids

8.1.1. Transport of fatty acids across the membrane

Exogenous long chain fatty acids are utilized by *E. coli* in two ways. Firstly, they can be incorporated into the membrane phospholipids by the acyltransferase system (PlsB and PlsC; Section 4). Secondly, they can be used as the sole carbon source for growth, and are in fact an important source of energy for *E. coli* in their normal habitat, the intestine [16]. The CoA thioester of the fatty acid is the substrate for both of these pathways. Fatty acids greater than 10 carbons in length require the *fadL* gene product to be taken up from the growth medium in sufficient quantities to support growth. FadL is a 46-kDa outer membrane protein produced following the cleavage of a 28-residue signal peptide from the propeptide. Fatty acid uptake is closely coupled to acyl-CoA formation, since very low levels of free fatty acid are found in the cells. The acyl-CoA synthetase, encoded by *fadD*, is a homodimer of 62-kDa subunits, and associates with the cytoplasmic leaflet of the inner membrane. Strains mutated in *fadD* cannot produce acyl-CoA and thus cannot grow on exogenous fatty acids, nor incorporate them into their membrane phospholipids. The esterification of the free fatty acid to CoA traps the fatty acid inside the cell, driving its transport across the inner membrane, and the net accumulation of fatty acid from the medium. Medium chain fatty acids do not require FadL to enter the cells, and may traverse the outer membrane by passive diffusion.

8.1.2. Degradation of fatty acids

Degradation of fatty acids proceeds via an inducible set of enzymes that catalyze the pathway of β -oxidation [16]. β -Oxidation occurs via repeated cycles of reactions that are essentially the reverse of the reactions of fatty acid synthesis (Fig. 8). However, three major differences distinguish the two pathways. Firstly, β -oxidation utilizes acyl-CoA thioesters, and not acyl-ACPs. Secondly, the β -hydroxy intermediates have the opposite stereochemistry (L in β -oxidation and D in synthesis). Finally, the enzymes of β -oxidation share no homology with those of synthesis.

The first step in the pathway is the dehydrogenation of acyl-CoA by the enzyme acyl-CoA dehydrogenase. While other organisms have several dehydrogenases with different chain length specificities (i.e. for short, medium or long acyl chains), it has been reported that *E. coli* has one enzyme active on all chain lengths. The dehydrogenase has been linked to two mutations in the 5 min region of the *E. coli* chromosome, *fadF* and *fadG*. However, the genome sequence suggests that a single gene in this region, *yahF*, encodes for a 92-kDa dehydrogenase. Thus, *fadF* and *fadG* probably represent different mutations in the same YahF dehydrogenase, and do not encode two distinct proteins. The acyl-CoA dehydrogenase is a flavoprotein, and is linked to a electron transferring flavoprotein (*fadE*). Further confusion exists in the literature as many acyl-CoA dehydrogenase (YafH) homologues are annotated as FadE homologues. Mutant strains of *E. coli* blocked in β -oxidation with *fadE* or *fadF* (*yahF*) mutations can accumulate acyl-CoA species, but cannot degrade them.

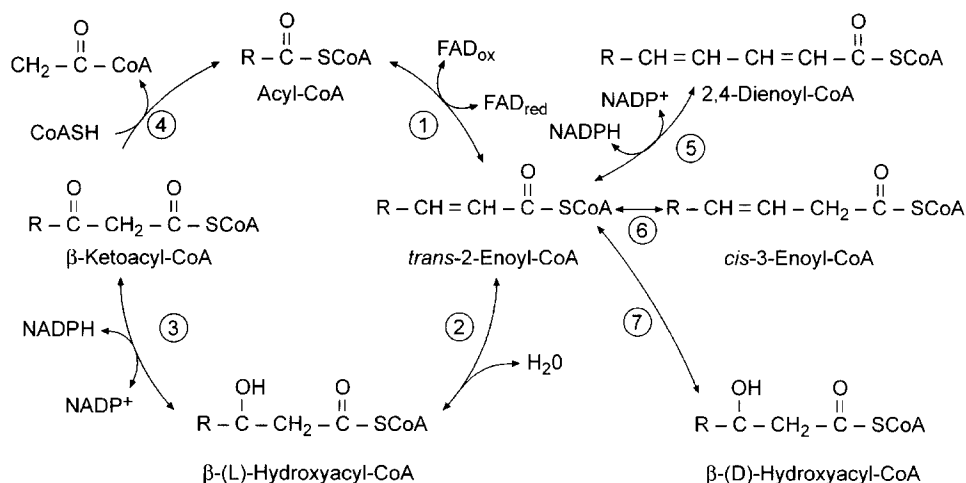


Fig. 8. β -Oxidation of fatty acids in *E. coli*. Long chain fatty acids are transported into the cell by FadL and converted to their CoA thioesters by FadD (not shown). The acyl-CoAs are substrates for the (1) acyl-CoA dehydrogenase (YafH) to form a *trans*-2-enoyl-CoA. The double bond is reduced by (2) *trans*-2-enoyl-hydratase (crotonase) activity of FadB. The β -hydroxyacyl-CoA is then a substrate for the NADP⁺-dependent dehydrogenase activity of FadB (3). A thiolase, FadA (4), releases acetyl-CoA from the β -ketoacyl-CoA to form an acyl-CoA for subsequent cycles. (5) Polyunsaturated fatty acids are reduced by the 2,4-dienoyl-CoA reductase (FadH). (6) FadB also catalyzes the isomerization of *cis*-unsaturated fatty acids to *trans*. (7) The epimerase activity of FadB converts D- β -hydroxy thioesters to their L-enantiomers via the *trans*-2-enoyl-CoA.

The second step in the cycle is enoyl-CoA hydratase, an activity commonly referred to as 'crotonase'. Traditionally, *in vitro* measurements of this activity utilize crotonoyl-CoA (*trans*-2-butenoyl-CoA) as the substrate. Crotonase activity in *E. coli* is one function present in a multifunctional protein encoded by *fadB*. The next step in the cycle is β -hydroxyacyl-CoA dehydrogenase, another function of the FadB enzyme. The β -ketoacyl-CoA produced in this reaction is a substrate for the monofunctional *fadA*-encoded β -ketoacyl thiolase, which cleaves acetyl-CoA from the acyl-CoA to produce an acyl-chain two carbons shorter than when it entered the cycle. The cycle is then repeated until the fatty acid is metabolized. The FadB protein is a homodimer of 78-kDa subunits, and is purified in complex with the homodimeric 42-kDa *fadA* gene product. The total complex is thus an $\alpha_2\beta_2$ heterotetramer with an apparent mass of about 260 kDa. The *fadA* gene encodes the β -subunit, while the *fadB* gene gives the α -subunit, the confusing nomenclature a remnant of the days of classical genetics.

Unsaturated fatty acids can also be degraded by the β -oxidation pathway. The FadB protein possesses *cis*- β -enoyl-CoA isomerase activity, which converts *cis*-3 double bonds to *trans*-2 (Fig. 8). A 2,4-dienoyl-CoA reductase encoded by *fadH* is also required for the metabolism of polyunsaturated fatty acids (Fig. 8). This protein is a 73-kDa monomeric, NADP⁺-dependent, 4Fe-4S flavoprotein. The FadH protein can utilize compounds with either *cis* or *trans* double bonds at the 4-position. An epimerase activity of FadB allows for the utilization of D-hydroxy fatty acids. The epimerase is actually a combination of a D- β -hydroxyacyl-CoA dehydratase and the

Table 2
Phospholipid degradative activities in *E. coli*

Enzyme	Gene	Location	Substrates
Phospholipase A ₁	<i>pldA</i>	Outer membrane	Phosphatidylethanolamine, phosphatidylglycerol, cardiolipin and lyso derivatives
Phospholipase A	<i>pldB</i>	Cytoplasm	Phosphatidylglycerol
Lysophospholipase L2		Inner membrane	Lyso-phosphatidylethanolamine
Lysophospholipase		Cytoplasm	Lyso-phosphatidylethanolamine, lyso-phosphatidylglycerol
Phospholipase C		Unknown	Phosphatidylethanolamine
Phospholipase D		Cytoplasm	Cardiolipin
Phospholipase D		Cytoplasm	Phosphatidylserine
Lipase		Membrane	Triacylglycerol
CDP-diacylglycerol hydrolase		Inner membrane	CDP-diacylglycerol
Phosphatidic acid phosphatase		Membrane	Phosphatidic acid
Thioesterase I	<i>tesA</i>	Periplasm	Acyl-CoA
Thioesterase II	<i>tesB</i>	Cytoplasm	Acyl-CoA

crotonase (hydratase) activities, resulting in the conversion of the D to the L enantiomer (Fig. 8).

The substrate specificities of the enzyme complex *in vitro* suggests that all of the enzymes can utilize all chain lengths of substrates, with the possible exception of the crotonase activity. This function of FadB appears somewhat limited to short chain substrates, and it has been suggested that a separate long chain enoyl-CoA hydratase may exist in *E. coli*. Two open reading frames in the *E. coli* genome, discovered by bioinformatics, are predicted to encode homologs of FadA and FadB. Thus, two complexes may be present with preferences for long or short chain acyl-CoAs.

8.2. Phospholipases

Based mainly on cell free assays, 10 enzymatic activities that degrade phospholipids, intermediates in the phospholipid biosynthetic pathway, or triacylglycerol have been reported (Table 2). The detergent-resistant phospholipase A₁ (encoded by *pldA*) of the outer membrane, characterized by Nojima and colleagues, is the most studied of these enzymes. This enzyme is unusually resistant to inactivation by heat and ionic detergents and requires calcium for maximal activity. The mature phospholipase has a subunit molecular mass of 31 kDa. Hydrolysis of fatty acids from the 1-position of phospholipids is the most rapid reaction, but the enzyme will also hydrolyze 2-position fatty acids, as well as both isomeric forms of lysophosphatides and mono- and di-acylglycerols. A detergent-sensitive phospholipase A₁ has also been described, although this activity has not been designated to a gene. This enzyme differs from the detergent-resistant protein in that it is located in the soluble fraction of the cell, is inactivated by heat and ionic detergents, and has a high degree of specificity for PG. The cytoplasmic phospholipase A also requires calcium for activity. There are also inner membrane and

cytoplasmic lysophospholipases. The best characterized of these is the inner membrane lysophospholipase L2 (*pldB*) which hydrolyzes 2-acyl-glycerophosphoethanolamine efficiently, but is barely active on the 1-acyl isomer. This lysophospholipase also catalyzes the transfer of fatty acids from 2-acyl-glycerophosphoethanolamine to PG to form acyl-PG.

The physiological role of these degradative enzymes remains unknown. Mutants lacking the detergent-resistant phospholipase (*pldA*), lysophospholipase L2 (*pldB*), or both enzymes do not have any obvious defects in growth, phospholipid composition, or turnover. Moreover, strains that overproduce the detergent-resistant enzyme also grow normally. It has been established that the detergent-resistant phospholipase is responsible for the release of fatty acids from phospholipids that occurs during infection with T4 and λ phages. However, phospholipid hydrolysis is not essential for the life cycle of these bacteriophages. One possible function for the hydrolytic activities with unassigned genes is that they are actually biosynthetic proteins (acyltransferases) that act as lipases in the absence of suitable acceptor molecules in the assay systems employed. An example of such an enzyme is PS synthase, which catalyzes both phospholipase D and CDP-diacylglycerol hydrolase reactions. PS synthase appears to function via a phosphatidyl-enzyme intermediate, and in the absence of a suitable acceptor such as serine or CMP, the phosphatidyl-enzyme complex is hydrolyzed by water; thus, the enzyme acts as a phospholipase D. Also, some of these enzyme activities may reflect a broad substrate specificity of a single enzyme rather than the presence of several distinct protein species. For example, the observed lipase activity that cleaves the 1-position fatty acids from triacylglycerols (a lipid usually not found in *E. coli*) may arise from the detergent-resistant phospholipase A₁ acting on triacylglycerol as an alternate substrate. Further, a lysophospholipase L(1) activity has been attributed to thioesterase I. Three open reading frames (*ybaC*, *yhjY*, and *yiaL*) present in the *E. coli* genome potentially encode proteins with lipase activity, but have not been studied to date.

8.3. Thioesterases

Thioesterases preferentially cleave the thioester bond of acyl-CoA molecules to produce CoA and free fatty acid. *E. coli* contains two well characterized thioesterases. Thioesterase I (encoded by *tesA*) is a periplasmic enzyme of 20.5 kDa, with a substrate specificity for acyl chains >12 carbon atoms. Thioesterase I hydrolyzes synthetic substrates used in the assay of chymotrypsin, which led to the initial conclusion that TesA was a protease ('protease I'). However, the purified protein does not cleave peptide bonds. Thioesterase I also appears to possess lysophospholipase L(1) activity. Thioesterase II is a cytosolic tetrameric protein composed of 32-kDa subunits encoded by the *tesB* gene. Thioesterase II cleaves acyl-CoAs of >6 carbons and β -hydroxyacyl-CoAs, but is unable to cleave acyl-pantetheine thioesters. The physiological function of thioesterases I and II is unknown. Null mutants have been constructed in both *tesA* and *tesB*, and the double mutant strain generated. None of these strains has an observable growth phenotype indicating that neither protein is essential. However, the *tesAB* double-null mutant still retains about 10% of the total wild-type thioesterase activity indicating the existence of a third unidentified thioesterase in *E. coli*.

9. Phospholipid turnover

9.1. The diacylglycerol cycle

The polar head group of PG is rapidly lost in a pulse-chase experiment, whereas that of PE is stable. The conversion of PG to CL, catalyzed by CL synthase, does not account for all the loss of ^{32}P -labeled PG observed in the pulse-chase experiments. A phosphate-containing non-lipid compound derived from the head group of PG was sought, which led to the discovery of the membrane-derived oligosaccharides (MDO) by Kennedy's group [17]. These compounds are composed of *sn*-glycerol-1-phosphate (derived from PG), glucose and (usually) succinate moieties, and have molecular weights in the range of 4000–5000. They are found in the periplasm of Gram-negative bacteria, an osmotically sensitive compartment. The synthesis of the MDO compounds is regulated by the osmotic pressure of the growth medium and decreased osmotic pressure gives an increased rate of MDO synthesis. Thus, MDO compounds seem to be involved in osmotic regulation.

In the synthesis of MDO, the *sn*-glycerol-1-phosphate polar headgroup of PG is transferred to the oligosaccharide, with 1,2-diacylglycerol as the other product (Fig. 9). The 85-kDa transmembrane protein that catalyzes this reaction is encoded by *mdoB*. Diacylglycerol kinase phosphorylates the diacylglycerol to phosphatidic acid, which reenters the phospholipid biosynthetic pathway (Fig. 6) to complete the diacylglycerol cycle (Fig. 9). In the overall reaction only the *sn*-glycerol-1-phosphate portion of the PG molecule is consumed; the lipid portion of the molecule is recycled back into phospholipid. MDO synthesis is responsible for most of the metabolic instability of the polar group of PG, since blocking MDO synthesis at the level of oligosaccharide synthesis by lack of UDP-glucose greatly reduces PG turnover. Moreover, the rate of accumulation of diacylglycerol in strains lacking diacylglycerol kinase (*dgk*) correlates with the presence of both the oligosaccharide acceptor and the osmolarity of the growth medium. *dgk* is immediately upstream of *plsB* in *E. coli*. The protein is a trimer of identical 13-kDa subunits, each with three predicted trans-membrane helices. Activity appears to be limited by diffusion of substrate across the membrane to the cytoplasmic active site. It should be noted that some species of MDO contain phosphoethanolamine. Although direct proof is lacking, it is likely that the ethanolamine moiety is derived from PE, as this is the only known source of ethanolamine.

9.2. The 2-acylglycerolphosphoethanolamine cycle

2-Acylglycerolphosphoethanolamine is a minor membrane lipid in *E. coli* generated from fatty acid transfer of the acyl moiety at the 1-position of PE to the outer membrane lipoprotein. 2-Acylglycerolphosphoethanolamine acyltransferase is an inner membrane enzyme that esterifies the 1-position of 2-acylglycerolphosphoethanolamine utilizing acyl-ACP (and not acyl-CoA) as the acyl donor. The acyltransferase was first recognized as a protein called acyl-ACP synthetase that catalyzes the ligation of fatty acids to ACP, hence the gene designation *aas*. ACP acts as a bound subunit for accepting the acyl intermediate in the normal acyltransferase reaction and high salt

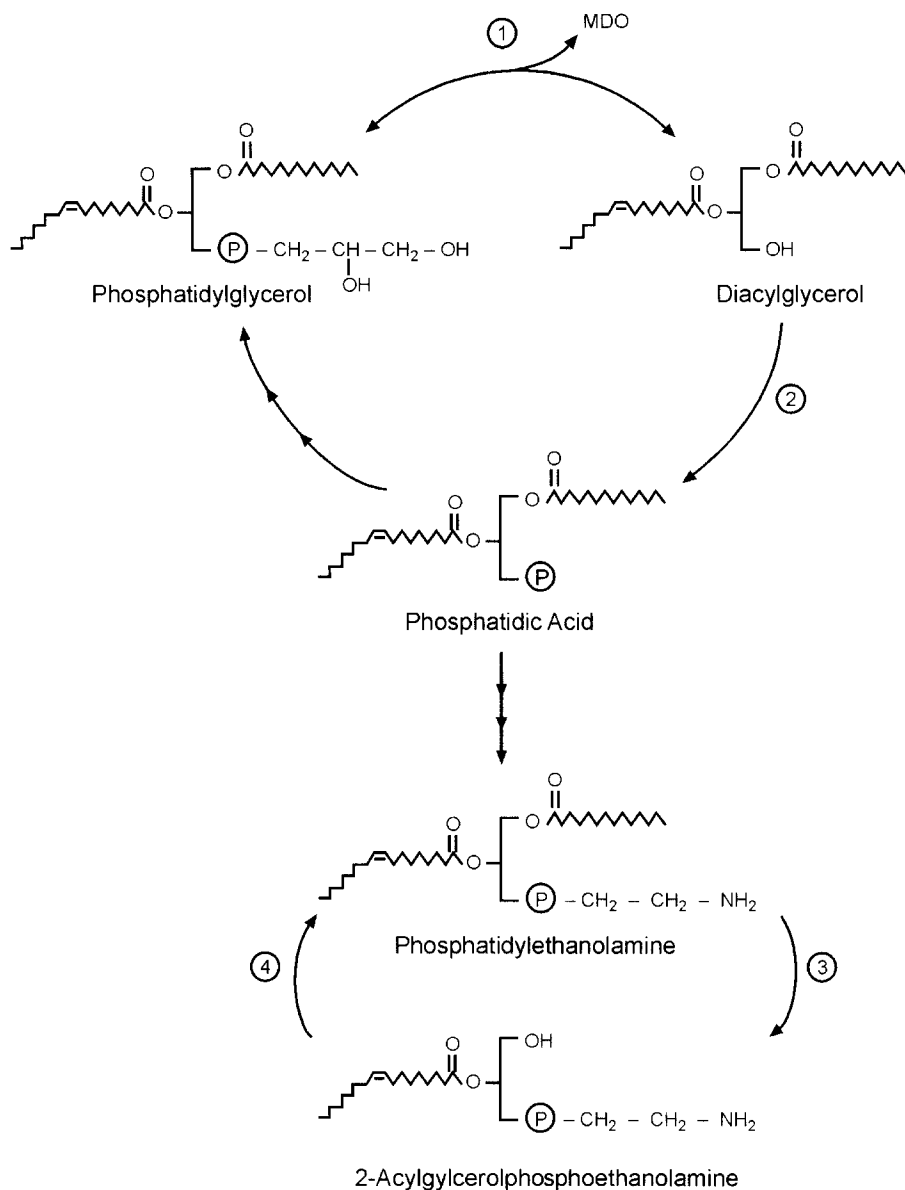


Fig. 9. Phospholipid turnover. The 1,2-diacylglycerol kinase cycle involves (1) the transfer of the *sn*-1-glycerol-phosphate moiety from phosphatidylglycerol to membrane derived oligosaccharides (MDO) by the enzyme MdoB. (2) Diacylglycerol kinase converts the diacylglycerol to phosphatidic acid, which can regenerate the phosphatidylglycerol (see Fig. 6). Phosphatidylethanolamine cycling involves (3) the transfer of an acyl chain to membrane lipoprotein and (4) re-esterification of the 1-position by 2-acylglycerolphosphoethanolamine (Aas).

concentrations are required to dissociate the acyl-ACP intermediate from the enzyme in vitro. However, 2-acylglycerolphosphoethanolamine acyltransferase is the only reaction catalyzed by Aas in vivo. *aas* mutants are defective in both acyl-ACP synthetase and 2-acylglycerolphosphoethanolamine acyltransferase activities in vitro. They do not accumulate 2-acylglycerolphosphoethanolamine in vivo unless they are also defective in the *pldB* gene which encodes a lysophospholipase that represents a second pathway for 2-acylglycerolphosphoethanolamine metabolism. The acyl-ACP synthetase reaction has proven extremely valuable in the preparation of acyl-ACPs for use as substrates and inhibitors for the enzymes of fatty acid synthase.

10. Regulation of lipid metabolism

10.1. Regulation of fatty acid chain length

Fatty acyl chains in the membrane phospholipids of *E. coli* are normally 16 or 18 carbons in length. This specificity is a result of a combination of two factors: the poor reactivity of the β -ketoacyl-ACP synthases for longer chains; and the high specificity of the acyltransferases for 16- and 18-carbon products. Overexpression of FabB leads to the overproduction of *cis*-vaccenate, which is incorporated into the membranes. Overexpression of FabH causes a decrease in the average fatty acid chain length and the appearance of significant amounts of myristic acid (14:0) in the phospholipids. This effect is attributed to an increased rate of fatty acid initiation, which leads to a deficiency in malonyl-ACP for the terminal elongation reactions. The fatty acid biosynthetic machinery has the capacity to produce longer chain fatty acids. Under normal conditions, the 16 or 18 carbon chains are removed from the cytoplasm by the action of the acyltransferases. However, when phospholipid synthesis is blocked at the acyltransferase step, the fatty acids that accumulate have abnormally long chain lengths (e.g. 20 and 22 carbons). Conversely, overproduction of the acyltransferase results in a somewhat decreased average chain length, represented mainly by an increase in myristic acid. Thus, competition among the elongation synthases, the supply of malonyl-ACP, and the utilization of acyl-ACPs by the acyltransferase are the most significant determinants of fatty acid chain length.

10.2. Temperature modulation of fatty acid composition

All organisms regulate the fluidity of their membranes to maintain a membrane bilayer in a largely fluid state. As temperatures are lowered, membranes undergo a reversible change from a fluid (disordered) to a non-fluid (ordered) state. In *E. coli*, the temperature of the transition point depends on the fatty acid composition of the membrane phospholipids [11]. At lower temperatures, the amount of *cis*-vaccenic acid is rapidly (within 30 s) increased due to the increased activity of FabF. Synthesis of mRNA and protein are not required. Overexpression of FabB also increases the amount of this fatty acid in the membrane, although in a temperature-independent manner. Mutants that lack FabF are unable to modulate their fatty acid composition in a temperature-

dependent manner. Thus FabF, and not FabB, is involved in the thermal regulation of the fatty acid composition of the membranes.

As a result of the specificity of the acyltransferases, palmitic acid occupies position 1 of the phospholipid backbone at 37°C, whereas palmitoleic and *cis*-vaccenic acids are found at position 2. As the growth temperature is lowered, *cis*-vaccenic acid competes with palmitic acid for position 1 of the newly synthesized phospholipids. This indicates that the specificity of the acyltransferases changes during this temperature shift also, although how this happens is not yet understood.

10.3. Transcriptional regulation of the genes of fatty acid synthesis and degradation

The known genes of fatty acid synthesis are scattered along the genome with only two clusters, the minimal *accBC* operon and the *fab* cluster. The *fab* cluster contains the *fabH*, *fabD*, *fabG*, *acpP*, and *fabF* genes and may have functional significance, since the work of Cronan has demonstrated that several genes are cotranscribed. However, most genes also appear to have a unique promoter, and the full relevance of this cluster is not yet fully understood.

For balanced production of each member of the acetyl-CoA carboxylase complex, one might expect each gene to be regulated in the same manner. However, while transcription of all four *acc* genes is under growth rate control, with the rate of transcription decreasing with decreased growth rate, the *accBC* operon seems to be regulated by a mechanism that differs from the regulation of the *accA* and *accD* genes. The *accBC* operon is transcribed from a promoter located unusually far upstream of the *accB* gene. The major *accA* promoter lies within the coding sequence of the upstream *polC* (*dnaE*) gene, although transcription through *polC* and perhaps other upstream genes also reads through the *accA* sequence. The *accD* gene is transcribed from a promoter located within the upstream *dedA* gene.

Transcriptional regulation of the other genes of fatty acid synthesis is no less complicated. The FadR protein, which was first identified as a repressor of transcription of genes in the *fad* regulon of β -oxidation and fatty acid transport, also positively regulates *fabA* transcription. FadR binds to DNA in the absence of acyl-CoA, to repress the β -oxidation regulon and activate *fabA*. Acyl-CoA (formed from exogenous fatty acids transported into the cell), binds FadR and the protein is released from the DNA. The molecular details of these interactions have been examined by the crystallization of FadR and the FadR–acyl-CoA and FadR–DNA complexes. Whether FadR activates or represses transcription depends on the location of its binding site within the promoter region. For repression, FadR binds in the –30 to +10 region of the promoter and prevents binding of DNA polymerase. For activation, the FadR operator site is located in a 17-bp region at –40 of the *fabA* promoter, and FadR binding promotes DNA polymerase binding. In *fadR* null mutants, the *fabA* gene is transcribed from two weak promoters of about equal strength whereas in wild-type strains, a 20-fold increase in transcription from the proximal promoter is seen in the absence of acyl-CoA. Thus, FadR monitors the intracellular concentration of acyl-CoA and coordinately regulates fatty acid synthesis and β -oxidation in response to these compounds.